

# The joint scaling limit of critical Ashkin–Teller currents

OpenAI

October 6, 2026

## Abstract

For each fixed point on the critical Ashkin–Teller line, including the four-state Potts endpoint, we prove the conjectured joint scaling limit of the height and both current-cluster collections in every bounded Jordan domain, for every admissible polygonal approximation. The height converges to a Gaussian free field with the predicted coupling constant, and the clusters converge to canonical recursive two-valued local sets of that same field. The joint limit includes all nesting depths and the distinguished wired boundary cluster.

## Contents

|          |                                                              |           |
|----------|--------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                          | <b>2</b>  |
| <b>2</b> | <b>The model and the limiting objects</b>                    | <b>4</b>  |
| 2.1      | Parameters and lattice domains . . . . .                     | 4         |
| 2.2      | The coupled currents and their height . . . . .              | 4         |
| 2.3      | An auxiliary representation . . . . .                        | 6         |
| 2.4      | Continuum normalization and two-valued sets . . . . .        | 6         |
| 2.5      | Topology and the theorem . . . . .                           | 7         |
| 2.6      | Stopping and limiting conventions . . . . .                  | 9         |
| 2.7      | Proof strategy . . . . .                                     | 9         |
| <b>3</b> | <b>Cuts, comparison, and spatial estimates</b>               | <b>11</b> |
| 3.1      | Exact cuts and the comparison order . . . . .                | 11        |
| 3.2      | Buffered circuits and rough walls . . . . .                  | 15        |
| 3.3      | Counts and comparison of complete patterns . . . . .         | 21        |
| 3.4      | Extending separation witnesses . . . . .                     | 28        |
| <b>4</b> | <b>Reflection positivity and the plane covariance</b>        | <b>33</b> |
| 4.1      | Four positive reflection frames . . . . .                    | 34        |
| 4.2      | The spectral measure and its Cauchy disintegration . . . . . | 35        |
| 4.3      | An angular defect in two frames . . . . .                    | 36        |
| 4.4      | Green covariance and reflection nullity . . . . .            | 37        |
| <b>5</b> | <b>Screening a Laplace insertion</b>                         | <b>39</b> |
| 5.1      | Uniform bounds under separated patterns . . . . .            | 40        |
| 5.2      | Moving arbitrary patterns . . . . .                          | 42        |
| 5.3      | Joining the gaps without revealing their colors . . . . .    | 44        |
| 5.4      | Transfer to flat domains . . . . .                           | 47        |

|          |                                                                    |            |
|----------|--------------------------------------------------------------------|------------|
| <b>6</b> | <b>Flat-domain Gaussian limits</b>                                 | <b>48</b>  |
| 6.1      | Limiting chambers and their Green kernels . . . . .                | 49         |
| 6.2      | Moments and boundary traces . . . . .                              | 50         |
| 6.3      | Harmonic moment kernels and their collision coefficients . . . . . | 54         |
| 6.4      | Wick’s rule in the limiting chambers . . . . .                     | 57         |
| 6.5      | Plane increment kernels and their collisions . . . . .             | 58         |
| 6.6      | Identifying the coefficient without circularity . . . . .          | 60         |
| 6.7      | Completion of the flat limit and conditioning at a cut . . . . .   | 61         |
| <b>7</b> | <b>Stopped local sets and canonical clusters</b>                   | <b>63</b>  |
| 7.1      | Stops and their cut neighborhoods . . . . .                        | 63         |
| 7.2      | The conditional field across a stopped cut . . . . .               | 66         |
| 7.3      | The stopped two-valued sets . . . . .                              | 69         |
| 7.4      | Fields at seed stops . . . . .                                     | 70         |
| 7.5      | Geometry of the candidate clusters . . . . .                       | 70         |
| <b>8</b> | <b>Local cable tests and the no-crossing principle</b>             | <b>74</b>  |
| 8.1      | Auxiliary cables and the tests . . . . .                           | 74         |
| 8.2      | The reverse buffer estimate . . . . .                              | 75         |
| 8.3      | A domain-independent local kernel . . . . .                        | 76         |
| 8.4      | Calibration on killed lattice graphs . . . . .                     | 79         |
| 8.5      | Forcing in the canonical parent . . . . .                          | 81         |
| 8.6      | The zero-boundary collar bound . . . . .                           | 85         |
| 8.7      | No crossing of a stopped component . . . . .                       | 86         |
| <b>9</b> | <b>Identification of the complete cluster collections</b>          | <b>88</b>  |
| 9.1      | Marked odd holes and their supplying clusters . . . . .            | 88         |
| 9.2      | A flat region cannot contain a revealed crossing . . . . .         | 92         |
| 9.3      | Open hole witnesses and the discovered mean . . . . .              | 94         |
| 9.4      | Complete matching, including invisible clusters . . . . .          | 97         |
| 9.5      | Completion of the joint limit . . . . .                            | 100        |
| <b>A</b> | <b>Conventions and elementary corrections in the inputs</b>        | <b>101</b> |
| A.1      | The path decomposition in the increment estimate . . . . .         | 101        |
| A.2      | Slope, free energy, and the Gaussian multiplier . . . . .          | 102        |
| A.3      | The conformal growth estimate . . . . .                            | 103        |

## 1 Introduction

The Ashkin–Teller model originates in the four-component lattice model of Ashkin and Teller [10]. Its current representation raises a geometric question beyond convergence of the associated height: what is the joint continuum law of all primal and dual current clusters, and how are these collections coupled to the field? We prove the joint limit in [Theorem 2.3](#) for every fixed point on the critical line, including the four-state Potts endpoint. This is the height and current-cluster assertion formulated by Alcalde López, Heeney, and Lis [4, Conjecture 1.2].

Fix  $J > 0$  and  $U \leq J$  with  $\sinh(2J) = e^{-2U}$ . Our zero-boundary Gaussian free field has covariance  $2\pi(-\Delta_D)^{-1}$ , with logarithmic singularity  $-\log|z - w|$ . In this normalization put

$$\lambda = \frac{\pi}{2}, \quad a = \lambda\sqrt{g}, \quad g = \frac{8}{\pi} \arcsin\left(\frac{\sqrt{1 + e^{4U}}}{2}\right).$$

Thus  $4/3 < g \leq 4$  and  $\lambda < a \leq 2\lambda$ . For the wired current, the first limiting cluster is the symmetric two-valued local set with labels  $\pm a$ . For the free current, first expose  $\text{CLE}_4$  holes and retain an asymmetric two-valued split inside each hole; repeat in all complementary components. Both constructions use the same field. The theorem retains every nesting depth and the distinguished wired boundary cluster. It includes tightness and identification of the entire compact supports in positive-diameter Hausdorff matching topology.

### Prior work and the scope of the result

The finite spin, percolation, and height representation used here is due to Lis [27, Sections 1.1–1.3 and 2.2]; his current expansion [26, Section 2] gives the precise odd/even-edge weights. A noncrossing primal–dual contour representation appears earlier in Rys [31, Section 3]. We recall and verify the finite coupling, including its boundary convention, in Section 2.

At  $U = 0$ , the currents are double random currents and  $g = 2$ . Duminil-Copin and Lis [13, Theorems 1.1–1.2] identified the finite signed nesting field with a dimer height. Duminil-Copin, Lis, and Qian [15] established conformally invariant limits for the nesting field and recursive cluster-boundary families. Their introductory statements cover Jordan domains, while the detailed joint boundary-family Theorems 6.2 and 6.4 assume a  $C^1$  boundary and use injective matching of loops. Alcalde López, Heeney, and Lis [4, Theorem 1.14 and Appendix B] state the joint limit of the height and both complete compact current-cluster collections in Jordan domains, using the noninjective positive-diameter matching topology considered here. Their Conjecture 1.2 asks for the critical-line extension obtained by replacing  $\sqrt{2}$  with  $\sqrt{g}$ .

Alcalde López and Sepúlveda [3, Theorems 1.4 and 9.5, and Lemma 9.1] study a stronger joint decomposition at the Ising point involving the XOR field, cluster measures, and signs, with the Gaussian field identified with the nesting height. Their introductory theorem is stated for Jordan domains; their detailed joint convergence theorem adds a boundary Minkowski-dimension hypothesis. Those additional field and measure limits are outside our statement. Our conclusion concerns the height and complete current supports for every fixed  $4/3 < g \leq 4$ .

The analytic input is related to the six-vertex model with weights  $(1, 1, c)$ , where  $c = \sqrt{1 + e^{4U}} \in (1, 2]$ . Duminil-Copin, Kozłowski, Lammers, and Manolescu [16] prove a height scaling limit for  $\sqrt{3} \leq c \leq 2$  and establish several inputs on the larger interval  $1 \leq c \leq 2$ : logarithmic bounds for even increment moments, one-parity spin-percolation circuits, a specified flat-exhaustion plane law, and a free-energy identification of the normalization conditional on a Gaussian plane limit. Odd plane correlations vanish by height reversal. We use these larger-range results. The covariance, screening, and Gaussianity arguments needed for the whole interval are given here; the free-energy formula is applied only after its convergence premise has been established. A related full-plane height theorem for the balanced six-vertex state is proved in [29, Theorem 1.1]. Its unit height corresponds to  $K = 2H$  here, and its multiplier  $1/\arcsin(c/2)$  agrees with ours after this normalization. This comparison does not replace the local Gaussianity proof or the random-domain and cluster identifications below.

For the reflection argument, coordinate and diagonal tests for six-vertex weights already appear in [21, Section 6]. We use the general reflection-positive framework of [20]; for the

reflected Hilbert-space construction and positive-transfer representation, see [33, Sections 3.1–3.3 and 4.1]. The height transfer in [16, Section 4.5.3] is a closer model-specific precedent and permits negative one-step spectrum. The transfer, screening, and collision strategy also has a related implementation for integer Lipschitz heights with two-arc boundary data in [30, Sections 5, 7, and 9–10]. We prove the square-frame and random-chamber arguments required here.

The continuum identification uses thin two-valued local sets [5, 6], first-passage sets [7, 8], and the canonical excursion decomposition [9]. The recursive exploration and thinness architecture also follows the double-current work [15]. In the present coupling, we must additionally identify which discrete odd holes belong to a single current cluster, exclude macroscopic pieces invisible to those holes, and compare passage tests after a second coupled random exploration. We prove these steps with the conditioning and approximation hypotheses stated in the corresponding propositions. A proof strategy follows the formal theorem in Section 2.7.

## 2 The model and the limiting objects

### 2.1 Parameters and lattice domains

Fix real numbers  $J > 0$  and  $U \leq J$  satisfying

$$\sinh(2J) = e^{-2U}.$$

Throughout the proof these parameters remain fixed. Put

$$t = (1 + e^{4U})^{-1/2}, \quad c = t^{-1}, \quad \theta = \arcsin(c/2), \quad g = \frac{8\theta}{\pi}. \quad (2.1)$$

Then  $1/2 \leq t < 1$ ,  $1 < c \leq 2$ , and  $4/3 < g \leq 4$ . The value  $U = 0$  gives  $g = 2$ ; the endpoint  $J = U = (\log 3)/4$  gives  $g = 4$ . Constants below may depend on the fixed parameter. We make no claim of uniformity as  $t \uparrow 1$ .

Let  $D \subset \mathbb{R}^2$  be a bounded Jordan domain. An admissible approximation is a family of simple nearest-neighbor polygons  $P_\delta$  in  $\delta\mathbb{Z}^2$  such that their boundaries have parametrizations converging uniformly to a parametrization of  $\partial D$ , and every compact subset of  $D$  eventually lies inside  $P_\delta$ . Let  $G_\delta = (V_\delta, E_\delta)$  contain all lattice vertices and edges on or inside the polygon, and let  $B_\delta$  be its boundary-cycle edges. The weak dual  $G_\delta^\dagger$  has a vertex at each bounded-face center and an edge across each edge of  $E_\delta \setminus B_\delta$ . We draw all primal and dual edges as straight segments. Boundary contraction will be used only to describe current weights, not when drawing clusters.

### 2.2 The coupled currents and their height

We use the spin–edge–height construction of [27, Sections 1.1–1.3], with its wired/free boundary specialization made explicit below. Sample spins

$$\tau : V_\delta \longrightarrow \{-1, 1\}, \quad \tau^\dagger : V(G_\delta^\dagger) \longrightarrow \{-1, 1\},$$

with  $\tau = 1$  on  $P_\delta$ . For each mutually dual pair  $e = uv$ ,  $e^\dagger = f'f''$ , impose

$$(\tau(u) - \tau(v))(\tau^\dagger(f) - \tau^\dagger(f')) = 0. \quad (2.2)$$

Give an admissible pair weight

$$t^{N(\tau) + N(\tau^\dagger)}, \quad (2.3)$$

where  $N$  counts nearest-neighbor disagreements on the respective graph. No boundary constraint is imposed on the dual spins.

Conditional on the spins, all edges of  $B_\delta$  are primal open. For each remaining mutually dual pair, independently of the other pairs, use the following rule:

| Spin information                        | Primal/dual open states  | Probabilities          |
|-----------------------------------------|--------------------------|------------------------|
| $\tau^\dagger(f) \neq \tau^\dagger(f')$ | $(1, 0)$                 | 1                      |
| $\tau(u) \neq \tau(v)$                  | $(0, 1)$                 | 1                      |
| Neither disagreement                    | $(1, 0), (0, 1), (0, 0)$ | $1 - t, 1 - t, 2t - 1$ |

The constraint makes the first two rows mutually exclusive. Denote the resulting edge sets by  $\hat{n}_\delta$  and  $\hat{n}_\delta^\dagger$ . Their open edges never cross. Both may be closed at a mutually dual pair; the two currents are not generally complementary.

After contracting the wired boundary and deleting its cycle edges, the primal odd edges are those crossing dual spin disagreements. The other primal open edges are even. In the current expansion of [26, Section 2, Equations (2)–(3) and Proposition 1], their law specializes to

$$\mathbb{P}(n_{\text{odd}}, n_{\text{even}}) \propto 2^{k(n_{\text{odd}} \cup n_{\text{even}})} x^{|n_{\text{odd}}|} y^{|n_{\text{even}}|}, \quad x = 1, \quad y = c - 1, \quad (2.4)$$

with  $n_{\text{odd}}$  Eulerian and disjoint from  $n_{\text{even}}$ . Here  $k$  includes isolated vertices. To check the weight directly, an odd edge contributes  $t$ , an even edge contributes  $1 - t$ , and a closed primal edge contributes  $t$ , after summing its possible dual states. The common factor is  $t^{|E_\delta \setminus B_\delta|}$ , leaving  $((1 - t)/t)^{|n_{\text{even}}|}$ . The primal colors are constant on open components and give  $2^{k-1}$  choices with the wired color fixed. An admissible Eulerian odd set determines the dual colors up to a common flip. These graph-dependent constant factors yield Equation (2.4). The same calculation gives free boundary conditions for the dual current. In the original parameters,

$$x = e^{2U} \sinh(2J) = 1, \quad y = e^{2U} \cosh(2J) - 1 = \sqrt{1 + e^{4U}} - 1.$$

It is the spin-and-edge construction above, not just Equation (2.4), that specifies the joint law.

Set  $H_\delta = 0$  on  $P_\delta$ , and define corner increments by

$$H_\delta(f) - H_\delta(v) = \frac{1}{2} \tau(v) \tau^\dagger(f) \quad \text{when } v \text{ is incident to } f. \quad (2.5)$$

The circulation around an interior primal edge is

$$\frac{1}{2} (\tau(u) - \tau(v)) (\tau^\dagger(f) - \tau^\dagger(f')) = 0.$$

All boundary primal increments are zero. Since the polygon is simple, these elementary circuits generate the corner graph's cycle space, so Equation (2.5) defines a unique height. It is integer-valued on primal vertices and half-integer-valued at face centers. We also use

$$K_\delta = 2H_\delta.$$

Thus a nearest corner increment of  $K_\delta$  has magnitude one. Conversely, the spins are recovered from its residues, up to a common sign  $\varepsilon$ , by

$$\tau(v) = \varepsilon(-1)^{K_\delta(v)/2}, \quad \tau^\dagger(f) = \varepsilon(-1)^{(K_\delta(f)-1)/2}.$$

Around a corner-lattice plaquette there are six possible increment patterns. The two patterns with both opposite diagonals flat have spin weight 1; the other four have weight  $t$ . Dividing by the common factor  $t$  gives the six-vertex weights  $(1, 1, c)$ , with  $c = 1/t$ ; see also [27, Section 2.2, Corollary 2.5]. The corner lattice is a fixed rotation and dilation of the square lattice. This local identification, together with the infinite-volume identification recalled in Section 4, is what permits the six-vertex estimates used below. As a distribution on  $D$ , the height is

$$H_\delta(\varphi) = \delta^2 \sum_{v \in V_\delta} H_\delta(v) \varphi(v), \quad \varphi \in C_c^\infty(D). \quad (2.6)$$

This is an atomic distribution, which belongs locally to  $H^s$  for  $s < -1$ . Interpolated heights used in intermediate estimates will be compared with this precise smearing.

Let  $\mathcal{C}_\delta^w$  be the collection of all connected components of the primal open subgraph, including its boundary component  $C_{\partial, \delta}$ , and let  $\mathcal{C}_\delta^f$  be the collection of all dual open components. A component includes its vertices and drawn edges. We retain components at every nesting depth.

### 2.3 An auxiliary representation

When analyzing one of the two currents, denote it by  $N$ , and denote its exact dual complement by  $W$ . This complement is an auxiliary spin percolation, not the other current of the joint sample. The spin constant along  $W$  is called  $s$ , and the spin constant along  $N$  is called  $d$ . At an ambiguous pair,  $W$  is open with probability  $t$ , whereas the other current is open with probability  $1 - t$ .

A *matching wall* means a revealed monochromatic open path or perimeter in the  $W$ -lattice. Its connections in the revealed region are retained as boundary connections; separate appearances of the same connected perimeter are not treated as unrelated wires. For a free current, the complementary  $W$ -percolation is wired. The finite cut law in Theorem 3.1 explains why the height in a component behind a matching wall has constant boundary data. All conditional comparisons below refer to this representation.

The same cut law gives the vocabulary for nesting. A hole of a free  $N$  component is *odd* when its adjacent  $s$  spin differs from the adjacent exterior  $s$  spin of that component. Crossing the component changes  $K_\delta$  by 2 times one relative sign in every odd hole, and by zero in every other hole. Conditional on  $(N, s)$ , these signs are independent and fair for distinct free components. After the wired boundary component is exposed, each residual chamber carries a free same-parity current; the value of  $K_\delta$  on its component-side  $W$  perimeter is  $+1$  or  $-1$  relative to the original boundary height. The edge factor and the jump identity proving these statements are given in Theorem 3.1.

### 2.4 Continuum normalization and two-valued sets

Let  $h$  be the zero-Dirichlet GFF in  $D$  with covariance

$$G_D = 2\pi(-\Delta_D)^{-1}.$$

Thus  $G_D(z, w) = -\log|z - w| + O(1)$  near the diagonal. Set

$$\lambda = \frac{\pi}{2}, \quad a = \sqrt{g} \lambda, \quad a^2 = 2\pi \arcsin(c/2). \quad (2.7)$$

In particular  $\pi/\sqrt{3} < a \leq \pi$ , so  $\lambda < a \leq 2\lambda$ . A zero-boundary GFF in an open subdomain is always extended by zero when viewed as a distribution on a larger domain.

We use the canonical thin two-valued local set  $A_{-\alpha,\beta}(f)$  of a zero-boundary GFF  $f$ , with  $\alpha, \beta > 0$  and  $\alpha + \beta \geq 2\lambda$ . It includes the domain boundary. The conditional harmonic function on each complementary component is either  $-\alpha$  or  $\beta$ ; after subtracting that value, the field in the component is an independent zero-boundary GFF. These sets are measurable functions of the field. We use their uniqueness, monotonicity, and Jordan-hole properties with the hypotheses stated in [Section 7](#); see [\[5–7\]](#). In particular the complementary loops of  $A_{-2\lambda,2\lambda}$  have law  $\text{CLE}_4$ .

**Definition 2.1** (Free cluster procedure). For a zero-boundary GFF  $f$  in a Jordan domain  $O$ , define  $\mathcal{F}_a(f, O)$  recursively as follows.

1. Expose  $A_{-2\lambda,2\lambda}(f)$ . In every complementary component  $V$ , write its label as  $\xi 2\lambda$ ,  $\xi \in \{-1, 1\}$ , and let  $f_V$  be the zero-boundary remainder.
2. If  $\xi = 1$ , retain  $A_{-2\lambda,2a-2\lambda}(f_V)$  as one cluster. If  $\xi = -1$ , retain  $A_{-(2a-2\lambda),2\lambda}(f_V)$ . The retained set includes its outer boundary  $\partial V$ .
3. In each complementary component of the retained set, subtract its conditional harmonic value and repeat the procedure with the zero-boundary remainder.

Keep all retained clusters from all recursive generations. The  $\text{CLE}_4$  set exposed in the first step is not itself retained.

If this procedure is invoked above an incoming flat value  $b$  of the original field, the CLE component has absolute label  $b + \xi 2\lambda$  and the retained split has absolute labels  $b$  and  $b + \xi 2a$ . We call the holes with the nonzero jump relative to this incoming value  $b$  *odd holes*. The use of this term in lattice explorations is consistent with the jump decomposition proved below.

Define

$$\mathcal{C}^f = \mathcal{F}_a(h, D).$$

For the wired collection, first retain

$$C_\partial = A_{-a,a}(h),$$

and apply  $\mathcal{F}_a$  to the zero-boundary remainder in every complementary component of  $C_\partial$ , retaining all resulting clusters together with  $C_\partial$ . Denote that collection by  $\mathcal{C}^w$ . Both constructions use the *same*  $h$ , including the canonical choices of all its local sets. This defines their joint coupling. The all-generation local finiteness needed below is proved in [Theorem 9.3](#), rather than assumed as part of the definition.

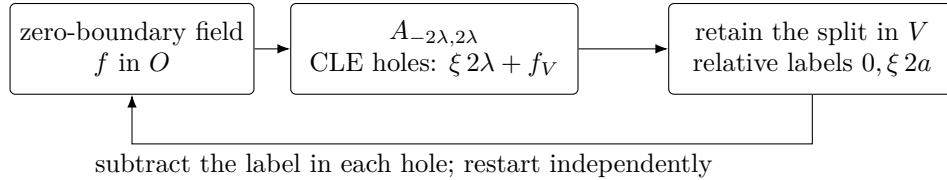


Figure 1: One recursive step of the free construction. Only the split inside a CLE hole is retained as a current cluster. For the wired construction, the boundary set  $A_{-a,a}$  is retained first.

## 2.5 Topology and the theorem

Fix a compact square  $X$  containing  $\overline{D}$  and all sufficiently fine polygonal approximations. Such a square exists by admissibility. Let  $\mathcal{H} = \mathcal{K}(X)$  be the space of nonempty compact subsets of

$X$ , with Hausdorff metric  $d_H$ , and let  $Z = \{\{x\} : x \in X\} \subset \mathcal{H}$  be its singleton subspace. A collection  $\mathcal{K} \subset \mathcal{H}$  is locally finite above positive diameter if

$$\#\{C \in \mathcal{K} : \text{diam}(C) > \varepsilon\} < \infty \quad \text{for every } \varepsilon > 0.$$

For such a collection put

$$F(\mathcal{K}) = Z \cup \{C \in \mathcal{K} : \text{diam}(C) > 0\}, \quad d_{\text{match}}(\mathcal{K}, \mathcal{K}') = d_H^{\mathcal{H}}(F(\mathcal{K}), F(\mathcal{K}')),$$

where  $d_H^{\mathcal{H}}$  is Hausdorff distance between compact subsets of  $\mathcal{H}$ . Local finiteness makes  $F(\mathcal{K})$  closed: any convergent sequence of distinct members has diameters tending to zero and therefore has a singleton limit. Thus  $d_{\text{match}}$  is a metric after singleton members and multiplicities are ignored.

This metric gives precisely the positive-diameter matching topology. To make the relation with the usual description explicit, write  $R_\varepsilon(\mathcal{K}, \mathcal{K}')$  for the following two requirements:

every  $C \in \mathcal{K}$  with  $\text{diam}(C) > \varepsilon$  has some  $C' \in \mathcal{K}'$  with  $d_H(C, C') < \varepsilon$ ,  
and the same statement holds with  $\mathcal{K}, \mathcal{K}'$  interchanged.

The matching need not be one-to-one. We retain the distinguished boundary component separately, with ordinary Hausdorff topology.

**Lemma 2.2** (Matching topology and compact containment). *As relations on locally finite collections,*

$$\{d_{\text{match}} \leq \varepsilon/2\} \subset R_\varepsilon \subset \{d_{\text{match}} \leq \varepsilon\}.$$

Let  $\mathfrak{X}$  be the compact space of closed subsets  $F \subset \mathcal{H}$  containing  $Z$ , and write  $N_t(F) = \#\{C \in F : \text{diam}(C) > t\}$ . If  $t_j \downarrow 0$  and every  $M_j$  is finite, then

$$\{F \in \mathfrak{X} : N_{t_j}(F) \leq M_j \text{ for every } j\}$$

is compact and consists of locally finite collections after singleton members are removed. Consequently, uniform tails for  $N_t$  at each fixed  $t > 0$  imply compact containment in the matching topology.

*Proof.* For every  $C \in \mathcal{H}$ ,

$$\frac{1}{2} \text{diam}(C) \leq \text{dist}_{\mathcal{H}}(C, Z) \leq \text{diam}(C).$$

The lower bound is the triangle inequality, and the upper bound follows by choosing a singleton in  $C$ . If  $d_{\text{match}} \leq \varepsilon/2$ , a member of diameter greater than  $\varepsilon$  must therefore match a positive-diameter member of the other family within  $\varepsilon/2$ . Conversely, under  $R_\varepsilon$  each member of diameter at most  $\varepsilon$  is within  $\varepsilon$  of  $Z$ , and each larger member has the required match. This proves the inclusions.

The hyperspace of nonempty compact subsets of a compact metric space is compact, and the condition  $Z \subset F$  is closed. For each  $t > 0$ ,  $N_t$  is lower semicontinuous: any finitely many distinct members of diameter greater than  $t$  persist in disjoint small hyperspace neighborhoods under Hausdorff convergence. Hence each constraint  $N_{t_j} \leq M_j$  is closed. Their intersection is compact, and choosing  $j$  with  $t_j < t$  proves finiteness above every  $t > 0$ . Finally, choose the  $M_j$  so that the respective tail probabilities are at most  $\eta 2^{-j}$ , and use the union bound. This proves compact containment.  $\square$

**Theorem 2.3** (Joint critical-line limit). *For every fixed pair  $J > 0$ ,  $U \leq J$  with  $\sinh(2J) = e^{-2U}$ , every bounded Jordan domain  $D$ , and every admissible polygonal approximation as defined in Section 2.1,*

$$(H_\delta, \mathcal{C}_\delta^w, \mathcal{C}_\delta^f, C_{\partial, \delta}) \xrightarrow[\delta \downarrow 0]{\text{law}} \left( \frac{h}{\pi\sqrt{g}}, \mathcal{C}^w, \mathcal{C}^f, A_{-a, a}(h) \right).$$

*The height converges in  $H_{\text{loc}}^s(D)$  for every  $s < -1$ . Each cluster collection converges in the matching topology just defined, and the distinguished component converges in Hausdorff topology. The continuum collections are locally finite above every positive diameter. All convergences are joint, and the same GFF determines both continuum collections.*

The height topology may equivalently be handled on a countable exhaustion of  $D$  and a countable sequence of exponents increasing to  $-1$ . Our proof first obtains joint tightness, then identifies every subsequential limit as the canonical construction of Theorem 2.3. This yields the full mesh limit without selecting a preferred approximation sequence. No claim about cluster area measures, magnetization fields, all spin-domain-wall curves, or an exact metric exponent is part of the theorem.

## 2.6 Stopping and limiting conventions

All lattice explorations reveal endpoint spins before prescribing an edge. An accessible chamber keeps the actual boundary connections created by the explored configuration. At a pinch, its boundary is the perimeter walk on the side of that chamber; unrelated interiors are not identified merely because their closures touch.

A *pointed limiting component* is specified by a fixed interior test point and compact subsets eventually contained in its discrete component. Several such open components may arise from a single discrete precursor. Statements about conditional fields in distinct limiting components therefore include their joint independence after the appropriate constants are removed. The needed uniform estimates are established before local-set identification.

Geometric scales, buffers, and diameter thresholds are fixed before letting  $\delta \downarrow 0$ . Subsequently a collar width or test scale may tend to zero, and a finite-count truncation may be removed. Rational polygons, rational balls, and a fixed countable dense family of interior points make all uses of these limits simultaneous. Whenever a random domain is used, its exact cut law specifies the residual boundary class. An estimate for the residual fields may be invoked conditionally on the stopping data when it is uniform over that class. Estimates controlling the production of the cut itself are instead averaged over the exploration, conditional only on the permitted initial flat data; they are not asserted after conditioning on a completed stopping history.

## 2.7 Proof strategy

The proof first identifies the height field and then recovers the complete current supports from the geometry coupled to that field.

Section 3 fixes one current  $N$  and its complementary spin percolation  $W$ . The exact edge factor makes an open monochromatic  $W$  perimeter a cut with a flat residual law. Ferromagnetic comparison then supplies circuit estimates in the permitted residual states. The collar argument controls long passages near a rough explored boundary by summing bridge failures over the gaps between ordered crossings. It yields macroscopic count tails and a no-shadowing estimate after a frozen indexed stop. For a complete prescription of every  $s$  spin and  $W$  edge in a small region,

the pattern comparison compares its probability under allowed exterior conditions with its plane probability, with a ratio bounded independently of the number of microscopic prescriptions. A mixed-sign witness argument then shows that full-spin events with vanishing plane probability also have vanishing probability uniformly under the allowed flat laws. This is the one-way contiguity used to transfer screening.

Section 4 gives Cauchy-mixture representations of the plane covariance in the four axial and diagonal reflection frames. At nonzero tangential frequency  $k$ , the normal Cauchy width  $s$  has defect  $D(k, s) = ((|k| - s)/(|k| + s))^2$ . Fourth angular harmonics in frames differing by 45 degrees cancel pointwise, making this defect integrable against the mixing measure. For fixed  $0 < b < B$ , its integral on  $b\delta \leq |k| \leq B\delta$  therefore tends to zero, so each rescaled mixing limit is supported on  $s = |k|$ . Agreement in two independent frames then fixes the remaining spectral weight in both directions, identifying the Green covariance without a 45-degree symmetry of the lattice law. In Section 5, the conditional mean of the neutral Laplacian average  $K_\delta(\Delta\phi)$ , given, outside a surrounding square, both spin fields, the  $W$  states, and computable height increments, tends to zero in  $L^1$ . This makes separated limiting moments harmonic in Section 6. Logarithmic bounds then exclude distributions supported on collision diagonals, and local neutral-pair mixing identifies the logarithmic collision coefficients, giving Wick's rule. For plane increment correlations the same argument gives compact-uniform convergence at every order, which verifies the six-vertex convergence criterion. Only after it yields the Gaussian plane limit does the free-energy correspondence fix the coefficient.

Section 7 identifies random cuts. The discrete carpet obtained by stopping each branch at its first odd jump converges to  $A_{-2a, 2a}$ , which is distinct from a retained cluster of Theorem 2.1. Exact finite cuts give independent flat laws in the residual chambers. A neighborhood argument removes the contributions of both the revealed mean and the centered remainder near the limiting cut. Conditional characteristic functions then identify the full zero-extended GFF remainder, while the piecewise constant labels give the full conditional mean. This proves the thin local-set law for carpets and the localized conditional decomposition away from the seed and the initial revealed wall for cluster stops. In Section 8, strong convergence of Gaussian projections and cutoff Cameron–Martin likelihoods give one local cable-test kernel shared by the canonical parent and a separately stopped component. Testing the canonical forcing event against the component's zero-boundary law shows that a canonical cluster in the parent cannot meet both an unsearched stopped component and its boundary.

Section 9 uses this no-crossing result for one inclusion of the visible partition: odd holes of one canonical cluster must come from the same discrete precursor. The fair-sign correlation identity excludes a precursor supplying holes of distinct canonical clusters. Finite witnesses and the lattice count tails then prove local finiteness across all recursive generations at once. To recover complete supports, we identify the full conditional mean of a stopped exploration. An oscillatory GFF test excludes a connected macroscopic cut where that mean is flat. The intersections of a connected limiting precursor with the canonical supports and the initial wall therefore form a countable pairwise disjoint closed cover. Sierpiński's theorem reduces this cover to one nonempty member. The rough-wall collar estimate excludes the initial-wall case. The indexed no-shadowing estimate and the discovery order exclude invisible duplicate precursors. This identifies every subsequential limit and proves Theorem 2.3.

### 3 Cuts, comparison, and spatial estimates

In Sections 3 to 6, heights are measured in the normalization  $K_\delta = 2H_\delta$ , whose nearest corner increments have magnitude one.

The field limit needs control of local patterns, whereas the cluster limit also needs estimates near the irregular boundaries left by an exploration. We obtain both from one edge representation. Fix one of the two currents and call it  $N$ . Let  $W$  be its exact edge complement on the other lattice: a  $W$  edge is open precisely when the crossing  $N$  edge is closed. Write  $s$  for the spin on the  $W$  lattice and  $d$  for the spin on the  $N$  lattice. Thus  $W$  is monochromatic in  $s$ , while  $N$  is monochromatic in  $d$ . The graph  $W$  is generally different from the other current in the coupling. At an ambiguous edge its opening probability is  $t$ , not  $1 - t$ .

A *flat chamber* means a component cut off by an open monochromatic  $W$  perimeter, with its height on that perimeter fixed. Boundary walks may repeat vertices. At a pinch, the two incident faces on the unsearched side remain separate unless they are actually connected there. One can equivalently use medial boundary polygons displaced by one lattice unit into the chamber. This convention changes Euclidean distances by  $O(\delta)$  and never joins two chambers across a pinch.

We use three kinds of conditional information. An *exact cut history* may contain the chosen current  $N$ , its colors, and the heights on the revealed side of completed  $W$  perimeters. The edge factor below removes that information from every residual chamber. A comparison or annular search *inside* a residual chamber reveals only  $(s, W)$ , and reveals an edge only after its endpoint spins. Finally, an auxiliary matching fill changes the removed side of a cut for a geometric circuit test; it is used through its accessible marginal and is not extra conditioning of the original law. The edge decoration of the other current has been summed out. It is never part of the stopping data on a cut or on its unsearched side: conditioning it on an open  $W$  edge can restore an interaction between the transverse spins. Data strictly on an already separated side may be retained by the exact cut factorization.

All constants below may depend on the fixed parameter  $t \in [1/2, 1)$  and on stated macroscopic buffer ratios. They do not depend on the mesh, on the number of microscopic edges of a prescribed pattern, or on the length or number of fingers of a retained perimeter. In comparisons involving several independently retained source wires, dependence on their number will always be stated.

#### 3.1 Exact cuts and the comparison order

**Lemma 3.1** (Finite cuts and colors). *For every finite domain with either of the model's flat boundary conditions, the joint weight of  $(s, W, d)$  is a product of the following factors, one for each mutually dual interior edge pair:*

$$\begin{cases} t \mathbf{1}_{\{s_i=s_j\}}, & W \text{ open,} \\ (t \mathbf{1}_{\{s_i \neq s_j\}} + (1-t) \mathbf{1}_{\{s_i=s_j\}}) \mathbf{1}_{\{d_i=d_j\}}, & W \text{ closed.} \end{cases} \quad (3.1)$$

The subscripts in each indicator refer to the endpoints on its own lattice. In particular:

1. Conditional on  $(s, W)$ , the free  $N$  components receive independent fair  $d$  signs. A prescribed wired  $N$  component retains its prescribed sign.
2. An open monochromatic  $W$  cut fixes a constant height on its perimeter and carries no transverse-spin interaction. Conditional on the cut and on an exact cut history on its other side, different unsearched chambers have independent flat laws, with free  $N$  colors.

3. *On the  $s$  lattice, height differences from an exterior wall are sums of jumps of size 2. Each free  $N$  cluster contributes one fair relative sign, common to all its odd holes, and contributes zero across its even holes. These signs are independent conditional on  $(N, s)$ . Here a hole is odd when its adjacent  $s$  spin differs from the adjacent exterior  $s$  spin of that cluster.*
4. *These statements hold at stopping explorations which reveal whole  $N$  clusters met by a connected growing seed attached to the exterior, or stop after a prescribed number of successive odd rims. In the wired  $N$  case the first exposed object is its boundary cluster; its holes have opposite-parity  $W$  rims at heights  $+1$  or  $-1$  relative to the initial height, and their same-lattice currents are free.*

*Proof.* If  $s$  disagrees, the crossing  $N$  edge is forced open and its  $d$  spins agree. If  $s$  agrees and  $W$  is closed, the weight is  $1 - t$  and the same agreement is required. Finally, if  $W$  is open, summing the possibilities for the other current gives weight  $t$  whether the transverse spins agree or disagree. This proves (3.1). Its dependence on  $d$  consists exactly of equality constraints on  $N$  edges, proving the first assertion. Fixing a wired component's sign removes the constant factor 2 associated with that component, independent of its geometry.

On an open  $W$  perimeter only the first line of (3.1) occurs. The transverse variables on its two sides have no interaction across it. The remaining product factors split by chamber. Along the perimeter  $s$  is constant, and the corner height rule gives constant height between successive same-parity vertices. This proves the second assertion, first for a simple perimeter and then for a boundary walk by keeping its incident faces separate.

For two  $s$  vertices  $v, v'$  incident to a transverse vertex  $f$ ,

$$K(v') - K(v) = \varepsilon_W d(f)(s(v) - s(v')), \quad (3.2)$$

where  $\varepsilon_W = 1$  when the  $s$  lattice is primal and  $\varepsilon_W = -1$  when it is dual. This fixed orientation sign has no effect on the independent fair relative signs. The rim facing any specified hole of a connected  $N$  component belongs to one incident  $W$  component, as does its exterior rim. Thus crossing that  $N$  component gives either zero or 2 times its  $d$  color times the fixed exterior  $s$  sign. All odd holes give the same relative jump. The third assertion now follows from the first.

For the stopping assertion, inspect no variable beyond a completed open  $W$  rim sealing an unsearched face. Whenever the search queries  $(s, W)$ , it exposes endpoints before edge states. For each possible finite history, its event is the cylinder of the values queried by the deterministic search. The product factors inside the residual chambers are precisely those already described, even if the rule deciding which completed cluster to examine next used its revealed  $d$  color. Thus the conditional residual kernels, and not just their average, are the stated independent flat laws. Skipping an even hole reveals no new jump; stopping at an odd rim retains its observed jump. A connected exterior-attached seed and all clusters it has met form one connected explored region  $S$ . Every edge from  $S$  to an unsearched vertex is  $N$ -closed, since every met cluster has been revealed in full. For one geometric component  $C$  of the unsearched vertices, its complement is connected through  $S$ ; every other unsearched component attaches to  $S$  in the underlying graph. Thus the edge boundary of  $C$  is a planar bond and its dual is one open  $W$  perimeter, with the stated medial resolution at pinches. It is monochromatic, and all its remote connections are retained. This argument concerns connectedness in the underlying graph; the added seed edges need not themselves be  $N$ -open. The seed may have visited several height levels, but this causes no varying boundary value in a residual chamber: every edge on its component-side  $W$  perimeter joins equal  $s$  spins, and (3.2) gives zero height increment along that edge. The

connected perimeter therefore has one constant height. The history determines that constant using only already revealed clusters and their colors; no interior or future information is used. For a wired current, the incident opposite-parity height differs from the boundary height by one corner increment. After its boundary cluster is removed, no interior  $N$  cluster is wired to the original boundary. This proves the last assertion.  $\square$

The same cut argument permits a single exploration to interleave several residual chambers behind an exposed wired layer. Attach each new seed vertex to the common exposed region through already visited vertices or known seed edges, with no unsearched free vertex in the attachment. The union remains connected, and the unsearched components retain their product of flat kernels. For example, a fixed global column sweep can visit every earlier column in every chamber before the next one; a missing neighbor is then in the exposed layer or is already visited. This is the composite exploration used later.

The useful order is the coordinatewise order on

$$(s, W^+, -W^-), \quad (3.3)$$

where  $W^+$  and  $W^-$  are the open edges of the two spin colors, and  $-1 < +1$  for vertex spins. An edge prescription is made only after its endpoint spins have been prescribed. A retained plus wire in a lower boundary condition must remain connected in the upper condition; for minus wires the inclusion is reversed. We call such boundary conditions *ordered*. All other already specified connections remain part of the conditional state, even if their paths run outside the window being examined.

**Proposition 3.2** (Comparison with pins and wires). *The finite  $(s, W)$  law is positively associated in (3.3), as is every conditional law obtained by fixing vertex spins and compatible edge states with both endpoints fixed. Increasing additional vertex and compatible edge prescriptions, or changing retained connections according to the ordered boundary rule, increases the residual law. The analogous statement holds with the two colors reversed.*

*These comparisons permit monotone circuit couplings: search from the changed data toward the region of interest, and stop at an open plus circuit in the lower configuration, or an open minus circuit in the upper configuration. Beyond such a common cut, the two configurations, including complementary cluster colors, can be sampled identically.*

*Proof.* Summing  $d$  in (3.1) contributes  $2^{k(N)}$ , up to the constant just noted for a fixed wired color. Planar Euler counting gives

$$2^{k(N)} = C 2^{k(W)+|W|}, \quad (3.4)$$

where  $C$  depends only on the underlying finite graph and its boundary convention. Indeed the components of the complementary dual graph are the faces of  $W$ , whose number is  $|W| - |V(W)| + k(W) + 1$ . For a local window, contract the actual same-color boundary contacts joined by exterior  $W$  paths and write  $k_\xi(W)$  for the resulting component count. The same identity holds with  $k_\xi(W)$ , up to a constant depending on this boundary convention. These identifications are retained in the law even when their paths lie far outside the window.

Put  $r = (1 - t)/t > 0$ . After division by the constant  $t^{|E|}$ , an open monochromatic  $W$  edge has weight 2, a closed monochromatic edge has weight  $r$ , and a disagreeing edge is closed with weight 1. Represent  $2^{k(W)}$  by auxiliary colors  $\rho \in \{-1, 1\}$ . Summing the  $W$  state on a monochromatic edge gives

$$r + 2\mathbf{1}_{\{\rho_i=\rho_j\}} = r + 1 + \rho_i\rho_j. \quad (3.5)$$

Consequently the  $s$  marginal is the product of a positive factor  $(1+r)^{|E_{\text{mono}}|}$  and the even-subgraph partition functions on the two monochromatic induced graphs, with edge parameter  $q = (1+r)^{-1} \in (0, 1)$ .

For completeness, let  $Z_F(A)$  denote such an even-subgraph partition function on an available edge set  $A$ , with ferromagnetic identifications  $F$  retained. It is increasing and log-supermodular in  $A$ . One way to see both facts is to write it as the zero-field ferromagnetic Ising partition function divided by its single-edge cosh factors. Adding an edge multiplies the normalized partition function by  $1 + q\mathbb{E}[\rho_i \rho_j]$ . The expectation is nonnegative and increases when other nonnegative couplings are added, by the two Griffiths inequalities [23]; a complete proof for nonnegative couplings is given in [19, Section 3.8.1, Theorem 3.49]. Indeed differentiating an expectation with respect to a coupling gives the corresponding nonnegative covariance. Identifications are limits of positive bonds: after dividing by the common maximal exponential, the finite-state factor is  $e^{-2KD_F}$ , where  $D_F$  counts violated identifications. As  $K \rightarrow \infty$  it tends to their joint indicator. Thus the same inequalities survive the finite-state limit. The single-edge factors cancel from the mixed inequality, proving

$$Z_F(A \cup B)Z_F(A \cap B) \geq Z_F(A)Z_F(B).$$

The same incremental-edge formula gives the boundary comparison that will be needed below: if  $A \subseteq B$  and  $F \subseteq F'$ , then

$$Z_{F'}(B)Z_F(A) \geq Z_{F'}(A)Z_F(B). \quad (3.6)$$

Indeed each multiplier  $1 + q\mathbb{E}[\rho_i \rho_j]$  increases when the bonds implementing  $F'$  are added. Integrating the Griffiths covariance inequality and then taking their infinite-bond limits proves (3.6). For plus vertex sets  $S, T$ ,  $E(S \cap T) = E(S) \cap E(T)$  and  $E(S \cup T) \supseteq E(S) \cup E(T)$ . Apply the preceding inequality and monotonicity to both the plus and minus induced graphs. The factor  $(1+r)^{|E_{\text{mono}}|}$  is itself ferromagnetic. This proves the spin lattice condition, to which the finite association and comparison theorems [17, 24] apply.

Conditional on  $s$ , the open edges on each monochromatic graph have the ordinary random-cluster law with cluster weight 2 and edge weight  $v = 2/r$ . Given the other edges, its opening probability is  $v/(v+2)$  if the endpoints are disconnected and  $v/(v+1)$  if they are already connected. The probabilities increase with the available connections, so common-uniform edge updates give the edge comparison. Together with the spin lattice condition, this gives positive association and the order (3.3). Fixed closed edges with prescribed endpoints give constants; fixed open edges give ferromagnetic identifications. Thus the same argument applies after all the allowed prescriptions. The ordered-wiring comparison follows from (3.6) for plus identifications and its reflected version for minus identifications. In particular, arbitrarily shaped exterior connections cannot be silently discarded.

Use these conditional comparisons at each step of a simultaneous exploration, always preserving each configuration's residual conditional law. A plus circuit in the lower configuration is also open plus in the upper. By Theorem 3.1, the laws on the sealed side then have the same constant boundary spin and no transverse interaction across the circuit. Couple those laws identically. No  $N$  component crosses the cut, so its free colors may also be coupled identically. The minus case is the reflected argument.  $\square$

There is also a useful bounded comparison when monotonicity is not the issue. Suppose two local conditional weights have exactly the same available edges, pins, and local edge factors, and differ only in the partition of  $k$  prescribed same-color source labels. Each label may have

many boundary contacts, which are kept identified in both laws. For every configuration the two component counts differ by at most  $k - 1$ . After normalization, (3.4) therefore gives

$$2^{-2(k-1)} \leq \frac{d\mu_\xi}{d\mu_{\xi'}} \leq 2^{2(k-1)} \quad (k \geq 1). \quad (3.7)$$

This statement does not permit a single source to be split into its many visible entrances, nor a change of local edge factors. A wire in this calculation is an identification in a probability law, not a geometric edge along which a circuit may travel.

At fixed mesh these conclusions also apply to bounded searches in the plane. Apply them first in finite flat exhaustions and take a joint subsequential limit of the configurations and of the finite search transcripts. The limit is the common plane law obtained from flat exhaustions in [16, Theorem 17.1]. Equality on the sealed side of a common cut holds on every finite subset and hence on the whole countable lattice in the limit, including the complementary colors. That theorem also gives no unbounded complementary component and the fair color law there. A failed circuit search stops at its prescribed buffer boundary; it never conditions the residual law on a future success.

### 3.2 Buffered circuits and rough walls

We specify the geometric completion used when a test window meets a removed region. A face of the  $W$  lattice is centered at a vertex of the dual  $N$  lattice. For a revealed set of  $N$  vertices and seed edges, take the union of these closed faces and the faces along the seed. We call this its *source cell region*. It is within  $O(\delta)$  of the drawn revealed graph. Holes occupied by unsearched chambers are retained; this operation does not fill the hull of a cluster. Its resolved complementary chambers and the medial perimeters above differ by  $O(\delta)$ , including at pinches.

Let  $F$  be such a removed cell region and  $U$  an accessible component. Suppose every frontier edge between  $U$  and  $F$  is a prescribed open  $W$  edge of one color  $\sigma$ . Define the *augmented matching graph* in a test window to consist of the actual  $W^\sigma$  edges on the accessible side and artificial matching  $W$  edges contained in  $F$ . The artificial edges are the lattice one-skeleton of the cell region; the frontier edges themselves retain their actual prescribed states. A circuit in this graph is an ordinary geometric circuit in the window. It may use the artificial edges inside  $F$ . An underlying dual path disjoint from  $F$  cannot cross it without crossing an actual  $W^\sigma$  edge. In particular, an abstract remote wire is never used as a geometric edge of this circuit.

There are two exact uses of this construction. For one component  $U$ , complete each removed source on its other side by matching spins and open edges. Keep the original conditions on any other boundary of  $U$ . The frontier factors in (3.1) are constants  $t$  with no transverse interaction. Integrating the removed variables therefore shows that the  $U$  marginal of this completion is exactly its original conditional marginal. This remains true when  $U$  has several boundary components, provided all the completed frontiers have the stated color and the test windows avoid any other prescribed pieces.

For the second use, suppose the residual chambers are the disjoint complementary components of one common connected source cell region  $F$  attached to the exterior. Conditional on an exact cut history they have a product of flat laws. Reflect  $s$  in each chamber, if needed, so that all its frontier spins have one color  $\sigma$ , and put matching spins and open edges in  $F$ . The same factorization realizes exactly this product as the accessible marginal of a single completed graph: every factor on the removed side cancels on normalization. There is no factor depending on the number of chambers. We call this the *product class with a common fill*. It includes

the chambers behind an exposed wired layer and subsequent composite seed explorations. An arbitrary collection of independent chamber copies is not asserted to have such a realization. Finally, when a completion changes only a partition of  $k$  indivisible source labels while preserving all local factors and pins, the comparison (3.7) gives the stated bounded substitute for exact equality. These are the only completion rules used below.

**Proposition 3.3** (Buffered circuits and attached-wall bounds). *Fix positive aspect and buffer ratios. There are  $p > 0$  and a fixed lattice cutoff  $c_* > 0$  with the following properties, uniformly for every physical test radius  $r \geq c_*\delta$  and every mesh. Test neighborhoods must lie in the ambient domain, apart from matching cell regions to which one of the preceding completion rules applies.*

1. *In a buffer containing no pins, a circuit of either prescribed  $W$  color in a fixed-aspect annulus has probability at least  $p$ , conditional on arbitrary admissible  $(s, W)$  data outside the buffer. Compatible pins of the requested color inside the buffer only improve the bound. Finite chains of overlapping circuit tests give a positive lower bound for a fixed-aspect tube connection.*
2. *After any admissible  $(s, W)$  stopping exploration, failure of matching circuits on  $m$  separated available scales has probability at most  $(1 - p)^m$ , when each scale is chosen from the current history before its interior is examined. In particular, suppose a prescribed connected matching wall reaches from  $B(z, r)$  to outside  $B(z, R)$ , with a fixed buffer around  $B(z, R)$ . A path avoiding that wall and all attached matching circuits, and joining  $B(z, r)$  to outside  $B(z, R)$ , has probability at most*

$$C \left( \frac{r + \delta}{R} \right)^\alpha \tag{3.8}$$

for some  $C, \alpha > 0$ .

3. *When a test buffer meets  $F$ , the circuit and tube events mean circuits and connections in the augmented matching graph just defined. They have the same lower bounds under the exact completion of one component or the exact completion with a common fill. A circuit attached to  $F$  blocks an accessible dual path by its actual part together with the forbidden region  $F$ . If only the  $k$ -label comparison is available, the probability bounds lose at most  $2^{O(k)}$  in total. The number of local entrances of one label does not enter this cost.*

The bulk assertion in item 1 also holds before removing the model's single wired  $N$  component. Conditioning in a search concerns  $(s, W)$ ; an exact cut history with exterior colors is allowed because its accessible marginal is already a fresh flat kernel.

*Proof.* The imported input is [16, Theorem 16.4]: for  $\mathbf{c} = 1/t \in [1, 2]$ , an opposite-color spin-percolation circuit in a lattice annulus of radius at least 4 has a positive lower bound, uniform in its radius, under the worst constant wired boundary condition. The decoration in that theorem opens an ambiguous edge with probability  $1/\mathbf{c} = t$ , so it is precisely  $W$ . Its exterior-conditioning version, [16, Corollary 16.6], allows arbitrary data of the same spin parity and its edges outside the buffered ball. We only use this statement for one parity. Its conditional FKG hypotheses agree with Theorem 3.2.

After the fixed change from source lattice units to physical units, the lower bound holds for  $r \geq c_*\delta$ . Comparison with the worst constant boundary condition gives item 1 for squares; a fixed finite cover gives the other stated aspect ratios and the tube version. Intersect its matching circuit events by positive association. Matching pins are favorable in (3.3). In the wired  $N$  law before removal, fixing the single wired  $d$  color removed the geometry-independent factor 2 in the

proof of [Theorem 3.2](#). Its  $(s, W)$  comparison is therefore the same. For the unpinned bulk test, choose a deterministic circuit in the unused part of the buffer between the annulus and the buffer boundary. Compare to the law with that circuit pinned to the worst  $W$  color. Its sealed side is a  $W$ -flat kernel by [Theorem 3.1](#), so the same lower bound applies there despite the arbitrary data beyond the buffer. Requested-color pins are then handled by monotonicity. This proves the claimed wired bulk case.

For separated scales, reveal earlier trial neighborhoods before testing the next one. The conditional failure probability remains at most  $1 - p$ , so multiplying conditional bounds proves item 2 without an independence assumption. Take a geometric sequence of scales between  $r + \delta$  and  $R$ , separated enough to have disjoint buffers. A circuit at one such scale meets the wall because the wall crosses the annulus. It separates the inner from the outer scale. This gives (3.8); the  $\delta$  term absorbs the finitely many scales below the lattice cutoff.

For item 3, apply item 1 in the completed graph. All additional pins in a test buffer have the requested matching color. The exact accessible marginal or (3.7) transfers the event to the original law. In this transfer an edge of the circuit inside  $F$  is explicitly artificial; every other edge of it is an actual matching edge. Thus a full circuit is available even when an accessible side is slit and has no full circuit by itself. The geometric blocking assertion follows from the definition of the augmented graph. For a connected source the boundary incident to one fixed complementary component is one boundary walk. It can enter the window arbitrarily often, but these entrances form one retained label. This verifies exactly the premise on labels when that bounded comparison is used.  $\square$

A useful consequence concerns a stopping set selected using revealed colors. Under the preceding flat kernel, if a cluster meeting a seed at distance at least  $R$  from  $B(z, r)$  reaches that ball, there is an  $N$  arm from  $B(z, r)$  to outside  $B(z, R)$ . Testing the  $W$  annuli in that preceding  $(s, W)$  marginal bounds this event by (3.8), regardless of the later rule selecting the cluster. Similarly, before revealing discovered colors, reaching a ball before a fixed number of odd rims entails failure to find that many alternating pairs of  $W$  circuits on the intervening scales. Conditional iteration gives a bound of the same polynomial form, with an exponent depending on the fixed number of rims. These are averaged bounds under the prior kernel. They are not bounds conditional on each completed stopping history containing colors. The same observation after exposing a wired layer uses its fresh product of flat kernels, and the bulk wired assertion handles the initial layer itself.

We next turn the circuit bound into an estimate for a whole collar. A *window* consists of a working rectangle  $R$ , a fixed intermediate enlargement  $R^+$ , and a larger rectangular buffer, with positive gaps between them. A traversal test is a path in  $R$  from left to right, staying a fixed distance from its top and bottom, or a rotated copy of this test. A local wall piece is a component of the cell region restricted to  $R$ ; pieces joined only outside  $R$  remain distinct locally. We say the region is *anchored* if every component in the buffer meeting  $R^+$  reaches the buffer boundary inside the cell region. This holds when the region is connected to an exterior seed outside the buffer. Pieces that touch are treated as one; otherwise the medial lattice convention separates their representative paths by lattice order. Write  $F^\eta = \{z : \text{dist}(z, F) \leq \eta\}$ .

**Lemma 3.4** (Uniform collar exclusion). *Fix a finite family of such windows and traversal tests. Let  $F_\delta$  be a prescribed matching cell region of color  $\sigma$ , anchored in each test. Use one of the following conditional laws: one actual flat chamber; one selected accessible component facing at most  $k$  connected prescribed sources, with all completed frontiers of color  $\sigma$ ; or the product class with a common fill. In the second case  $k$  is fixed and any partition change must satisfy the*

premise of (3.7). Test buffers contain no other pins. There is no bound on the number of local pieces of one source.

Consider paths in the full underlying dual graph, whether or not their edges are  $N$ -open. Let  $\mathcal{X}_\delta(\eta)$  be the event that such a path makes one of the prescribed traversals, lies in  $F_\delta^\eta$ , and is disjoint from  $F_\delta$  and from every augmented matching circuit in the buffer which is attached to  $F_\delta$ . Then, uniformly over the stated walls and positive probability allowed histories,

$$\lim_{\eta \downarrow 0} \limsup_{\delta \downarrow 0} \sup \mathbb{P}[\mathcal{X}_\delta(\eta)] = 0. \quad (3.9)$$

An exact cut history may contain colors on its removed side; the residual search law has the meaning specified above. It is never conditioned on arbitrary interior  $d$  colors.

The same estimate applies simultaneously to paths constrained to lie in any closed disk bounded by an actual opposite-color  $W$  loop, if that disk is disjoint from  $F_\delta$ . Such a path automatically avoids every matching circuit attached to  $F_\delta$ . The windows and their buffers may all be chosen inside an open annular band. In particular, (3.9) holds along every deterministic sequence  $\eta_\delta \downarrow 0$ .

*Proof.* These tests detect every connected passage of a fixed positive diameter  $b$  in a fixed box. Choose an inner square of side a small fixed multiple of  $b$  from a fixed grid around one endpoint, and stop the connecting path on its first exit from a larger square of diameter less than  $b/2$ . The exit side fixes an orientation, and a fixed enlargement of the vertical extent gives a traversal with the required margin. There are finitely many grid squares and orientations. Their working rectangles, intermediate enlargements, and buffers are all chosen before applying the lemma; anchoring is required on those intermediate enlargements. The same construction inside an open band detects a full traversal of the band.

It suffices to prove one left-to-right test. Use the crossing segment from the last left-side hit before its first right-side hit, and then erase loops. The result is a simple proper crosscut: its interior misses both vertical sides. This trimming is needed because a simple path that returns to its starting side can leave a third pocket. A proper crosscut has just a lower and an upper side. Every local obstacle piece disjoint from it is assigned to one of those sides. Include the top and bottom as virtual barriers, which the traversal margin already forbids. Here the pieces are components after restriction to this working rectangle. Connections outside it retain their wire labels in the law but do not identify pieces on opposite sides of an endpoint.

We first reduce these assignments, except for a uniformly small event, to a fixed finite number of possibilities. An anchored piece meeting a narrower central rectangle contains a path to the boundary of the working rectangle: take the part of an anchored path up to its first exit. A piece reaching the top or bottom is already assigned to that barrier. For a piece reaching the left, in the fixed-width left end strip, retain a segment from its last hit of the outer vertical to its first hit of the inner vertical and erase loops. It is a proper crosscut of that strip. Make a deterministic choice if a piece has several such paths. Representatives of distinct pieces are disjoint and can be ordered from bottom to top. The analogous construction is used at the right side and in the rotated windows. Connections of a piece between different sides later impose consistency; they add no choices.

Here is the quantitative reduction in one end strip. Choose a middle horizontal interval  $I$  of length  $\ell > 0$  and take  $\rho > 0$  below both its horizontal clearance from the vertical ends and the fixed clearance of  $R$  inside the larger window buffer. For consecutive representative arms, let  $g_j(x)$  be the difference of their topmost intersection heights with the vertical at  $x$ , and put  $d_j = \ell^{-1} \int_I g_j(x) dx$ . These heights have the same order as the arms: the upper crosscut separates

the lower one from the top of the strip, so a vertical ray from the top to the topmost point of the lower one first meets the upper one. The arms need not be graphs. In particular,

$$\sum_j d_j \leq H, \quad (3.10)$$

where  $H$  is the fixed strip height. The set  $E_j = \{x \in I : g_j(x) \leq 2d_j\}$  has length at least  $\ell/2$ .

At lattice scale the separation convention gives  $d_j \geq a_0\delta$  for a fixed  $a_0 > 0$ . Put  $h_j = d_j + \delta \leq \beta d_j$ , and choose a fixed large  $K$  so that  $2Kh_j$  exceeds the cutoff in [Theorem 3.3](#). If  $4K\beta d_j < \rho/2$ , a maximal selection of points in  $E_j$  at horizontal separation greater than  $16Kh_j$  has at least  $\ell/(64K\beta d_j)$  points, up to a fixed endpoint loss. Center a disk at the midpoint of the two arm intersections at each selected point. Both arms meet its radius- $2Kh_j$  disk and leave its radius- $4Kh_j$  disk, since they continue to the strip ends. The radius- $8Kh_j$  buffers are disjoint by construction. Since  $8Kh_j < \rho$ , they stay inside the allowed window buffer and lie between the end strip's vertical sides. Every augmented matching circuit in the intervening annulus meets both proper arms. Orient such a circuit, stop at its first upper-arm hit after a chosen lower-arm hit, and retain the arc from its last preceding lower-arm hit. Its interior misses both arms. The two proper arms divide the infinite vertical strip into lower, middle, and upper components. Only the closure of the middle component meets both arms, so the arc's connected interior lies in that component. Its closure is contained in the finite end strip. The retained arc therefore joins the arms geometrically inside the end strip, even if the rest of the circuit extends beyond its horizontal sides. In particular, a remote wire alone is not counted as the joining connection.

Successive conditional circuit tests in the completed graph, followed when needed by the single density comparison (3.7), now give

$$\mathbb{P}[\text{the two arms have no such local joining circuit}] \leq C_k \exp(-c/d_j). \quad (3.11)$$

Here and below  $C_k$  is the fixed density factor for the allowed completion; it is an absolute constant for either exact completion, including the product class with any number of chambers. Gaps too large for the preceding packing are simply retained. The locations of all tests depend only on the prescribed obstacles, not on the sought traversal. If a matching circuit joins two arms, an avoiding crosscut cannot assign them to opposite sides.

For any  $u > 0$ , the total probability lost by discarding the gaps  $d_j \leq u$  is at most

$$\sum_{d_j \leq u} C_k e^{-c/d_j} \leq C_k \left( \sum_j d_j \right) \sup_{0 < d \leq u} \frac{e^{-c/d}}{d} \rightarrow 0 \quad (u \downarrow 0). \quad (3.12)$$

At most  $H/u$  larger gaps remain, together with a bounded number adjacent to virtual barriers. A proper traversal can split the ordered representatives at only one of these ranks. Repeating this on the finitely many sides leaves a finite product of deterministic rank lists, bounded in terms of  $u$  and the windows. A piece joined around another side must receive the same assignment there, which only removes rank choices. Trials for different gaps may overlap; the union bound in (3.12) does not require their independence.

We finish by a compactness argument, keeping its choices deterministic. If (3.9) failed, there would be deterministic allowed histories and walls at meshes  $\delta_n \downarrow 0$ , widths  $\eta_n \downarrow 0$ , and a fixed positive lower bound for the traversal probability. First fix  $u$  so that (3.12), summed over the sides, is less than one quarter of that bound. One of the finitely many surviving rank choices then has a fixed positive event mass along a subsequence. We do not condition the law on that

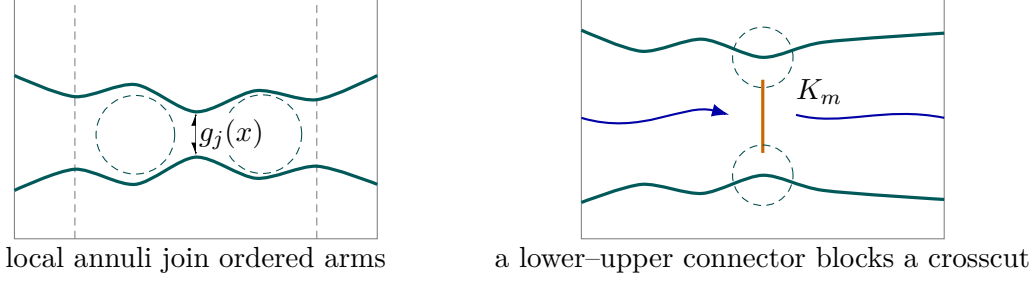


Figure 2: The two geometric steps in collar exclusion, drawn schematically. Representative arms can double back; their topmost vertical encounters remain ordered. The dashed matching circuits are local geometric circuits, with any artificial part contained in the forbidden cell region. In the right panel the compact gap segment  $K_m$  is forbidden to a sufficiently thin collar.

event. Take Hausdorff limits, in the central buffered rectangle, of the unions assigned below and above, including the virtual barriers.

Suppose first that the two limits touch there. Unless a virtual barrier is involved, there are deterministic points on two oppositely assigned pieces at distance  $r_n \downarrow 0$ . Anchoring makes both pieces cross the annuli from scale  $r_n + O(\delta_n)$  to a fixed outer scale. Any augmented matching circuit in such an annulus joins them. By (3.8), the probability of a traversal with this assignment tends to zero. If one limiting piece is a virtual barrier, the fixed traversal margin rules out the opposite assignment: a connected piece reaching that margin lies on the barrier's side of the proper crosscut.

It remains that the two closed limits are disjoint in the central rectangle. On a central vertical line choose a consecutive lower encounter  $a$  and upper encounter  $b$ . Such a transition exists because the virtual bottom and top have those assignments. The open segment  $(a, b)$  misses both limiting unions. Fix an integer  $m$ . At each actual endpoint choose  $m$  separated circuit radii, all less than  $|a - b|/8$  and the available buffer clearance; call their smallest radius  $r_m > 0$ . If both endpoints are virtual, take any fixed sufficiently small  $r_m$  in their traversal margins instead. Only after this choice, trim the gap by less than  $r_m/2$  at its ends and call the remaining compact segment  $K_m$ . It has positive distance from the whole limiting obstacle union, not just from its intersections with the vertical. This clearance holds for the discrete obstacles for all large  $n$ , and then  $K_m \cap F_{\delta_n}^{\eta_n} = \emptyset$ .

Center the endpoint tests at discrete points converging to  $a$  and  $b$ . An attached circuit at any of the chosen radii meets  $K_m$ : it surrounds its center, and the ray into the gap crosses the circuit between that radius and twice that radius. Its other intersection is with the endpoint's anchored piece. For a virtual endpoint the short segment in the fixed traversal margin supplies the connection instead. Consequently a successful endpoint circuit at each actual endpoint, together with  $K_m$ , is a connected set from a lower-assigned piece or barrier to an upper-assigned one, wholly in the rectangle. Every proper left-to-right crosscut with that assignment meets this set. It cannot do so: it avoids the pieces and the attached circuits, and the collar condition excludes  $K_m$ .

For fixed  $m$ , the probability that either required endpoint connection fails has limit superior at most  $2C_k(1 - p)^m$ , by the separated-scale estimate and, if needed, one density comparison. Letting  $m$  increase contradicts the retained positive event mass. The order of choices here matters: the finite endpoint radii precede the trimmed segment, so every requested circuit really reaches

that segment. This proves uniformity in (3.9).

Finally let a closed disk be bounded by an actual  $W^{-\sigma}$  loop and disjoint from  $F_\delta$ . A connected augmented matching circuit attached to  $F_\delta$  cannot enter this disk. Its artificial edges are in  $F_\delta$ , outside the disk, and its actual  $W^\sigma$  edges cannot cross or share a vertex with the opposite-color boundary. A dual path inside the disk therefore automatically avoids all those circuits. The existence event already ranges over every underlying dual path, so the bound holds simultaneously over the possible disks, without conditioning on a future choice of one. Choosing all the windows with fixed margin inside an open band gives the last localization assertion. Monotonicity in  $\eta$  gives the conclusion for  $\eta_\delta \downarrow 0$ .  $\square$

### 3.3 Counts and comparison of complete patterns

**Proposition 3.5** (Cluster and transmission counts). *The following bounds hold in fixed buffered boxes, uniformly in the mesh and in allowed conditional histories. A joint count of actual clusters across chambers in items 1 and 3 requires the product class with a common fill. Item 2 has the interior scope stated there.*

1. *For fixed  $b > 0$ , the number of unsearched free  $N$  clusters having a connected passage of diameter at least  $b$  in a fixed box has an exponential tail. Its constants depend on  $b$  and the fixed box geometry, not on the roughness of the surrounding walls or on the number of chambers in the realization with a common fill.*
2. *For one chamber, let a lattice-aligned rectangle and its joining buffer lie wholly in the unprescribed accessible interior. Use the relative dual restriction with separate side ports defined in the proof. The number of distinct components of the restricted  $N$  graph crossing between two specified opposite sides has a geometric tail. Consequently, if  $Q_0 \subseteq Q_1$ , the number of restricted components in  $Q_1$  which meet  $Q_0$  and a vertex incident to an edge outside the restriction has a geometric tail, provided the fixed covering rectangles in  $Q_1 \setminus Q_0$  and every joining buffer have the same unprescribed interior clearance. Arbitrary admissible  $(s, W)$  prescriptions outside a larger buffer are allowed. The same interior assertion holds in the original wired law before removal of its boundary cluster.*
3. *Counts of distinct free  $N$  clusters of the residual kernels enclosing or passing through a fixed aspect-ratio annular shell have moments of every fixed order, uniformly after rescaling the shell down to lattice order. Under a stopping history these are the unsearched clusters. A distinguished wired cluster adds at most one. For arbitrary independent chambers with pairwise disjoint interiors this last statement is available separately in each chamber; it asserts no bound for their total count without a common fill. The precise no-shadowing consequence at indexed discoveries is stated in [Theorem 3.8](#) below.*

*Proof.* For item 1, start with the connected cell region of the given exact cut history; for a single flat chamber use its exterior side. Explore with a deterministic connected seed attached to that region. Add one seed vertex at a time and immediately reveal the entire previously undiscovered  $N$  cluster it meets. The seed eventually visits every vertex of the box. Record only discoveries with the specified passage, choosing one such passage by a deterministic rule. At every recorded stop, the initial region, seed, and all clusters already discovered form one connected cell region. By [Theorem 3.1](#), the residual laws are exactly the flat kernels, jointly in the class with a common fill when there are several chambers.

A finite family of smaller collar windows detects every passage of diameter at least  $b$ . Freeze the cell region  $F_i$  at the  $i$ th recorded stop, with  $F_0$  the initial region, before continuing the search. An actual unsearched  $N$  path is disjoint from  $F_i$  and cannot cross the actual part of an augmented matching circuit. Thus [Theorem 3.4](#) gives an  $\eta > 0$  such that, conditionally on any such frozen history, the probability that *some* still unsearched qualifying passage lies wholly in  $F_i^\eta$  is at most a fixed  $p < 1$ . This event is existential over future passages; no future location or visibility is conditioned upon. It also bounds the event that the next recorded cluster exists and its selected passage lies wholly in  $F_i^\eta$ .

On the complementary event, the next passage contains a point at distance greater than  $\eta$  from  $F_i$ . Such points from successive records are mutually  $\eta$ -separated, because each earlier cluster is included in every later frozen region. A bounded box contains at most  $M = M(\eta)$  such points. There is no cost for the number of intervening microscopic discoveries: they only enlarge the same connected region. In the product case reflect the residual chamber spins to one matching color and use the exact realization with a common fill for this single existential collar event. In particular, no union bound over the number of holes is taken. A test buffer meeting the initial wall is covered by the same augmented graph.

If at least  $n$  records occur, at most  $M$  of the first  $n$  can be of the separated-point kind. Conditional iteration for the other indices, including the initial stop, gives

$$\mathbb{P}[\text{at least } n \text{ recorded discoveries}] \leq \sum_{j=0}^M \binom{n}{j} p^{n-j} \leq C e^{-cn}. \quad (3.13)$$

The same proof starting from a later allowed stop bounds its unsearched count. It does not bound the number already fixed by that history. Meshes above the small threshold used by the collar lemma are absorbed by the deterministic number of vertices in the fixed box.

For item 2 we use a crossing exploration followed by a finite-domain boundary comparison. First fix the relative planar dual of the given restricted  $N$  lattice quadrilateral. Its medial boundary follows the four sides of the rectangle. In the auxiliary  $W$  graph, truncate a dual edge at that boundary when it exits, and give each boundary incidence its own degree-one port. There are no edges in an exterior side column and no identifications between distinct ports. Every retained edge or half-edge inherits the state of its actual  $W$  edge. This is the matched dual restriction used for every  $W$  connectivity test below; its placement differs from the geometric rectangle by  $O(\delta)$ , which is absorbed by the joining buffer.

If  $J_N$  distinct restricted  $N$  components cross the quadrilateral, choose one simple crossing in each, ordered from bottom to top. Between each two successive crossings, relative planar duality gives a parallel open crossing of this matched  $W$  graph. The separators in two successive gaps belong to different components of that graph: a connection inside it would cross the intervening  $N$  path, and the separate ports give no route around an endpoint. Consequently

$$J_N \leq J_W + 1,$$

where  $J_W$  counts crossing components in precisely the matched  $W$  graph. The probabilistic search uses only these  $W$  crossings; it never certifies a separator against an unrevealed upper  $N$  crossing.

Distinct  $W$  crossing components have a consistent vertical order: one component cannot have crossing paths on both sides of another, because its joining path would cross the latter. In each component choose a lowest simple proper crossing, with a deterministic tie rule; the lower boundary of the union of its crossing paths supplies such an open crossing. Enumerate these

representatives from below, with all connectivity restricted to the matched graph. For a candidate  $\gamma$ , the event that it is the next distinct crossing is determined by the  $(s, W)$  configuration in its closed below envelope. All lower candidate crossings lie there, and any connection from a lower component to  $\gamma$  first meets  $\gamma$  from below. A proposed connection between lower pieces through the region above  $\gamma$  would first join  $\gamma$  as well. This is the stopping-line property for the enumeration.

After selecting  $\gamma$ , reveal its whole closed below envelope, with endpoint spins before edge states. Reflect the colors if necessary so that  $\gamma$  is plus. We use test rectangles with sides parallel to the coordinate axes of the  $W$  lattice; the primal and dual square lattices have the same axes. Complete only the two endpoint half-edges of  $\gamma$  to their actual  $W$  edges. Their states are already known: each half-edge inherited the state of that actual edge, and its being open forces the additional endpoint to have the same spin. Thus this operation assumes no unexamined connection. The actual endpoints of all left-exiting half-edges lie on one  $W$ -lattice column, and the analogous statement holds on the right. Let  $B$  have these two columns as its lateral boundary and extend its top and bottom a fixed distance into the unprescribed joining buffer. The completed  $\gamma$  is a proper crosscut of  $B$ : its only vertices on the lateral boundary are its two endpoints, because the matched graph contains no exterior side-column edges. Write  $U_\gamma$  for the component above this crosscut and  $F_\gamma$  for the closed region below it. All the earlier crossing-search transcripts in  $B$  are below  $\gamma$ .

Analytically disintegrate the conditional law over the entire configuration outside  $U_\gamma$ , including the unused parts of the buffer, and over its boundary contact spins before interface edge states. This extra disintegration is used only for the estimate; it is not added to the next search transcript. On every positive exterior fiber the finite weight on  $U_\gamma$  is the weight of [Theorem 3.2](#), with the exterior same-color connections retained. All contacts on  $\gamma$  are plus and belong to a single class, since  $\gamma$  itself is open. The other contacts lie on the upper boundary arc of  $B$ .

Replace this fiber's boundary condition by the following adverse one. Keep  $\gamma$ , including its endpoints, plus and open. Make all other contacts on the upper boundary arc minus and wire them together; retain only the one plus class supplied by  $\gamma$ . The two boundary edges at a change of spin are closed. This is an ordered decrease for the plus order: it retains every required lower plus connection, removes only additional plus connections, and merges all remaining minus connections. Changes of contact spins and their compatible incident edge states use the same order. Therefore the original fiber dominates this fixed adverse law on  $U_\gamma$ , by [Theorem 3.2](#). In particular, no event belonging to the original history has been removed from a conditional probability.

Realize the adverse law on the full box  $B$  by filling  $F_\gamma$  with plus spins and open  $W$  edges, including the lower outer boundary arc, while keeping the upper outer boundary arc minus and wired as above. Every edge of the separating crosscut remains an actual open plus edge. The finite edge factors in [\(3.1\)](#), or equivalently the component-count formula in [\(3.4\)](#), show that this completion has exactly the adverse law as its  $U_\gamma$  marginal: factors on the filled side cancel, the plus contacts form one class, and the other boundary contacts form the one minus class. There is no factor depending on the length or roughness of  $\gamma$ .

The completed law is obtained from the law in  $B$  with the worst constant minus wired boundary by increasing the lower boundary data and imposing compatible plus pins in  $F_\gamma$ . [Theorem 3.2](#) therefore makes it dominate that constant-boundary law for every increasing plus-circuit event. The ordinary buffered circuit estimate used in [Theorem 3.3](#), namely [\[16, Theorem 16.4\]](#), now gives a uniform positive lower bound for each circuit test in a fixed central strip of  $B$ . In transferring a circuit to  $U_\gamma$ , edges in  $F_\gamma$  are artificial matching edges; all its other edges are actual plus

edges. The estimate holds on every original exterior fiber, so averaging over those fibers preserves the same lower bound. This argument uses only the ordinary full-circuit estimate, not the arbitrary-history partial-circuit assertion of [16, Lemma 19.4].

Choose a fixed chain of buffered circuit tests in this central strip, starting below the counted rectangle and ending above it. All test buffers stay away from the lateral sides of  $B$ , and their number depends only on the fixed aspect and buffer ratios. The usual tube gluing from [Theorem 3.3](#), followed by positive association in the completed law, gives a uniform positive probability  $p$  that the augmented matching graph joins the filled lower part to above the rectangle. This tube event is increasing, so the same boundary comparison and fiberwise averaging give probability at least  $p$  under the original search history. Its last exit from  $F_\gamma$  is on  $\gamma$  in the central strip. From there until its first exit through the top of the counted rectangle the connection is actual and lies in the matched graph. A later parallel crossing of the same color would meet this connection and hence join  $\gamma$  inside that graph, whereas a crossing of the opposite color cannot cross it. The tube therefore rules out another distinct  $W$  crossing.

These tube tests are used to bound the conditional probability; their upper data are not added to the next search transcript. For every transcript at the  $i$ th representative, the conditional chance of a further one is at most  $1 - p$ . Iteration gives a geometric tail for  $J_W$  and hence for  $J_N$ . Connections outside the rectangle play no role because both counts use restricted components. The original wired law has the same interior  $(s, W)$  comparison by [Theorem 3.3](#).

A fixed finite rectangle cover of the shell between  $Q_0$  and  $Q_1$  assigns each transmitting restricted component to a crossing of one of these rectangles. The small lattice layer in the incidence definition is absorbed by the fixed clearance. This proves the transmission assertion.

For the shell assertion, the clusters counted are unsearched global free clusters of the residual law, rather than distinct restricted components. At shell radius  $r$ , a fixed finite cover assigns every enclosing or passing cluster a connected passage of diameter at least  $c_2 r$  in one of its boxes. Apply item 1 after rescaling by  $r$ . Its collar proof is uniform when  $\delta/r$  is small, including boxes met by matching walls in the realization with a common fill. When  $\delta/r$  is bounded below there are only a bounded number of vertices in the rescaled boxes. The resulting exponential tail gives every fixed moment, uniformly down to lattice order. For disjoint independent chambers apply this assertion separately in each one. A pair of vertices in different chambers shares no cluster, which is the only cross-chamber fact needed for the later moment estimates.  $\square$

**Corollary 3.6** (Counts in an enclosing box). *Fix a bounded box  $L$  and  $b > 0$ . Suppose the finite lattice domain and the supports under consideration lie in  $L$ , and place  $L$  in a fixed larger regular ambient box. For one free flat  $N$  law, or the residual product law with a common fill conditional on its initial exact cut data, let*

$$\mathcal{N}_\delta(b) = \#\{\text{free } N \text{ clusters } C : \text{diam } C \geq b\}.$$

*There are constants  $C, c > 0$ , depending on  $L, b, t$  but not on the initial walls or the mesh, such that*

$$\mathbb{P}[\mathcal{N}_\delta(b) \geq n \mid \text{initial cut data}] \leq C e^{-cn}. \quad (3.14)$$

*The unconditioned free dual current is included. For the original wired current, expose its boundary cluster first; its free descendants satisfy this estimate conditionally, and adding the distinguished boundary cluster increases the count by at most one. The boxes and all cover buffers are fixed before the mesh limit.*

*There is also a pathwise consequence for a seed exploration. Suppose a compact set  $Q$  is at distance at least  $r > 0$  from the seed and initial removed support. List the discovered free clusters*

whose support meets  $Q$ , or whose union of odd holes meets  $Q$ . If  $\mathcal{L}_\delta(\tau; Q, r)$  is the number listed up to any stop, then, for all sufficiently small  $\delta$ ,

$$\mathcal{L}_\delta(\tau; Q, r) \leq \mathcal{N}_{\delta, \text{initial}}(r/3). \quad (3.15)$$

Add one on the right if the distinguished wired cluster is listed. This is domination by the initial count, not a tail conditional on the completed stop.

*Proof.* Cover  $L$  by finitely many inner squares of side a small fixed multiple of  $b$ . Around each choose a passage square and a larger buffer, with fixed ratios, in the regular ambient box. If a connected cluster has diameter at least  $b$ , choose two of its points at that separation and an inner square containing the first. The path to the second has a first exit from the passage square; its initial portion is a connected passage of diameter at least  $c_1 b$  there, for a fixed  $c_1 > 0$ . Let  $\mathcal{N}_{\delta, i}^{\text{passage}}(c_1 b)$  be the item 1 count of unsearched global free clusters of the initial residual law having such a passage in the  $i$ th square. Every global cluster is counted in at least one of these counts. Consequently

$$\mathcal{N}_\delta(b) \leq \sum_{\text{cover squares } i} \mathcal{N}_{\delta, i}^{\text{passage}}(c_1 b).$$

The cover is fixed before  $\delta$ . Its passage buffers may cross the original boundary or the revealed wired layer: extend the initial cell region to the ambient box and use the augmented matching completion of [Theorem 3.3](#). Item 1 of [Theorem 3.5](#) and a finite union bound give (3.14). For the original free dual law the matching primal boundary is the initial fill. Behind the wired  $N$  boundary cluster, [Theorem 3.1](#) gives precisely the product law whose common initial fill is that connected cluster joined to the exterior.

For (3.15), every discovered cluster meets the seed. If its support meets  $Q$ , it has diameter at least  $r - O(\delta)$ . If instead one of its odd holes meets  $Q$ , that hole lies in the convex hull of the cluster's cell region: a bounded complementary component is contained in that convex hull. A point of the hole is therefore at distance at most  $\text{diam } C + O(\delta)$  from the seed contact on  $C$ . This again gives  $\text{diam } C \geq r - O(\delta)$ . The listed clusters are distinct members of the initial free collection, proving the pathwise inequality. This argument covers a cluster which surrounds  $Q$  without meeting it.  $\square$

**Lemma 3.7** (Distance to a chamber-side rim). *Let  $U_\delta$  be a resolved residual chamber of an exact cut exploration, let  $S_\delta$  be the revealed support including the initial boundary and seed as needed, and let  $\Gamma_\delta$  be its own chamber-side  $W$  perimeter. For any subset  $C_\delta$  of that chamber,*

$$\sup_{x \in C_\delta} \text{dist}(x, \Gamma_\delta) \leq \sup_{x \in C_\delta} \text{dist}(x, S_\delta) + O(\delta). \quad (3.16)$$

*The constant is the lattice resolution constant. In particular, if  $C_\delta \rightarrow C_*$  and  $S_\delta \rightarrow S_*$  in Hausdorff distance with  $C_* \subset S_*$ , the left side tends to zero.*

*Proof.* For  $x \in U_\delta$ , take a segment to a nearest point of  $S_\delta$ . Its first exit from the resolved chamber meets that chamber's own medial perimeter, within  $O(\delta)$  of the corresponding  $W$  edge. The distance to this first exit is at most the length of the segment. Taking suprema proves the inequality. This uses the own perimeter even if the nearest support point belongs to another nearby part of a pinched wired layer. For the original free chamber the same argument uses its polygonal boundary as  $S_\delta$ . The Hausdorff conclusion follows from the one-sided inclusion  $C_* \subset S_*$ .  $\square$

**Corollary 3.8** (No shadowing at recorded stops). *Fix a buffered box  $Q$  and  $b > 0$ . At a stop  $\tau$  of an exterior-attached seed exploration, let  $F_\tau$  be its frozen connected cell region, including the initial walls, all visited seed vertices, and all clusters already revealed. Let*

$$\Xi_\delta(\tau; b, \eta) = \left\{ \begin{array}{l} \text{some unsearched } N \text{ cluster has a connected} \\ \text{passage } \gamma \subset Q, \quad \text{diam } \gamma \geq b, \quad \gamma \subset F_\tau^\eta \end{array} \right\}.$$

*For allowed exact cut or  $(s, W)$  stopping histories  $\mathcal{H}_\tau$  with the supported residual product class,*

$$\mathbb{P}[\Xi_\delta(\tau; b, \eta) \mid \mathcal{H}_\tau] \leq \omega_{Q,b}(\eta, \delta), \quad \lim_{\eta \downarrow 0} \limsup_{\delta \downarrow 0} \omega_{Q,b}(\eta, \delta) = 0, \quad (3.17)$$

*uniformly over positive probability histories. Exterior colors rendered irrelevant by a completed cut are allowed. Arbitrary interior  $d$  pins, or conditioning on a later visibility event, are not part of this statement.*

*For an exploration in the enclosing box of Theorem 3.6, index stops by their rank among discoveries of clusters of diameter at least  $b$ , and let  $\tau_0$  be the initial stop. All intervening smaller discoveries are included in each frozen  $F_{\tau_i}$ . If  $D_\delta(b)$  is the number of these records, then, for every integer  $K$ ,*

$$\mathbb{P}[\exists 0 \leq i \leq D_\delta(b) : \Xi_\delta(\tau_i; b, \eta) \mid \text{initial cut data}] \leq Ce^{-cK} + (K+1)\omega_{Q,b}(\eta, \delta). \quad (3.18)$$

*The indices here are recorded macroscopic ranks, not raw discovery numbers.*

*If a random width  $\eta_\delta$  tends to zero only on a limiting event  $E$ , the usable deterministic inclusion is, for fixed  $\eta$ ,*

$$E \cap \Xi_\delta(\tau; b, \eta_\delta) \subseteq \Xi_\delta(\tau; b, \eta) \cup (E \cap \{\eta_\delta > \eta\}). \quad (3.19)$$

*Thus one first fixes  $\eta$ , takes the mesh limit (and Fatou's inequality when passing to a limit event), and then sends  $\eta$  to zero. No estimate is asserted after conditioning on  $E$ .*

*Proof.* An actual unsearched  $N$  passage avoids  $F_\tau$  and the actual edges of every attached augmented matching circuit. A finite collar-window cover of  $Q$  therefore gives (3.17) from Theorem 3.4, including windows met by the initial wall. The exact cut factorization identifies the residual marginal for every permitted history. The bound is for the existential future event under that marginal; it does not reveal a particular future passage.

On  $D_\delta(b) \leq K$ , there are at most  $K+1$  frozen indexed stops. Apply (3.17) at each one and use a union bound. The complementary event is bounded by (3.14), which yields (3.18). Finally, (3.19) follows from monotonicity of neighborhoods. It gives the stated order of limits even when the width was chosen from a future Hausdorff containment.  $\square$

For subsequent analytic arguments an additive mixing estimate is not sufficient: the prescribed microscopic patterns may themselves have very small probability. We therefore record a multiplicative comparison. A *complete pattern* in a finite vertex-and-edge set specifies all its  $s$  spins and all its  $W$  edge states. Every specified edge includes its endpoints.

**Proposition 3.9** (Pattern comparison). *Fix regular compact sets  $Q_i$  and filled buffered neighborhoods  $B_i$ ,  $1 \leq i \leq m$ , with the  $B_i$  pairwise disjoint and a fixed positive separation between  $Q_i$  and  $\partial B_i$ . Assume each  $B_i$  is contained in the domain, with no prescribed data there other than*

the pattern being tested in  $Q_i$ . Exterior data may be arbitrary admissible  $(s, W)$  data and their retained connections. There is a constant  $C$  such that, for complete patterns  $p_i$ ,

$$C^{-m} \prod_{i=1}^m \pi_i(p_i) \leq \mu[p_i \text{ on every } Q_i] \leq C^m \prod_{i=1}^m \pi_i(p_i). \quad (3.20)$$

Here  $\mu$  is any such conditional flat law, and  $\pi_i$  is the plane pattern law in  $Q_i$ . The constant depends on the fixed buffer geometry and  $t$ , not on the patterns or on their number of microscopic prescriptions. Copies under parity-preserving lattice translations and reflections satisfy the same assertion.

In particular, the single-box pattern laws have uniformly bounded Radon–Nikodym density ratios, and the same remains true locally after conditioning on any collection of separated patterns in the other boxes.

*Proof.* Consider first a single  $Q$  inside its buffer  $B$ . Prescribe a complete admissible pattern in four batches: its plus vertices, its minus vertices, its increasing compatible edge prescriptions, and its decreasing compatible edge prescriptions. After the first two batches every edge endpoint is fixed. A disagreeing edge is already determined closed. On monochromatic edges the two remaining batches mean open plus or closed minus, and closed plus or open minus, respectively. Each batch is monotone in one of the two orders of [Theorem 3.2](#). The number of batches is four, independently of the pattern size.

Fix the preceding batches and consider an increasing current batch  $E$ . Choose two deterministic annular buffers between  $Q$  and  $\partial B$ , and fix a plus circuit on the farther one. Let  $\lambda$  be the law inside that deterministic circuit with the preceding prescriptions retained. Comparison gives

$$\mu(E) \leq \lambda(E). \quad (3.21)$$

In the nearer annulus, conditional on the preceding prescriptions, a random plus circuit has probability at least  $p > 0$ . To see this even if those prescriptions contain both signs, cover that annulus by buffered test boxes missing  $Q$ ; the exterior bound for one parity and FKG supply its circuit with the same positive constant. Search outside-in, inspecting nothing on the sealed side, and stop at the first such circuit in a fixed exploration. Its interior law is plus wired by [Theorem 3.1](#), and hence dominates the restriction of  $\lambda$ , with the same preceding prescriptions. Consequently

$$\mu(E) \geq p\lambda(E). \quad (3.22)$$

The decreasing batch uses minus circuits and the reversed order.

The law  $\lambda$  is common to any two exterior conditions under comparison. Applying (3.21)–(3.22) to each of the four conditional batch probabilities yields bounded ratios for the complete pattern probabilities. Zero-probability patterns are exactly the common local incompatibilities and are harmless. These constants remain valid when the comparison law is a flat exhaustion of the plane. The common plane spin/percolation law obtained from flat exhaustions in [16, Theorem 17.1] therefore gives the comparison with  $\pi_i$  by taking that exhaustion limit. For fixed  $W$  parity this law is invariant under parity-preserving lattice translations and reflections, with the compatible global residue action. An automorphism exchanging the lattices acts also by exchanging which parity is called  $W$ ; it is not invariance of the same fixed edge decoration under that automorphism. The height-gradient marginal has the full corner-grid symmetries used later.

For several boxes, condition successively on the patterns in the other boxes. These constraints lie outside the current filled buffer, so the single-box estimate applies with unchanged constants.

The chain rule gives (3.20). Conditioning on separated patterns and repeating the same argument proves the last assertion. Notice that the proof is multiplicative at every stage; it is not an inference about rare events from a total variation error.  $\square$

### 3.4 Extending separation witnesses

The preceding proposition concerns one spin parity and its edge decoration. The other spins color complementary components, so their admissible assignments can depend on connections outside a tested pattern. We now localize that dependence. In this subsection choose  $W$  on the parity carrying the flat wall. The complementary  $N$  components in the chamber are all free. This restriction is needed only for the color statement here and below; the wired bulk circuit case of Theorem 3.3 remains available.

Fix regular boxes  $Q \Subset Q' \Subset B$  with fixed positive gaps. For a geometric closed set, its complete pattern includes every  $W$  edge meeting the set and both endpoint spins; the resulting one-mesh boundary layer is part of the pattern. Form the restricted  $N$  graph using precisely the dual edges whose  $W$  states that pattern determines. A restricted component is a *terminal* if it meets  $Q$  and contains a vertex incident to an unprescribed  $N$  edge. Every component meeting  $Q$  that is not a terminal is sealed by the pattern, so its color is independent of the exterior. This incidence definition avoids losing an edge in the lattice boundary layer. Select one marked vertex of each terminal by a deterministic rule. A vector of signs on these marks is *compatible* with an exterior configuration if no two oppositely assigned marks become connected in the full  $N$  graph.

**Lemma 3.10** (Buffered separation witnesses). *Fix  $p > 0$  and an integer  $k$ . Conditional on any complete pattern in  $Q'$  with at most  $k$  terminals, suppose a specified terminal assignment has compatibility probability at least  $p$  in a  $W$ -flat domain containing  $B$ . There is  $c = c(p, k, Q, Q', B, t) > 0$  such that, for every sufficiently small mesh, with conditional probability at least  $c$ , every opposite pair of marks is separated by an actual open monochromatic  $W$  loop contained in  $B$ . Thus compatibility is witnessed without using the configuration beyond  $B$ . The estimate is uniform in the microscopic pattern and in the enclosing domain.*

*Admissible  $(s, W)$  exterior data beyond a fixed larger buffer are allowed. No transverse  $d$  colors are conditioned upon in this statement.*

*Proof.* In a finite  $W$ -flat domain, planar cut-cycle duality says that two marks disconnected in  $N$  are separated by a simple monochromatic  $W$  loop, allowing portions of the matching outer perimeter. We will replace possibly long loops by loops in successively enlarged cores. Write  $\mathcal{O}$  for the set of required opposite pairs and  $m = |\mathcal{O}| \leq k(k-1)/2$ . If  $m = 0$  there is nothing to prove. Choose regular closed cores

$$Q' = K_0 \Subset K_1 \Subset \cdots \Subset K_m \Subset B.$$

In each gap fix a middle annular band and a wider band compactly inside the gap, with further positive margins. At lattice scale the core includes every  $W$  edge meeting the closed core and its endpoints. This convention counts a crossing on its boundary; all such edges of  $K_m$  lie in  $B$  for small mesh.

For a complete pattern  $g$  on  $K_j$ , let  $\mathcal{S}(g) \subseteq \mathcal{O}$  be the pairs separated by some actual monochromatic loop in that lattice core. This set is determined by  $g$  and can only increase when the core pattern is enlarged in the same configuration. A pair outside  $\mathcal{S}(g)$  is called missing. We prove the following one-step statement. If, under the original law conditional only on  $g$ ,

compatibility has probability at least  $a > 0$  and some pair is missing, then

$$\mathbb{P}[\text{compatibility and } |\mathcal{S}(P_{j+1})| > |\mathcal{S}(g)| \mid P_j = g] \geq r(a, k) > 0. \quad (3.23)$$

The constants use the fixed bands of this step and are uniform in  $g$  and the mesh. Here  $P_j$  denotes the complete core pattern.

First choose a finite type of witness configuration. On compatibility, among simple loops separating a missing pair choose one whose enclosed set  $I$  of marks is inclusion-minimal, with a deterministic rule for ties. It contains a nonempty proper subset of the marks. No missing pair is wholly inside its disk. To prove this, let  $C$  be the selected loop and suppose a missing pair  $a, b$  lies inside. Choose a witness loop  $D$  for that pair. If it meets  $C$ , their colors agree, since opposite-color open  $W$  paths cannot meet. Consider the finite planar graph

$$H = (C \cup D) \cap \overline{\text{int } C}.$$

Every underlying dual path from  $a$  to  $b$  leaving  $\text{int } C$  crosses  $C$ , and every such path staying inside crosses  $D$ . Thus  $H$  separates  $a$  from  $b$  in the full plane. Planar cut-cycle duality extracts a simple cycle of  $H$  separating them. Its disk lies in  $\overline{\text{int } C}$  and contains one of  $a, b$  but not the other, so its marked set is a nonempty proper subset of  $I$ . All its edges have the common color. If  $C$  and  $D$  are disjoint, the disk of  $D$  must lie inside that of  $C$  and gives the same contradiction. This proves the claim, including shared-edge and tangency cases.

Call  $C$  the target. For every missing pair wholly outside  $I$ , choose a witness loop and record its color. Record also the target color and  $I$ . Reflect the colors for the description so that the target is plus. A chosen minus witness is disjoint from the target; its disk is either disjoint from the target disk or contains that disk. Of its two marked endpoints record the one on the side away from the target disk. The whole type consists of these data, including the ordered away-endpoint vector and the fixed mark  $i_0 = \min I$ . There are at most  $L(k) < \infty$  types, independently of the shapes of all loops. If the compatibility probability is at least  $a$ , one type event  $\mathcal{T}$  has probability at least  $a_0 = a/L(k)$ . Fix that type now. The sampling law remains the original law conditional on  $g$ ; it is not conditioned on  $\mathcal{T}$ .

From its fixed away marks perform a deterministic exploration of the full components of the underlying dual graph whose edges do not cross an open  $W^-$  edge. These are components of the complement of  $W^-$ , not components of  $N$ . Explore through already known core edges as well as new ones, revealing endpoint spins before each new edge state, until each component is exhausted. There are at most  $k$  distinct source components. The transcript  $h$  is the cylinder of the queried values, so its residual law  $\mu_h$  is an allowed pin law in [Theorem 3.2](#). In particular it remains positively associated. The interior frontier of a full source component consists of actual  $W^-$  edges; it is not a frontier artificially stopped at the old core. Original outer-boundary conditions are retained, and the middle-band test windows are away from the old core and the outer boundary.

Let  $F_h^0$  be the union of the source cell regions of these full components, without filling any complementary pocket. On  $\mathcal{T}$ , each chosen minus witness prevents the source on its away side from reaching the target disk. Thus the entire  $F_h^0$ , including its frontier, misses that disk. The transcript also certifies the selected minus separations: the other endpoint of each such pair is outside its explored complement-of- $W^-$  component. Every  $N$  edge belongs to that complement graph, so these certificates imply the corresponding  $N$  disconnections. Denote them by  $\mathcal{C}(h)$ .

In the residual law, let  $\mathcal{U}_h$  be the event that there exists an actual plus target loop enclosing exactly  $I$ , whose closed disk misses  $F_h^0$ , together with actual plus witnesses for the specified missing outside pairs of plus type. This event is increasing in (3.3). The minimality rule and the

identities of the previously selected loops have been dropped; they would not be increasing. If  $\nu_g$  is the law of the transcript under the original law conditioned on the core, the implication from  $\mathcal{T}$  gives

$$a_0 \leq \int \mathbf{1}_{\mathcal{C}(h)} \mu_h(\mathcal{U}_h) \nu_g(dh).$$

It follows that the histories satisfying  $\mathcal{C}(h)$  and  $\mu_h(\mathcal{U}_h) \geq a_0/2$  have  $\nu_g$  mass at least  $a_0/2$ . We work separately under each such  $\mu_h$ . There is no FKG or circuit estimate under the mixture conditioned on this choice of histories.

For this fixed history choose the accessible component  $U_h$  of the complement of the raw region  $F_h^0$  containing the already fixed mark  $i_0$ . Histories with  $i_0 \in F_h^0$  have  $\mu_h(\mathcal{U}_h) = 0$  and do not occur in the retained family. Every target disk in  $\mathcal{U}_h$  is connected, contains  $i_0$ , and misses  $F_h^0$ , so it lies in this one component  $U_h$ . Only after choosing  $U_h$ , add the other complementary pockets to the forbidden cell region and call the resulting region  $F_h$ . This addition is for the marginal collar test alone. It does not discard any actual residual variable from  $\mu_h$ ; plus witnesses for other pairs may still lie in those pockets.

For each individual connected source, the connected set  $U_h$  lies in one component of its complement and hence faces one incident boundary walk of that source. It may face that walk at many entrances, but there are at most  $k$  source labels in total. Against  $U_h$  the frontier is actual minus in the middle band. The exact completion of one component of [Theorem 3.3](#) therefore applies there; any use of a partition comparison has the same  $k$  indivisible labels. The added pockets attach to these sources, so every local piece in the band is anchored back to a source mark in the old core or to the outer boundary. A target disk remains disjoint from  $F_h$ .

Put  $b_0 = a_0/2$ . Choose a collar width  $\eta > 0$ , uniformly in the retained histories, so that [Theorem 3.4](#) bounds by  $b_0/2$  the existence of a full underlying dual traversal of the middle band which lies in  $F_h^\eta$  and avoids  $F_h$  and its attached minus circuits. Use the selected  $U_h$  marginal for this estimate. The full residual law  $\mu_h$  is retained for all other events. The virtual-disk clause makes this a simultaneous bound for traversals inside any target disk in  $\mathcal{U}_h$ . In particular, we subtract this error from  $\mu_h(\mathcal{U}_h)$ ; we do not condition the law on absence of the collar event.

We spell out the increasing event supplied by this subtraction. Write  $K$  for the enlarged lattice core,  $A$  for its fixed middle band, and  $A^+$  for the wider band. Choose a fixed regular outer dual graph  $R$  inside  $B$ , containing  $K$  with a complete connected annular shell outside  $K$ . In the path tests below every underlying dual edge of  $R$  is available, regardless of its  $N$  state or whether it crosses a minus edge or the forbidden region. For each required pair across  $I$ , orient paths from its  $I$  mark, and require every simple underlying path  $\gamma$  between the two marks to satisfy

$$\begin{aligned} &\gamma \text{ crosses an actual } W^+ \text{ edge in } K, \\ &\quad \text{or} \\ &\gamma \text{ has a subpath } \beta \subset A \text{ from the inner to the outer boundary of } A \\ &\quad \text{which visits a point } z \text{ with } \text{dist}(z, F_h) > \eta. \end{aligned} \tag{3.24}$$

Intersect this requirement with the actual plus separations for the specified outside pairs, and call the event  $\mathcal{V}_h$ . The path family is finite and fixed by  $R$ ;  $F_h$  is fixed by  $h$ . Only the first alternative changes with  $W^+$ . Thus  $\mathcal{V}_h$  is increasing. Neither  $F_h$  nor any minus edge is a real barrier in this definition.

On  $\mathcal{U}_h$  outside the collar event, (3.24) holds. Indeed a path avoiding the actual  $W^+$  edges in  $K$  must exit  $K$  before its first crossing of the target loop. An edge meeting the closed core is counted in  $K$ , so a first crossing on its boundary cannot be lost. The path prefix before that target crossing lies in the target disk. Before its first exit from  $K$ , take the segment from its last

hit of the inner band boundary to its first subsequent hit of the outer one. This is a full subpath  $\beta \subset A$ , even if the original path made earlier excursions. It lies in the target disk and therefore avoids  $F_h$  and all attached minus circuits. Absence of the collar event forces it to visit a point at distance greater than  $\eta$ . Hence

$$\mu_h(\mathcal{V}_h) \geq b_0/2. \quad (3.25)$$

We replace the second alternative by actual plus circuits. Choose a deterministic grid in  $A$  of mesh much smaller than  $\eta$  and than the separation of its two boundary curves. For each cell whose filled buffer is disjoint from  $F_h$  and contained in  $A^+$ , request an actual plus circuit surrounding the cell in that buffer. Choose the grid and buffers so that every point of  $A$  at distance greater than  $\eta$  from  $F_h$  lies in an inner cell surrounded by one requested circuit, and the disk of every such circuit has diameter smaller than the distance between the two boundary curves of  $A$ . This is possible by taking the cell size a sufficiently small fixed multiple of  $\eta$  and of the band width. There are at most  $M = M(\eta)$  cells. Their buffers miss all source queries and the old core; for small mesh they exceed the lattice cutoff.

Under the full residual pin law  $\mu_h$ , each requested actual plus circuit has probability at least  $p_1 > 0$  by Theorem 3.3. Positive association intersects all these increasing events with  $\mathcal{V}_h$ , giving probability at least  $(b_0/2)p_1^M$ . On their intersection, no across- $I$  path can avoid the actual plus edges in  $K$ . Otherwise (3.24) supplies a full band subpath visiting the interior of one requested circuit. A connected path avoiding that circuit and visiting its interior must stay inside its disk. Its two band-boundary endpoints cannot both lie there, by the diameter choice. This contradiction realizes the temporary alternative using actual plus edges only.

For completeness, separation in the graph  $R$  gives actual closed witness loops in  $K$ . Any path using a more distant exterior can have each exterior excursion replaced in the complete connected shell of  $R$  outside  $K$ . Thus separation in  $R$  is separation in the full underlying dual graph by  $W^+ \cap K$ . Planar cut-cycle duality yields a simple plus loop in  $K$  for each across pair; a boundary-to-boundary cut is impossible because these blocking edges lie strictly inside  $R$ . Artificial fill and minus edges play no role in this final separation.

The history certificates protect the outside minus pairs. The outside plus pairs are protected by  $\mathcal{V}_h$ , the across pairs by the new local plus loops, and there was no missing pair wholly inside  $I$ . Previously short loops are fixed in  $g$ . Therefore compatibility holds and at least one missing pair has joined  $\mathcal{S}(P_{j+1})$ . The retained histories have  $\nu_g$  mass at least  $b_0$ , so the construction proves (3.23) with, for example,

$$r(a, k) = \frac{b_0^2}{2} p_1^{M(\eta)} > 0, \quad b_0 = \frac{a}{2L(k)}.$$

The collar width, grid, and mesh threshold were chosen from  $a, k$  and the fixed bands, not from the particular core pattern. Replace  $r(a, k)$  by its minimum over the finitely many band groups.

We finally iterate without retaining the long source histories. Let  $\mu$  be the original law conditional on the initial pattern, and set

$$\mathcal{G}_j = \{\text{compatibility and } |\mathcal{S}(P_j)| \geq j\}.$$

Initially  $\mu(\mathcal{G}_0) \geq p$ . If  $\mu(\mathcal{G}_j) \geq q_j > 0$ , disintegrate *this original law* only by the complete pattern  $P_j$ . Patterns with  $|\mathcal{S}(P_j)| \geq j$  and posterior compatibility at least  $q_j/2$  have total mass at least  $q_j/2$ : the other patterns contribute at most  $q_j/2$  to  $\mu(\mathcal{G}_j)$ . If such a pattern already has  $j+1$  short pairs, its conditional contribution is at least  $q_j/2$ ; otherwise apply (3.23). Hence

$$\mu(\mathcal{G}_{j+1}) \geq \frac{q_j}{2} \min \left\{ \frac{q_j}{2}, r(q_j/2, k) \right\} =: q_{j+1} > 0.$$

At the next step all source transcripts and selected type events have been integrated out. The new conditional law is again the original law given only its complete core pattern, so its unused buffers have the pin/FKG law required above. After  $m$  steps all pairs have short witnesses in  $K_m \Subset B$ , with probability at least  $q_m$ . All widths and the finitely many mesh thresholds are chosen before taking the mesh limit. This proves the lemma.  $\square$

**Proposition 3.11** (Full-spin contiguity). *Fix compactly nested regular boxes  $Q \Subset B$  in the interior of a  $W$ -flat domain, with fixed positive buffer ratios. Thus the complementary  $N$  components are free. Let  $\mu_\delta$  be its law and  $\pi_\delta$  the plane law, restricted to the full spin pair in  $Q$ . The domains and any allowed exterior  $(s, W)$  data may depend on  $\delta$ , but their walls and prescriptions stay outside  $B$ . There is a uniform contiguity modulus: for every  $\varepsilon > 0$  there is  $\zeta > 0$  such that, for every sufficiently small mesh and every event  $E$  determined by the full spin pair in  $Q$ ,*

$$\pi_\delta(E) < \zeta \implies \mu_\delta(E) < \varepsilon. \quad (3.26)$$

*The conclusion also holds if the tested data include  $W$  in  $Q$ . In particular it applies to mesh-dependent tests. Either flat parity can be used by assigning  $W$  to the parity carrying that flat wall. This statement does not include a fixed-color wired  $N$  component before its removal.*

*Proof.* Choose  $Q \Subset Q' \Subset B'$  with  $B' \Subset B$ , leaving the fixed buffers of [Theorem 3.10](#). Let  $P$  be the complete  $(s, W)$  pattern in  $Q'$ , and let  $T(P)$  be its number of terminals. By [Theorem 3.5](#), uniformly in both domain and plane laws,

$$\mathbb{P}[T(P) > M] \leq Ce^{-cM}. \quad (3.27)$$

Choose  $M$  so that the right side is less than  $\varepsilon/4$  for the domain law.

On  $T(P) \leq M$ , write  $b$  for an assignment of signs to its terminals, and let  $v_\mu(b \mid P)$  be its conditional exterior compatibility probability. By [Theorem 3.1](#), the actual terminal-color probability  $q_\mu(b \mid P)$  satisfies

$$2^{-M}v_\mu(b \mid P) \leq q_\mu(b \mid P) \leq v_\mu(b \mid P). \quad (3.28)$$

Indeed, given a compatible exterior partition, specifying these terminal signs costs  $2^{-j}$  for some  $j \leq M$  distinct free global components. The same calculation applies in the plane because [\[16, Theorem 17.1\]](#) gives no unbounded complementary component and independent fair transverse colors. Components of the restricted pattern not incident to an unprescribed edge carry independent fair colors. Their distribution is the same under the two laws once  $P$  and the terminal signs are given. The lower bound in [\(3.28\)](#) uses freeness; a terminal wired to a prescribed incompatible color would not satisfy it.

Choose  $p > 0$  with  $2^M p < \varepsilon/4$ . The total domain probability of assignments with  $v_\mu(b \mid P) < p$  is at most  $2^M p$ , by the upper bound in [\(3.28\)](#). For every remaining assignment, [Theorem 3.10](#) gives short witnesses in  $B'$  with conditional probability at least  $c(p, M) > 0$ . Fix a realized pattern  $g$  for  $P$ , and let  $\mathcal{W}_b$  be this short-witness event, determined by a complete pattern in  $B'$ . If  $C_{B'}$  and  $C_{Q'}$  are the unconditioned pattern density constants in [Theorem 3.9](#), summing over the larger patterns and then dividing gives

$$\pi_\delta(\mathcal{W}_b \mid P = g) = \frac{\pi_\delta(\mathcal{W}_b \cap \{P = g\})}{\pi_\delta(P = g)} \geq (C_{B'} C_{Q'})^{-1} \mu_\delta(\mathcal{W}_b \mid P = g).$$

This uses only the original laws on one parity on the two pattern sets, not a comparison after conditioning on  $d$  or on the witness event. Hence there is  $c_0 = c_0(p, M) > 0$  such that

$$v_\pi(b \mid P) \geq c_0, \quad q_\pi(b \mid P) \geq 2^{-M} c_0 \quad (3.29)$$

whenever  $v_\mu(b \mid P) \geq p$  and  $T(P) \leq M$ . There is no conditioning on  $d$  in this comparison: the short witness event concerns only  $W$  and  $s$ .

Let  $C_0$  be the uniform density-ratio constant for  $P$  in [Theorem 3.9](#). Conditional on  $P$  and a terminal assignment, integrate the identical independent internal colors. On the retained assignments, (3.29) bounds the ratio of terminal-color probabilities by  $2^M/c_0$ . Therefore every tested event satisfies

$$\mu_\delta(E) \leq \frac{\varepsilon}{2} + C_0 \frac{2^M}{c_0} \pi_\delta(E). \quad (3.30)$$

Choosing  $\zeta < \varepsilon c_0/(2C_0 2^M)$  proves (3.26). All constants were chosen from  $\varepsilon$ , the fixed geometry, and  $t$  before the mesh limit, so the modulus is uniform in the stated domain laws. Enlarging the tested data by  $W$  does not change the conditioning argument.  $\square$

The distinction between [Theorem 3.9](#) and [Theorem 3.11](#) is useful later. The former supplies uniform multiplicative estimates even for extremely rare patterns on one parity. The latter supplies domination in probability for both spin parities, without asserting a bounded density ratio for every possible microscopic color assignment.

## 4 Reflection positivity and the plane covariance

This section identifies every subsequential plane covariance, without assuming either Gaussianity or rotation invariance of a scaling limit. The positive transfer method of [30, Section 5] suggests the argument, but its sixfold rotational cancellation is unavailable here. Two square-lattice reflection frames at angle  $\pi/4$  provide the required replacement.

We temporarily choose Euclidean units in which the corner grid is  $\mathbb{Z}^2$ . This is a fixed rotation and dilation of the original embedding. The local plaquette calculation above identifies its height weights with the six-vertex weights  $(1, 1, c)$ . The spin and decoration law obtained from flat exhaustions is the measure of [16, Theorem 17.1]; [16, Lemma 18.7(iii)] identifies its height-gradient marginal with the slope-zero six-vertex plane measure. By [16, Theorem 2.2], that marginal is stationary under all corner-grid translations and invariant under the lattice reflections. The geometric height reflection used below sends  $K$  to its composition with the spatial reflection, up to an additive constant. In arrow variables this combines spatial reflection with global arrow reversal; it also preserves the finite plaquette weights directly. These statements apply for every  $1 \leq c \leq 2$ . A uniform simultaneous choice of the two spin signs is understood when residues rather than increments are observed. In particular an odd corner-grid translation  $T$  exchanges the parities with gauge  $(\tau', \tau'^\dagger) = (\tau^\dagger \circ T, -\tau \circ T)$ : its height is  $K'(x) = K(Tx) - K(T0)$ . This preserves the incident-corner rule and accounts for the change of reference parity. The additive constant of  $K$  is immaterial. For a smooth compactly supported function  $f$  of integral zero, use bilinear nodal interpolation on each corner-grid square to define  $K_\delta(f)$ . This interpolation commutes with all four row reflections and with grid translations. Write its finite array of coefficients as  $b_{\delta,x}$ ; their sum is exactly zero. For a  $2m$ th moment, choose smooth mass-one reference densities in  $2m$  disjoint fixed disks outside the common support of the tests. Let  $r_{\delta,j,y}$  be nonnegative

discretizations of these densities, normalized so that  $\sum_y r_{\delta,j,y} = 1$ . For each occurrence  $j$  the exact finite-sum identity is

$$K_\delta(f) = \sum_{x,y} b_{\delta,x} r_{\delta,j,y} (K(x) - K(y)).$$

Use a separate reference index  $j$  in each factor of the moment. The regularity estimate [16, Theorem 4.5] then applies to increment correlations with separately summed reference variables. Its collision singularities are products of logarithms, which are locally integrable; the lattice cells of exact coincidence contribute at most a lattice area times a fixed power of  $1 + |\log \delta|$ . Integrating the estimate gives

$$\sup_{0 < \delta < \delta_0} \mathbb{E} |K_\delta(f)|^{2m} < \infty, \quad m \in \mathbb{N}. \quad (4.1)$$

The bounds are uniform for tests in a bounded smooth family with common compact support. The estimate holds throughout  $1 \leq c \leq 2$ , not just the range of the convergence theorem in that paper. The elementary path-factor correction in its proof is supplied in [Section A.1](#).

#### 4.1 Four positive reflection frames

**Lemma 4.1** (Positive normal transfer). *The plane law is reflection positive across a row of corner-grid sites in each of the directions  $0, \pi/4, \pi/2, 3\pi/4$ . In its reflection Hilbert space, translation by two successive row spacings in the inward normal direction is a positive self-adjoint contraction. For the height-increment law, tangential translation by one row period is a commuting unitary. Reflection positivity and positive normal transfer remain valid for local spin and edge decorations strictly on one side of the row, and for independent uniform phase lifts of the height. With a fixed one-parity edge decoration, the tangential unitary assertion is restricted to parity-preserving translations.*

*Proof.* First use a finite reflection-symmetric flat exhaustion. Across an axial row, plaquette interactions on the two sides factor after conditioning on the row heights. Across a diagonal row, a straddling plaquette has the following conditional weight matrix. If its two row heights are equal, the other heights are independently one unit above or below that value. Equal transverse heights have weight  $c$ , and unequal ones have weight one. Consequently the matrix, after multiplication by  $t$ , is

$$\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

It is positive semidefinite because  $0 < t \leq 1$ . If the row heights differ by two, the transverse heights are forced and there is only a scalar factor. Factoring each positive matrix as a Gram matrix proves positivity after summing the row data. Independent local decorations are included in the factors on their own sides. Spin residues are determined by heights and one compatible global sign choice, which can be fixed at an anchor on the row. For a phase  $U \exp(i\alpha(K(x) - K(o)))$ , also condition on its value at that anchor;  $U$  has independent uniform law. Thus these augmentations preserve the same Gram representation. Circuit comparison passes each reflected exhaustion to the same plane law.

Let  $\Theta$  be reflection in the row and set  $\langle F, G \rangle = \mathbb{E}[\overline{\Theta F} G]$  for functions on its inward half-plane. Denote the pure normal translation through two row spacings by  $T$ . Stationarity gives symmetry of  $T$  for this form. Reflection positivity at the intermediate row gives  $\langle F, TF \rangle \geq 0$ . To prove boundedness before quotienting, put  $b_j = \|T^j F\|$ . Reflection Cauchy–Schwarz gives

$b_j^2 \leq b_{j-1}b_{j+1}$ , whereas ordinary Cauchy–Schwarz and stationarity give  $b_j \leq \|F\|_{L^2}$  for all  $j$ . A bounded nonnegative log-convex sequence cannot increase at its first step. Hence  $\|TF\| \leq \|F\|$ , also when  $\|F\| = 0$ . Thus  $T$  descends to a positive self-adjoint contraction. Tangential translation preserves the form, has a form-preserving inverse, and commutes with  $T$ , whenever it preserves the chosen state space. For the height-increment law this includes one row period. With a fixed one-parity decoration it includes two axial periods or one diagonal period; the normal transfers used above already preserve that parity.  $\square$

In a frame with unit tangent  $u$  and normal  $v$ , write the rectangular sublattice just obtained as

$$\Lambda_0 = \ell\mathbb{Z}u + d\mathbb{Z}v.$$

For an axial frame one may take  $(\ell, d) = (1, 2)$ ; for a diagonal frame take  $(\ell, d) = (\sqrt{2}, \sqrt{2})$ . Each has finite index in the corner grid. All frequency coordinates below are Euclidean:  $p = ku + lv$ . In particular, the normal spectral scale and  $|k|$  have the same physical units. The spectral argument uses the undecorated height-increment space; the decorated argument in [Section 5](#) uses only normal transfers.

## 4.2 The spectral measure and its Cauchy disintegration

**Lemma 4.2** (Increment spectral measure). *There is a nonnegative measure  $C$ , locally finite on the punctured reciprocal torus of the corner grid, such that every finitely supported real array of zero sum satisfies*

$$\mathbb{E} \left( \sum_x b_x K(x) \right)^2 = \int \left| \sum_x b_x e^{ip \cdot x} \right|^2 C(dp). \quad (4.2)$$

*There is no additional random linear tilt. For sufficiently small  $r$ ,*

$$C\{r \leq |p| \leq 2r\} \leq C_0. \quad (4.3)$$

*Consequently  $\int_{0 < |p| < r_0} |p|^\eta C(dp) < \infty$  for every  $\eta > 0$ .*

*Proof.* Let  $V_x$  be ordinary translation on the plane probability space,  $E(dp)$  its spectral resolution [\[34, Theorem 5.6\]](#), and  $\xi_x = K(x) - K(0)$ . The cocycle identity gives

$$(V_y - I)\xi_x = (V_x - I)\xi_y.$$

On any spectral neighborhood where  $e^{ip \cdot y} - 1 \neq 0$ , define

$$C(dp) = \frac{\langle \xi_y, E(dp)\xi_y \rangle_{L^2}}{|e^{ip \cdot y} - 1|^2}.$$

The identity makes the definitions agree on overlaps; two grid directions cover the punctured torus. The invariant part is additive and has the form  $A \cdot x$  for a random vector  $A \in L^2$ . Smearing at scale  $L$  against a neutral test of first moment  $m$  gives invariant part  $LA \cdot m$ . Its norm is uniformly bounded by [Equation \(4.1\)](#), since orthogonal projection cannot increase norm. Two independent choices of  $m$  give  $A = 0$ .

Choose two real neutral smooth tests whose Fourier transforms have no common zero on  $1 \leq |p| \leq 2$ ; derivatives in the two coordinate directions of a sufficiently narrow smooth bump suffice. Their scale- $r^{-1}$  interpolation transforms converge uniformly on this annulus to those transforms. The sum of their squared absolute values is therefore bounded below on  $r \leq |p| \leq 2r$ . Apply [Equation \(4.2\)](#) and the uniform second moment bound. Summing over dyadic annuli proves the last assertion.  $\square$

Project  $C$  to the reciprocal rectangle  $[-\pi/\ell, \pi/\ell) \times [-\pi/d, \pi/d)$  of  $\Lambda_0$ , and call the result  $C'$ . Besides the physical origin, the preimages of its origin are finitely many nonzero reciprocal points, called aliases. Their neighborhoods have finite mass; atoms at aliases are allowed. Put  $P = 2\pi/d$ . For  $0 < s < \infty$ , define the probability law

$$Q_s(dl) = \sum_{j \in \mathbb{Z}} \frac{s \, dl}{\pi(s^2 + (l + jP)^2)}, \quad -P/2 \leq l < P/2,$$

with  $Q_0 = \delta_0$  and  $Q_\infty$  uniform on the circle.

**Lemma 4.3** (Positive Cauchy mixture). *For  $k \neq 0$  there is a nonnegative mixing measure  $\nu$  such that*

$$C'(dk, dl) = \int_{[0, \infty]} Q_s(dl) \nu(dk, ds). \quad (4.4)$$

*It is finite on horizontal bands away from zero and satisfies*

$$\int_{0 < |k| < \kappa} |k| \nu(dk, ds) < \infty \quad (4.5)$$

*for small fixed  $\kappa$ .*

*Proof.* The row increment  $F = K(\ell u) - K(0)$  belongs to the reflection Hilbert space. The joint spectral theorem for its commuting tangential unitary and positive normal contraction [34, Theorem 5.6] gives, for  $j \geq 0$ ,

$$\mathbb{E}[FV_{n\ell u + jdv}F] = \int e^{in\ell k} \lambda^j \gamma(dk, d\lambda), \quad 0 \leq \lambda \leq 1.$$

Reflection gives the same expression with  $|j|$ . Set  $\lambda = e^{-ds}$ , including  $s = \infty$  when  $\lambda = 0$ . These are precisely the vertical Fourier coefficients of  $Q_s$ . Uniqueness of Fourier coefficients for finite measures identifies the ordinary increment measure as this mixture. That measure is  $|e^{i\ell k} - 1|^2 C'$ . Division for  $k \neq 0$  proves the formula. Since every  $Q_s$  has mass one, the integral in Equation (4.5) equals  $\int |k| C'(dk, dl)$  over the same band. Near the physical origin it is finite by Equation (4.3); all other contributions have finite mass.  $\square$

### 4.3 An angular defect in two frames

Define

$$\chi(k, l) = \cos(4 \arg(k + il)) = 1 - \frac{8k^2 l^2}{(k^2 + l^2)^2}, \quad D(k, s) = \left( \frac{|k| - s}{|k| + s} \right)^2,$$

where  $D(k, \infty) = 1$ . Uniformly in  $s \in [0, \infty]$ ,

$$\int \chi(k, l) Q_s(dl) = D(k, s) + O(|k|), \quad k \rightarrow 0. \quad (4.6)$$

For completeness, with  $b = |k|$ , the bounded analytic function  $((b + iz)/(b - iz))^2$  on the upper half-plane has boundary real part  $\chi(b, l)$ . Its Poisson average at  $is$  is  $((b - s)/(b + s))^2$ . To periodize, use

$$1 - \chi(b, l) = \frac{8b^2 l^2}{(b^2 + l^2)^2}, \quad \int_{\mathbb{R}} (1 - \chi(b, l)) \, dl = 4\pi b.$$

The sum of noncentral Cauchy densities over one period is uniformly bounded in  $s$ . Indeed split its sum at  $|j|$  comparable to  $s/P$  and use  $1/s$  for the nearer terms and a summable square tail for the rest. The omitted unfolded tail of  $1 - \chi$  is  $O(b^2)$ . This proves Equation (4.6), including its two limiting laws.

**Proposition 4.4** (Summable dispersion defect). *In each of the four reflection frames,*

$$\int_{0 < |k| < \kappa} D(k, s) \nu(\mathrm{d}k, \mathrm{d}s) < \infty, \quad \nu\{r \leq |k| \leq 2r\} \leq C_1. \quad (4.7)$$

*The second bound holds for every  $r > 0$ .*

*Proof.* Integrate Equation (4.6) over  $\varepsilon < |k| < \kappa$ . Its error is bounded independently of  $\varepsilon$  by Equation (4.5). Replacing  $C'$  outside the small physical disk by the original measure costs bounded outer and alias mass. In the radial annulus  $\varepsilon < |p| < \kappa$ , the omitted endcap  $|k| \leq \varepsilon$  has negative fourth harmonic only where  $|p| \leq C\varepsilon$ : negativity implies  $|l/k| < 1 + \sqrt{2}$ . Its negative part is bounded by a fixed number of the annuli in Equation (4.3). Thus, in a frame of angle  $\theta$ ,

$$\int_{\varepsilon < |k| < \kappa} D \, \mathrm{d}\nu \leq \int_{\varepsilon < |p| < \kappa} \cos(4(\arg p - \theta)) C(\mathrm{d}p) + C_2. \quad (4.8)$$

Choose the same physical inner and outer radii for frames  $\theta$  and  $\theta + \pi/4$ , making the fixed outer radius smaller if necessary. Their radial integrands are negatives pointwise. Adding the inequalities therefore bounds the sum of the two nonnegative defect integrals. Monotone convergence proves finiteness for each. Apply the same argument to the other frame pair. No rotation of the measure  $C$  through  $\pi/4$  has been used.

On  $r \leq |k| \leq 2r$ , the part with  $s \notin [r/2, 4r]$  has  $D \geq 1/9$ , so its mass is bounded. For  $s \in [r/2, 4r]$ , a fixed fraction of  $Q_s$  lies in  $|l| \leq 2r$ . The remaining mixing mass is bounded by a constant times  $C'\{r \leq |k| \leq 2r, |l| \leq 2r\}$ , which is bounded by the annular estimate and finite alias mass. This proves the horizontal-band bound for sufficiently small  $r$ . For the remaining  $r$ , the bands lie in a fixed horizontal region bounded away from zero, whose  $\nu$  mass is finite by Theorem 4.3; bands beyond the reciprocal interval are empty. Increasing  $C_1$  gives the bound for every  $r > 0$ .  $\square$

**Lemma 4.5** (Spectral endcaps). *For every frame, sufficiently small  $r$ , and  $0 < \varepsilon < r/10$ ,*

$$C\{r < |p| < 2r, |k| \leq \varepsilon\} \leq C_3 \varepsilon / r.$$

*In particular the punctured small disk has no mass on a frame's line  $k = 0$ .*

*Proof.* Describe the same region in a frame at angle  $\pi/4$ . Its new coordinates  $k', l'$  both have size comparable to  $r$ . At fixed  $k'$ , the old restriction  $|k| \leq \varepsilon$  confines  $l'$  to an interval of length  $O(\varepsilon)$ . On this interval each central Cauchy density is at most  $C/r$ , since  $s/(s^2 + l'^2) \leq 1/(2|l'|)$ , and the folded tails are bounded. The  $s = 0$  atom is outside the interval; the uniform law also satisfies the bound. The horizontal-band mass in the new frame is bounded by Equation (4.7). Projection only adds nonnegative alias mass, so its bound bounds the original physical measure. Letting  $\varepsilon$  decrease to zero gives the last statement.  $\square$

#### 4.4 Green covariance and reflection nullity

For neutral smooth tests write

$$\mathcal{L}(f, g) = \iint f(x) \log \frac{1}{|x - y|} g(y) \, \mathrm{d}x \, \mathrm{d}y.$$

**Proposition 4.6** (Subsequential plane Green covariance). *Every mesh sequence has a subsequence and a finite positive number  $\beta$  such that, for all neutral smooth compactly supported tests,*

$$\mathbb{E}[K_\delta(f)K_\delta(g)] \longrightarrow \beta \mathcal{L}(f, g). \quad (4.9)$$

*The possible values of  $\beta$  are bounded above and bounded away from zero. The convergence also holds for smoothly converging tests in a common compact set. At this stage  $\beta$  is allowed to depend on the subsequence.*

*Proof.* Rescale the physical measure by  $p \mapsto p/\delta$ . The annular bound gives vague subsequential compactness off zero; write  $M$  for one such limit. Fix  $0 < b < B$  in one reflection frame. Let  $\nu_\delta^{b,B}$  be the pushforward of  $\mathbf{1}_{\{b\delta \leq |k| \leq B\delta\}} \nu$  under  $(k, s) \mapsto (\xi, \sigma) = (k/\delta, s/\delta)$ , with  $\infty/\delta = \infty$ . The horizontal-band bound in Equation (4.7) bounds its total mass uniformly. Homogeneity of  $D$  and its integrability there give

$$\int D(\xi, \sigma) \nu_\delta^{b,B}(\mathrm{d}\xi, \mathrm{d}\sigma) = \int_{b\delta \leq |k| \leq B\delta} D(k, s) \nu(\mathrm{d}k, \mathrm{d}s) \longrightarrow 0.$$

Take a weakly convergent further subsequence on the compact space  $\{b \leq |\xi| \leq B\} \times [0, \infty]$ , where the last interval is compactified at infinity. On this space  $D$  is continuous, equals one at  $\sigma = 0$  and  $\sigma = \infty$ , and vanishes exactly at  $\sigma = |\xi|$ . Every limiting mixing measure is therefore supported on that graph. In particular, the defect estimate excludes mass at both ends of the rescaled normal parameter.

Let  $\widehat{Q}_{\delta,\sigma}$  be the pushforward of  $Q_{\delta\sigma}$  under  $l \mapsto l/\delta$ . On a compact vertical interval, uniformly for  $\sigma$  in a compact subset of  $(0, \infty)$ ,

$$\widehat{Q}_{\delta,\sigma}(\mathrm{d}l) = \left( \frac{\sigma}{\pi(\sigma^2 + l^2)} + O(\delta) \right) \mathrm{d}l.$$

The first term is the rescaled central Cauchy density; the bound on the folded tails used in Equation (4.6) gives the error. The vanishing defect makes the mixing mass outside any fixed neighborhood of  $\sigma = |\xi|$  tend to zero. We may therefore pass this kernel formula against compactly supported vertical tests and tangential tests in the interior of the closed band. This restriction avoids any issue from atoms on its boundary.

For these tests the projected measure  $C'$  and the physical measure have the same limit. Indeed, the preimage near an alias  $p_a$  lies in  $B(p_a, c_*\delta) \setminus \{p_a\}$  for a constant  $c_*$  depending on the test support: the nonzero scaled tangential coordinate excludes  $p_a$  itself. The measure  $C$  is finite near  $p_a$ , so its mass on this shrinking punctured neighborhood tends to zero. There are only finitely many aliases. Exhausting  $k \neq 0$  by interiors of the bands above now gives

$$M(\mathrm{d}k, \mathrm{d}l) = \frac{|k|}{\pi(k^2 + l^2)} \eta(\mathrm{d}k) \mathrm{d}l$$

for a nonnegative measure  $\eta$ . The tangential measures agree on overlapping bands, since integrating the displayed positive density over a fixed vertical interval recovers them uniquely.

Thus  $|p|^2 M$  is locally invariant under normal translations where  $k \neq 0$ , without assuming that  $\eta$  has a density. At every nonzero point at least two of the four frames have nonzero tangential coordinate and independent normals. The two corresponding distributional derivatives of  $|p|^2 M$  vanish in a neighborhood. Convolving locally with a smooth kernel shows it is a constant multiple of Lebesgue measure there. Constants agree on overlaps in the connected punctured plane. We obtain  $M(\mathrm{d}p) = b_0 |p|^{-2} \mathrm{d}p$ , with  $0 \leq b_0 < \infty$ .

To pass from vague convergence to tests, let  $A_{\delta,f}$  be the transform of the interpolation coefficients. They have exactly zero sum. Near zero, for every fixed integer  $q \geq 3$ ,

$$|A_{\delta,f}(p)| \leq C_q \min\{|p|/\delta, (1 + |p|/\delta)^{-q}\}.$$

The first bound uses the first absolute moment of the coefficients; the second follows by  $q$  lattice summations by parts against the smooth test convolved with its nodal hat. Away from zero the same summations give  $O(\delta^q)$ . Thus the low-frequency tail is  $O(\varepsilon^2)$ , the high-frequency tail within the small disk is  $O(R^{-2q})$ , and the remaining tail vanishes by local finiteness. On the intervening compact annulus use vague convergence. The Fourier convention  $\hat{f}(p) = \int e^{-ip \cdot x} f(x) dx$  gives

$$b_0 \int \frac{\hat{f}(p) \overline{\hat{g}(p)}}{|p|^2} dp = 2\pi b_0 \mathcal{L}(f, g).$$

This proves the assertion with  $\beta = 2\pi b_0$ , including varying tests.

Finally choose two disjoint small disks and a difference of nonnegative mass-one tests in them. Choose two fixed nested annuli enclosing the first disk and avoiding the second, with positive margins. For all sufficiently small meshes, the nodal supports remain in their respective disks. Apply [Theorem 3.3](#) successively in the two annuli. With probability bounded below, they contain  $W$  circuits of opposite prescribed colors. This uses the conditional one-parity bound, so it does not require positive association between the two opposite-color events.

Reveal the outer circuit by an outside-in stopping search and condition on it and its exterior. No  $N$  cluster crosses this cut. Conditional also on the complete  $(s, W)$  data in its finite interior, [Theorem 3.1](#) gives independent fair colors to the enclosed  $N$  clusters. The opposite  $s$  colors of the two circuits imply that an odd number of their intervening cluster jumps are nonzero. At least one such free cluster has an odd hole containing the first support, while the second support is outside the outer cut. Flipping its fair color shifts the first average by two in one direction or the other relative to the second, and hence changes their difference by four. Its conditional variance is four. The resulting strictly positive lower variance bound is uniform in mesh. It bounds every possible  $\beta$  away from zero. The upper bound follows from [Equation \(4.1\)](#) applied to these tests.  $\square$

**Corollary 4.7** (Reflection-null Laplace insertion). *Let  $\phi \in C_c^\infty(\mathbb{R}^2)$  be supported a positive distance to one side of a line in any of the four reflection frames. For site rows converging to that line, the reflection norm of  $X_\delta = K_\delta(\Delta\phi)$  tends to zero. The assertion includes converging placements and the local augmentations in [Theorem 4.1](#).*

*Proof.* Every subsequence has a further covariance limit in [Theorem 4.6](#). The squared reflection norm then tends to the Green pairing of  $\Delta\phi$  with its reflected copy, times  $\beta$ . This pairing is zero: after integration by parts it is a constant times the integral of the two gradients, whose supports are disjoint. The coefficient need not yet be unique. Positivity and the subsequence argument prove convergence to zero along the full mesh limit. Decorations not used by  $X_\delta$  do not change this covariance, and converging placements are covered by the varying-test part of [Theorem 4.6](#).  $\square$

## 5 Screening a Laplace insertion

Reflection nullity is an unconditional plane identity. We now strengthen it to an identity conditional on all exterior spins. The distinction matters: conditioning on one spin and its edge

decoration does not specify the other spin. The proof has three steps. We move separated, arbitrarily rare block patterns by positive normal transfers; we include the finitely many colors that can transmit information through their buffers; and we join the blocks by short open wires, protected by surrounding circuits.

All lengths in this section are physical lengths. Every layout, including its smallest gaps, is fixed before the mesh tends to zero. We use the neutral nodal discretization of [Section 4](#). In particular,

$$X_\delta = K_\delta(\Delta\phi), \quad \phi \in C_c^\infty(B), \quad (5.1)$$

has coefficients of exactly zero sum and is a function of increments in a fixed arbitrarily small enlargement of  $B$ . The plane law is denoted by  $\mathbb{P}_\delta$ . Its two spins and the decoration are denoted by  $(s, d, W)$  as in [Theorem 3.1](#);  $N$  is the dual complement of  $W$ .

**Proposition 5.1** (Exterior orthogonality). *Let  $Q$  be a square and let  $B$  be a concentric disk whose radius is less than one thousandth of the side length of  $Q$ . For  $\phi \in C_c^\infty(B)$ , let  $\mathcal{E}_\delta(Q)$  be the sigma-field generated by the full pair of spins outside  $Q$ , the  $W$  states on mutually dual edge pairs whose incident corner-grid plaquette is disjoint from  $Q$  (and hence the corresponding  $N$  states), and all height increments computable there. Then*

$$\|\mathbb{E}_{\mathbb{P}_\delta}[X_\delta \mid \mathcal{E}_\delta(Q)]\|_1 \longrightarrow 0. \quad (5.2)$$

*The same assertion holds under any sequence of flat-domain laws having a fixed larger square  $Q^+$  compactly in their interiors, with  $\bar{Q} \subset Q^+$ . It is uniform over those domains, including the component laws supplied by an earlier exact flat cut. In the latter case it is applied to the conditional law at that cut, before any later data are revealed. Additive height constants do not affect the assertion.*

## 5.1 Uniform bounds under separated patterns

A block pattern specifies all  $s$  spins and all  $W$  edges of a finite regular block, including both endpoints of every specified edge. Blocks will have pairwise disjoint filled isolating neighborhoods; a slightly smaller compact core in each block is reserved for observing  $d$  colors. The blocks and their neighborhoods can have finitely many predetermined indentations. Constants may depend on this geometry, but never on the number of microscopic values in a pattern. Write  $P_i$  for a specified pattern in block  $i$ , and  $p_i = \mathbb{P}_\delta(P_i) > 0$ .

**Lemma 5.2** (Neutral insertion moments under patterns). *Suppose a smooth neutral insertion  $Y_\delta = K_\delta(f)$  has a filled pin-free neighborhood with positive clearance from finitely many pattern neighborhoods. For every finite  $p$ ,*

$$\mathbb{E}[|Y_\delta|^p \mid P_1 \cap \cdots \cap P_m] \leq C_p. \quad (5.3)$$

*The estimate is uniform over admissible patterns, over a compact family of positively separated placements, and under the flat laws of [Theorem 5.1](#). It remains true after adjoining reflected copies of the patterns when their neighborhoods have the same positive separations.*

*Proof.* We give the count argument, so that this estimate does not depend on the Gaussian limit proved later. Write the coefficients of the insertion as  $b_v$ , where  $\sum_v b_v = 0$ ,  $|b_v| \leq C\delta^2$ , and  $\sum_v |b_v| \leq C$ . If both corner parities occur, transfer the coefficients on the other parity to nearest  $s$  sites. Each corner increment has absolute value one, so this changes the insertion by a uniformly

bounded remainder  $R_\delta$  and preserves exact neutrality. Conditional on the complete  $(N, s)$  data, the finite cut identity gives

$$Y_\delta = R_\delta + 2 \sum_C \epsilon_C q_C, \quad q_C = \sum_{v \in O_C} b_v. \quad (5.4)$$

Here  $O_C$  is the union of the odd holes of  $C$ , and the signs  $\epsilon_C$  are independent and fair. An additive constant disappears because the coefficients sum to zero. If a fixed wired color is present, its single contribution is bounded by  $2 \sum_v |b_v|$  and is treated separately. The same argument can first be made in finite exhaustions; the estimates below are uniform in the exhaustion.

Fix a connected neighborhood of the support with positive clearance inside a pin-free buffer. For a cluster of diameter  $r$  contained in that buffer, its filled hull lies in a disk of radius  $r + O(\delta)$ , so  $|q_C| \leq C(r + \delta)^2$ . A cluster with  $q_C \neq 0$  must meet the connected neighborhood: otherwise its odd-hole indicator is constant there and neutrality gives  $q_C = 0$ .

We localize the count of such clusters before applying pattern comparison. Fix a sufficiently large constant  $L$ . For  $L\delta \leq r$  below a fixed buffer scale, cover the one-mesh enlargement of the insertion neighborhood by  $O(r^{-2})$  inner squares of side  $r/20$ . Assign each relevant cluster of diameter in  $[r, 2r)$  to one such square containing a deterministically chosen vertex of the cluster. Some other vertex is at distance at least  $r/2$  from that vertex, so a path in the cluster exits the concentric square of side  $r/2$ . Its first exiting segment gives a component of the relative restricted  $N$  graph meeting the inner square and a vertex incident to an edge outside the restriction. Distinct global clusters give distinct restricted components. The fixed rectangle cover in item 2 of [Theorem 3.5](#) therefore gives

$$A_{r,j} \leq \sum_{\ell=1}^{L_0} T_{r,j,\ell},$$

where  $A_{r,j}$  is the assigned global count in square  $j$ , and each  $T_{r,j,\ell}$  counts restricted crossings of one rectangle, using the separate side ports of that proposition. The number  $L_0$  is fixed. Each  $T_{r,j,\ell}$  is determined by a complete local pattern in an enlarged box of size  $O(r)$ . Choose the buffer scale so that these boxes and all their joining buffers lie inside the pin-free neighborhood.

The transmission bound gives moments of every fixed order for these local counts. Since they are local-pattern measurable, the density comparison in [Theorem 3.9](#) transfers their moments under all the remote patterns, with constants uniform in  $r$ . It follows that the conditional  $L^q$  norm of each  $A_{r,j}$  is bounded by  $C_q$ . For  $\delta \leq r < L\delta$ , use boxes of side  $L\delta$  instead: each contains only a bounded number of lattice vertices, so its assigned cluster count is bounded deterministically. For every  $q \geq 1$ , Minkowski's inequality now gives, under the law conditioned on the remote patterns,

$$\left\| \sum_{C: r \leq \text{diam } C < 2r} |q_C|^2 \right\|_q \leq C_q r^2, \quad r \geq \delta,$$

as long as  $r$  is smaller than a fixed buffer scale. Summing dyadic scales is harmless.

The remaining clusters with  $q_C \neq 0$  meet the insertion neighborhood and either have diameter at least the fixed buffer scale or leave the buffer. In either case they transmit across a fixed intermediate scale. The same first-exit argument bounds their number by a fixed finite sum of local restricted crossing counts, with joining buffers inside the pin-free neighborhood, regardless of how their pieces join farther out. Transfer these local counts by the same density comparison. Each coefficient is at most  $\sum_v |b_v|$ , so these contributions also have bounded moments. Conditioning on  $(N, s)$  and applying the elementary moment bound for a sum of independent signs now proves

(5.3). All comparisons used only  $(s, W)$  patterns, not  $d$ -color prescriptions. The same buffer and counting proof works in a flat domain, and after reflection of the entire separated arrangement.  $\square$

Let  $\Theta$  be a lattice-row reflection. For functions on its positive side, put

$$\langle A, B \rangle_{\Theta} = \mathbb{E}[\overline{\Theta A B}], \quad \|A\|_{\Theta}^2 = \langle A, A \rangle_{\Theta}.$$

These are the reflection spaces of [Theorem 4.1](#).

**Lemma 5.3** (Normalized reflection bound). *Suppose all the pattern neighborhoods and the support of  $A$  lie strictly on the positive side of a reflection row. Assume that  $A$  is bounded by one, or is a bounded factor times one smooth neutral insertion with a disjoint pin-free neighborhood. Then*

$$\left\| A \prod_{i=1}^m \frac{\mathbf{1}_{P_i}}{p_i} \right\|_{\Theta} \leq C. \quad (5.5)$$

*Bounded factors may depend on both spins and the edge decoration. The constant is uniform over the patterns and over compact families of placements preserving all the stated clearances.*

*Proof.* Set  $P = \bigcap_i P_i$  and  $J = \prod_i p_i$ . The square of the left side is

$$\frac{\mathbb{E}[\mathbf{1}_{P \cap \Theta P} \overline{\Theta A A}]}{J^2}.$$

The reflected and original pattern neighborhoods are disjoint. [Theorem 3.9](#) bounds their joint probability by  $CJ^2$ . If  $A$  contains a neutral insertion, apply conditional Cauchy–Schwarz and [Theorem 5.2](#) to the insertion and its reflection. Otherwise the numerator is bounded directly by the pattern probability. Thus the *squared* denominator is canceled. No lower bound on any individual  $p_i$  is required.  $\square$

## 5.2 Moving arbitrary patterns

We record the analytic argument in full. It is the positive-transfer mechanism of [\[30, Section 7\]](#); the arbitrary-pattern bounds required here are supplied by the preceding lemmas, rather than by the pin estimates for that different model.

**Lemma 5.4** (Analytic translation). *Partition finitely many local factors and pattern neighborhoods into two groups separated by a strict projection gap along one of the four reflection normals. There is at most one unbounded factor, a smooth neutral insertion with its pin-free buffer. Move the high group by  $tn$  and divide the expectation by  $\prod_i p_i$ . Denote the result at lattice-normal-step translations by  $M_{\delta}(t)$ . If  $M_{\delta}(t_{\delta}) \rightarrow 0$  whenever  $t$  belongs to some nonempty open interval of positive numbers and  $t_{\delta} \rightarrow t$  is lattice rounded, then  $M_{\delta}(0) \rightarrow 0$ . The assertion is uniform along sequences of patterns, bounded factors and convergent positively separated placements.*

*Proof.* Choose  $\eta > 0$  smaller than the projection gap. Back the high group up by  $\eta_{\delta} = N_{\delta} \tau_{\delta} \rightarrow \eta$ , where  $\tau_{\delta}$  is the positive normal step in [Theorem 4.1](#). Choose a reflection row inside the remaining gap. Reflect and conjugate the normalized low-side factor, obtaining a positive-side vector  $A_{\delta}$ ; call the backed-up high-side factor  $B_{\delta}$ . By [Theorem 5.3](#), both reflection norms are at most a constant independent of the patterns. If  $T_{\delta}$  is the positive normal contraction, define

$$F_{\delta}(z) = \langle A_{\delta}, T_{\delta}^{(z+\eta_{\delta})/\tau_{\delta}} B_{\delta} \rangle_{\Theta}, \quad \operatorname{Re} z > -\eta_{\delta}.$$

For  $\operatorname{Re} w > 0$ , the spectral power is  $T_\delta^w = \int_{(0,1]} e^{w \log \lambda} dE_\delta(\lambda)$ , with value zero on the zero spectral subspace. This is a holomorphic contraction by bounded spectral calculus [34, Proposition 5.3 and Corollary 5.7]: on  $\operatorname{Re} w \geq b > 0$ , each derivative is bounded by  $\sup_{0 < \lambda \leq 1} \lambda^b |\log \lambda|^m < \infty$ . Hence  $F_\delta$  is bounded on the common half-plane  $\operatorname{Re} z > -\eta/2$  for all sufficiently small  $\delta$ . At nonnegative lattice-step arguments it equals  $M_\delta$  exactly. Cauchy's estimate makes lattice rounding negligible on compact sets. Every subsequence has a locally uniformly convergent further subsequence; its holomorphic limit vanishes on the given interval and hence everywhere by the identity theorem. In particular it vanishes at zero. The norm bounds and the common backoff remain valid for all the sequences in the statement. Moving the low group in direction  $-n$  is equivalent.  $\square$

Arrange thick blocks in a square annulus inside  $Q$ , leaving narrow gaps where two horizontal lines, one above and one below  $B$ , cross its two side strips. The top and bottom blocks are denoted by  $T$  and  $D$ , and the middle left and right blocks by  $L$  and  $R$ . The middle band is so narrow that  $L$  and  $R$  are strictly below and above  $B$ , respectively, in projection on  $u = (1, 1)/\sqrt{2}$ . The blocks may subsequently acquire small predetermined indentations at their gap-facing boundaries; none of these changes the strict projection inequalities. They have disjoint filled isolating neighborhoods avoiding  $B$ .

**Lemma 5.5** (Gapped pattern identity). *For this fixed arrangement, for arbitrary patterns  $P_i$  in the blocks and arbitrary factors  $F_i$  of absolute value at most one supported in their respective blocks,*

$$e_\delta := \sup_{(P_i), (F_i)} \frac{|\mathbb{E}[X_\delta \prod_i \mathbf{1}_{P_i} F_i]|}{\prod_i p_i} \longrightarrow 0. \quad (5.6)$$

*The factors may involve both spins. Their choices and their supports inside the fixed blocks may depend on the specified patterns.*

*Proof.* Use the normals  $e_y = (0, 1)$ ,  $u = (1, 1)/\sqrt{2}$ ,  $e_x = (1, 0)$ , and  $v = (-1, 1)/\sqrt{2}$ . First translate  $T$  upward and  $D$  downward until  $T \cup R$  and  $D \cup L$  are strictly above and below the insertion in  $u$ -projection. Next translate  $T \cup R$  in direction  $u$  and  $D \cup L$  in direction  $-u$ . Choose these distances large enough that the first group is above the insertion and wholly to its right, and the second group wholly to its left. Move the latter farther left until it is strictly above every other support in  $v$ -projection. This additional horizontal move is needed because  $u \cdot v = 0$ . Finally move that group in direction  $v$  until all factors are above the insertion.

Every move is rigid on a complete side of a strict projection gap. Within-group clearances are preserved, and between-group gaps increase. The distances can be chosen in open intervals with positive margins. At the final arrangement, reflection Cauchy–Schwarz in a horizontal row gives the bound  $C\|X_\delta\|_\Theta$ , which tends to zero by Theorem 4.7. Undo the moves in reverse order using Theorem 5.4, each time first varying the last distance over an open interval. There is no motion through overlapping supports. This proves convergence along every sequence of patterns and factors. If the displayed supremum did not tend to zero, a sequence within half its positive limsup would contradict precisely this conclusion.  $\square$

**Lemma 5.6** (Adding core colors). *Let  $S_\delta$  be the complete patterns in the blocks, and observe the  $d$  spins on arbitrary pattern-measurable subsets of their compact cores. Write  $C_\delta$  for those observations and  $Y_\delta = (S_\delta, C_\delta)$ . Then*

$$\|\mathbb{E}[X_\delta \mid Y_\delta]\|_1 \longrightarrow 0. \quad (5.7)$$

*Proof.* For each block pattern, form the restricted  $N$  graph from precisely the dual edges whose  $W$  states that pattern specifies, as in [Section 3](#). An observed restricted component with no vertex incident to an unprescribed  $N$  edge is sealed. Its  $d$  color is independent and fair by the color law there; its filled hull lies in the block's filled isolating neighborhood, so that color cannot affect  $X_\delta$ . Integrate these colors out. Every remaining observed component meets the core and contains a vertex incident to an unprescribed edge; call these components terminals.

Under  $\mathbb{P}_\delta$ , before fixing a realization of  $S_\delta$ , the fixed core and block clearances meet the unprescribed-interior hypotheses of item 2 of [Theorem 3.5](#). A finite union of its transmission bounds gives a geometric tail for the total number of terminals.

Truncate this number at  $M$ . For each pattern, choose one representative in each terminal, in a deterministic order. An assignment of their at most  $M$  signs is the product of bounded local indicators, one factor in each block. It need not have a positive probability bounded below: global joining outside the blocks can forbid assignments. The quantity in (5.7) is the sum of absolute *unnormalized* expectations over the data. Thus [Theorem 5.5](#) bounds its retained part by

$$2^M e_\delta \sum_{P_1, \dots, P_m} \prod_i \mathbb{P}_\delta(P_i) = 2^M e_\delta.$$

Integrating the independent internal colors contributes a factor one. On the discarded event, Cauchy–Schwarz and [Theorem 5.2](#) bound the contribution by a constant times the square root of the count-tail probability. First let  $\delta \downarrow 0$  at fixed  $M$ , then let  $M \rightarrow \infty$ .  $\square$

### 5.3 Joining the gaps without revealing their colors

We next construct an actual separator. The following details distinguish an averaged shielding estimate from a generally false estimate uniform over every fully specified pattern.

For each of the four horizontal gaps, place  $M$  disjoint buffered trial squares of side comparable to  $R$ . In the center of each trial leave a square hole of side comparable to  $w = qR$ , straddling the gap line; the horizontal gap itself has width much smaller than  $w$ . The holes produce the predetermined indentations in the adjoining blocks. Within each trial ask for two open plus  $W$  arms, one on each side of the gap, running from the middle portions of the two facing hole edges, with fixed corner clearance, to distance comparable to  $R$  inside the adjoining blocks. Choose their outer ends in compact cores. All cores, arms and holes have fixed positive mutual clearances where required. In particular the outer ends and core paths avoid disks of radius  $R/10$  about every hole.

The choices are made in the order

$$\begin{aligned} q > 0, \quad M < \infty, \quad R > 0 \text{ small enough to fit the trials,} \\ w = qR, \quad \text{gap width,} \quad \delta \downarrow 0. \end{aligned} \tag{5.8}$$

Thus an arbitrarily large finite number of trials fits in a fixed annulus. Shrinking  $R$  changes the fixed core geometry, which is allowed before applying [Theorem 5.6](#).

**Lemma 5.7** (Selected-arm shielding). *For fixed  $q$ , each trial has conditional two-arm success probability at least  $p(q) > 0$ , given all earlier trial data. If the first successful trial is selected, the probability that no trial succeeds is at most  $(1 - p(q))^M$ . The probability that the selected hole has no surrounding plus  $W$  circuit between scales  $Cw$  and  $R/10$  is at most  $\eta(q)$ , where  $\eta(q) \rightarrow 0$  as  $q \downarrow 0$ . These bounds hold after averaging over the complete block patterns.*

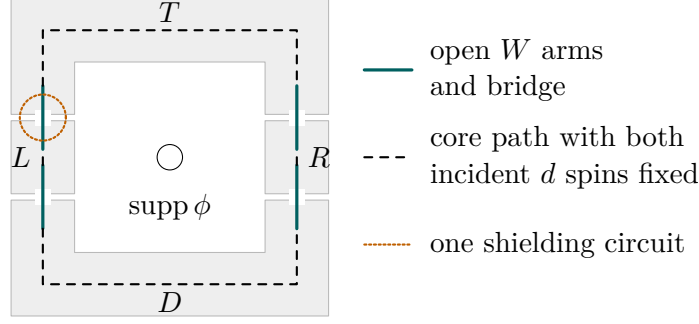


Figure 3: The separator after successful trial selection. Gray regions carry complete  $(s, W)$  patterns. Short bridges join the selected open arms across the gaps; their transverse colors need not be observed. On the dashed core portions both incident transverse spins are fixed. A surrounding circuit protects the outside data during each bridge insertion. Only one shielding circuit and one selected trial per gap are shown.

*Proof.* Reveal previous trials completely before testing the next. Their buffered neighborhoods are disjoint, so this is exterior  $(s, W)$  data for the current trial. The tube version of [Theorem 3.3](#) gives  $p(q)$ ; the aspect ratios are fixed once  $q$  is fixed. A stack of separated annuli between  $Cw$  and  $R/10$  gives circuit failure probability at most  $\eta(q) \leq Cq^\alpha$  for some  $\alpha > 0$ .

Write  $A_j$  for the increasing two-arm event,  $G_j$  for the surrounding plus-circuit event, and  $\mathcal{H}_j$  for the full previous trial history. Positive association in the remaining law gives

$$\mathbb{P}(A_j \cap G_j^c \mid \mathcal{H}_j) \leq \eta(q) \mathbb{P}(A_j \mid \mathcal{H}_j). \quad (5.9)$$

If  $J$  is the first successful trial, sum this inequality over the disjoint events  $\{J = j\}$ . Their total probability is at most one; there is no factor  $M$  in the resulting shielding error. Iterating the conditional arm bound gives the failure probability. Revealing all remaining block patterns afterward does not change either averaged estimate. We do not assert (5.9) after conditioning on each individual complete pattern.  $\square$

Retain the original marginal law of  $S_\delta$ . Given this pattern, select the first successful trial at every gap using a fixed order, and select its two arms by a deterministic rule. For each selected pair of arms, choose by a fixed deterministic rule a  $W$ -lattice path inside the hole joining their endpoints and meeting the prescribed pattern only at those plus endpoints. The fixed hole geometry provides such a path for all sufficiently small meshes. On success, replace the conditional law by the upper law obtained by pinning these paths open and plus. The new conditional law is therefore well defined, and its bridge pins are functions of  $S_\delta$ . On unsuccessful patterns leave the conditional law unchanged, couple it identically to the original law, and take no core-color observations. Call the resulting law  $\mathbb{Q}_\delta$ .

**Lemma 5.8** (Bridge coupling and exact separation). *The laws  $\mathbb{P}_\delta$  and  $\mathbb{Q}_\delta$  can be coupled with their block patterns identical. Except on probability at most  $4\eta(q)$ , their insertion, full exterior data, and all  $d$  colors at core sites outside the shielding disks agree, on the event that all four gaps have successful trials. Under  $\mathbb{Q}_\delta$ , on that success event one can choose pattern-measurable core paths and condition on their incident  $d$  sites so that the insertion and the full exterior are conditionally independent given  $Y_\delta = (S_\delta, C_\delta)$ .*

*Proof.* Fix a block pattern  $S$ , and write  $H(S)$  for success at all four gaps. On this event let  $G_i(S)$  be the plus-circuit event around the selected hole at gap  $i$ , and let  $\mu_i^S$  be the conditional law after the first  $i$  selected bridges have been pinned, with  $\mu_0^S$  the original conditional law. On unsuccessful patterns put  $\mu_i^S = \mu_0^S$  and  $G_i(S) = \Omega$ . For two successive laws use the monotone stopping coupling of [Theorem 3.2](#), searching outward from the newly pinned hole inside its shielding disk. Stop at the disk's outer boundary and declare this adjacent coupling unsuccessful if no surrounding plus circuit has been found. All previously inserted pins are identical in these two laws. In the lower law the first plus circuit surrounding that hole is also open in the upper law. All changed data are inside this common cut. By [Theorem 3.1](#), the configurations outside it can be sampled identically, including complementary-component colors. In particular their exterior color connections are unchanged. The exact neutrality of the insertion removes any change in the chosen additive height anchor. If  $G_i(S)$  occurs in the lower law, this cut is found inside the shielding disk, so the required insertion, exterior data, and core colors outside that disk agree.

To realize these adjacent couplings in the plane, first run each disk-stopped search in finite flat exhaustions at fixed  $\delta$  and  $S$ , retaining a binary failure marker. Each conditioning event involves finitely many prescriptions and has positive plane probability. To verify positivity, place  $S$  and the compatible bridge pins in a larger buffered flat box with an open plus  $W$  perimeter. Keep every prescribed state, put plus on all remaining  $s$  sites, close all remaining interior  $W$  edges, and take all  $d$  spins equal. Every factor in [Equation \(3.1\)](#) is positive because  $t \in [1/2, 1)$ . Let  $\mu_{\text{box}}$  denote the flat law in this box. The restriction  $p$  of this filling to a regular inner core containing the prescriptions is a complete pattern with  $\mu_{\text{box}}(p) > 0$ . The upper bound  $\mu_{\text{box}}(p) \leq C\pi(p)$  in [Theorem 3.9](#) gives  $\pi(p) > 0$ , and hence positive plane probability for the conditioning event containing it.

The conditional marginals therefore converge on cylinder events. Extract a joint limit on the countable product of spin and edge coordinates, together with the failure marker. Its probability bound passes to the limit, as does equality of each fixed exterior coordinate on success. Taking their countable intersection gives equality of the full exterior configuration; its computable height increments agree as well. Thus this limit has the plane marginals and the coupling properties just proved in finite volume.

For fixed  $S$ , the event  $G_i(S)$  is now fixed and increasing. The previous plus pins can only increase its probability. With  $\mathbb{E}_S$  denoting expectation over the original pattern marginal, this gives

$$\mathbb{E}_S \left[ \mathbf{1}_{H(S)} \mu_{i-1}^S (G_i(S)^c) \right] \leq \mathbb{E}_S \left[ \mathbf{1}_{H(S)} \mu_0^S (G_i(S)^c) \right] \leq \eta(q).$$

The first inequality is the comparison at each fixed pattern; the second is [Theorem 5.7](#) after averaging over the selected trial. At a fixed mesh the pattern space is finite, so choose these adjacent conditional couplings measurably in  $S$ . Apply the displayed estimate to each of their marginal failure probabilities, then glue the four couplings and use a union bound. Their searches stay in disjoint buffers, and every intermediate law retains the original pattern marginal. This proves the asserted joint coupling.

On a successful pattern, join the outer arm ends through compact cores of the blocks. The core portions together with the open arms and the new bridges give a closed walk winding once around  $B$  and lying inside  $Q$ . Extract a simple winding cycle, with its lattice-width incident sites, by deterministic loop erasure. On every non-open core edge, condition on both incident  $d$  sites; include the incident sites at the transitions. Along the arms and bridges no  $d$  observation is needed. Indeed the local factor there is  $t \mathbf{1}_{s_i=s_j}$ , independent of transverse colors. Along a remaining core edge all its  $s, W$  data and the two incident  $d$  values are fixed. Every local factor

crossing the cycle is consequently either fixed or independent of the transverse variables. The Gibbs weight factors into an inside weight and an outside weight after these observations. This proves conditional independence. This factorization is made with colors as actual spin variables, before summing them into cluster-count factors; no hidden connectivity weight remains across the separator. All observed  $d$  sites lie in the compact cores and outside the shielding disks.  $\square$

*Proof of Theorem 5.1 in the plane.* Truncate the insertion to  $[-T, T]$ , writing the result as  $X_\delta^{(T)}$ , and set

$$r(T) = \sup_{\delta} \mathbb{E}_{\mathbb{P}_\delta} |X_\delta - X_\delta^{(T)}|.$$

The moment bound gives  $r(T) \rightarrow 0$ . Suppose the coupling mismatch probability is  $\varepsilon$ , and the unsuccessful-pattern probability is  $\zeta$ . Duality against bounded functions gives

$$\begin{aligned} \|\mathbb{E}_{\mathbb{P}_\delta}[X_\delta^{(T)} \mid \mathcal{E}_\delta]\|_1 &\leq \|\mathbb{E}_{\mathbb{Q}_\delta}[X_\delta^{(T)} \mid \mathcal{E}_\delta]\|_1 + 2T\varepsilon \\ &\leq \|\mathbb{E}_{\mathbb{Q}_\delta}[X_\delta^{(T)} \mid Y_\delta]\|_1 + 2T\varepsilon + 2T\zeta \\ &\leq \|\mathbb{E}_{\mathbb{P}_\delta}[X_\delta^{(T)} \mid Y_\delta]\|_1 + 4T\varepsilon + 2T\zeta. \end{aligned} \tag{5.10}$$

For the middle inequality use conditional independence on success and bound the unsuccessful histories by their probability times  $2T$ . The outer inequalities use agreement of the joint insertion and conditioning data. They do not require equal laws of  $Y_\delta$ .

By Theorem 5.6, at each fixed layout the final conditional norm has limsup at most  $r(T)$ . The coupling and trial lemmas give  $\varepsilon \leq 4\eta(q)$  and  $\zeta \leq 4(1 - p(q))^M$ . Returning from  $X_\delta^{(T)}$  to  $X_\delta$  therefore gives

$$\limsup_{\delta \downarrow 0} \|\mathbb{E}[X_\delta \mid \mathcal{E}_\delta]\|_1 \leq 2r(T) + 16T\eta(q) + 8T(1 - p(q))^M.$$

Given an error tolerance, first fix  $T$  large, then  $q$  small, then  $M$  large, and finally fix a layout as in (5.8). Taking its mesh limit proves (5.2). Only original-law moments have been used; there is no moment assumption for the artificially pinned law.  $\square$

## 5.4 Transfer to flat domains

*Completion of the proof of Theorem 5.1.* Choose a regular, fixed-width spin ring  $A$  between the insertion and  $\partial Q$ , with positive clearance from both. Let  $R_\delta$  denote both spins on a lattice-width enlargement of this ring. For a ring assignment  $r$  admitting an interior filling, let  $K_\delta^r$  be the finite interior spin Gibbs kernel obtained from Equations (2.2) and (2.3), with  $r$  fixed on the ring and all factors meeting the interior retained. Define

$$g_\delta(r) = \int X_\delta \, dK_\delta^r,$$

and put  $g_\delta(r) = 0$  when no interior filling is admissible. The neutral insertion is determined by the interior height increments, so its value in this formula is independent of an additive height anchor. The finite product weights give this same conditional interior kernel in every flat domain; the identity also passes from flat exhaustions to the plane on cylinder events. Thus  $g_\delta(R_\delta)$  is a version of the conditional mean of  $X_\delta$  under each of these laws, including on admissible ring assignments that may have zero plane probability. All these expectations are finite at each mesh.

Apply the separated-block proof to a slightly smaller surrounding square, whose exterior contains the ring. All its strict projection clearances persist under this small change of the surrounding square. Since the ring data belong to that exterior sigma-field, conditional Jensen gives  $\mathbb{E}_{\mathbb{P}_\delta}[g_\delta(R_\delta)] \rightarrow 0$ , hence convergence in plane probability. The uniform full-spin contiguity of [Theorem 3.11](#), on a fixed larger box, transfers this convergence in probability to the flat laws, even for mesh-dependent tests of the ring data. The neutral moment estimate in [Theorem 5.2](#), with no patterns, gives uniform integrability of these conditional means by conditional Jensen. Consequently their absolute means tend to zero uniformly over the flat laws. Finally the Markov property across the ring gives

$$\mathbb{E}[X_\delta \mid \mathcal{E}_\delta(Q)] = \mathbb{E}[g_\delta(R_\delta) \mid \mathcal{E}_\delta(Q)],$$

and conditional Jensen proves the asserted exterior estimate. Uniformity can equivalently be checked by taking an arbitrary contradicting sequence of flat domains; all buffers remain fixed and the same argument applies. At a random exact cut, the statement is used for the realized conditional flat law. Subsequent conditioning is not included in this claim.  $\square$

In particular, if finitely many spectator height differences or heights anchored on an exterior wall are evaluated along paths outside  $Q$ , their bounded functions are legitimate exterior observables. Unbounded products are handled by the moment estimates of the next section. This is the conditional harmonicity input, with no box shrinking at the lattice scale.

## 6 Flat-domain Gaussian limits

We prove the Gaussian limit for the unit-increment height  $K_\delta = 2H_\delta$ , first in deterministic flat chambers and then conditionally in chambers produced by a stopping exploration. The uniformity in the chamber is what permits the second use.

Every chamber in this section is an accessible component behind a connected matching  $W$  perimeter, with the medial resolution at pinches from [Theorem 3.1](#). Its open face domain is simply connected. The perimeter carries its actual wire, and incident faces are not joined across a pinch. We subtract its constant height. For the moment arguments choose  $N$  to be the free current complementary to this wall percolation  $W$ ; all its interior components then have fair colors. The initial flat domain is included by taking  $N$  to be the free dual current. Either corner parity may carry the wall.

A collection of chambers will always have pairwise disjoint open interiors; their closures may touch. Sample their centered flat laws independently, patch the fields on these disjoint cell interiors, and set the field to zero on the exposed complement. Let  $U_\delta$  denote their open union. Fix an open disk  $B_0$  containing all chamber closures with a fixed outer margin, and put

$$F_\delta = \overline{B_0} \setminus U_\delta.$$

Thus  $F_\delta$  is the closed complement in a common compact ambient set. Hausdorff convergence of these  $F_\delta$  is equivalent here to local Hausdorff convergence of the closed complements in the plane. Constants may depend on  $B_0$ , the fixed parameter, and the moment order, but not on the shapes or the number of chambers.

**Theorem 6.1** (Flat-domain limit). *Let  $U_\delta$  be a single deterministic flat chamber, or a finite mesh-dependent union of independent chambers with pairwise disjoint interiors, as above. Suppose*

$F_\delta$  converges in Hausdorff distance to  $F$ , and set  $U = B_0 \setminus F$ . Then

$$K_\delta \implies h_U/a \quad \text{in } H_{\text{loc}}^s(U), \quad s < -1,$$

where the restrictions of  $h_U$  to the components of  $U$  are independent zero-Dirichlet GFFs with logarithmic singularity  $-\log|x-y|$ . All finite joint smooth-test moments with supports in  $U$  converge. Different limiting components are independent even if their discrete precursors were the same chamber. Either wall parity and either atomic or interpolated smearing gives the same limit. The fields extended by zero off  $U_\delta$  are uniformly tight in  $H_{\text{loc}}^s(B_0)$  for every  $s < -1$ .

The proof first bounds all moments and their boundary traces. Screened orthogonality then makes the limiting moments harmonic away from collisions; a local neutral-pair comparison identifies their logarithmic poles and hence their Wick recursion. We determine its coefficient from the plane limit before completing the theorem. The last proposition turns the deterministic result into the conditional statement needed for stopped explorations.

## 6.1 Limiting chambers and their Green kernels

For an open set  $V$  whose components are simply connected, write  $G_V$  for the Dirichlet kernel with logarithmic singularity  $-\log|x-y|$  in each component, extended by zero between different components and when either argument lies outside  $V$ . For tests put

$$G_V(f, g) = \iint f(x) G_V(x, y) g(y) \, dx \, dy.$$

The planar connected-complement criterion shows that  $F_\delta$  above is connected: an open subset of the sphere has simply connected components exactly when its complement is connected. This uses disjoint interiors, not disjoint closures. The next lemma supplies the boundary facts used twice below, first in Wick's rule and then in the conditional limit.

**Lemma 6.2** (Connected complements and Green convergence). *Let  $F_n$  be connected compact subsets of  $\overline{B_0}$  containing  $\partial B_0$ , and suppose  $F_n \rightarrow F$  in Hausdorff distance. Put  $V_n = B_0 \setminus F_n$  and  $V = B_0 \setminus F$ . Every component of  $V_n$  and  $V$  is simply connected, every compact subset of  $V$  is eventually contained in  $V_n$ , and*

$$G_{V_n}(x, y) \longrightarrow G_V(x, y)$$

locally uniformly for distinct  $x, y \in V$ , including when they lie in different components of  $V$ . The zero-extended kernels satisfy  $0 \leq G_{V_n}, G_V \leq G_{B_0}$ . As a function of the closed complement and its two arguments, this zero-extended Green kernel is Borel measurable.

For any proper simply connected domain  $W$ , with  $d_W(x) = \text{dist}(x, \partial W)$ , one has

$$0 \leq G_W(x, y) \leq 4 \sqrt{\frac{d_W(x)}{|x-y|}} \quad \text{if } d_W(x) \leq \frac{|x-y|}{16}. \quad (6.1)$$

In particular these Green kernels have zero boundary trace without a boundary smoothness assumption.

*Proof.* Adjoin the exterior of  $B_0$  in the sphere. The complements remain connected, and so does their Hausdorff limit. The planar criterion just stated gives simple connectivity of every open component. Hausdorff convergence of compact sets gives uniform convergence of their distance functions, which proves compact containment.

For the estimate, choose a conformal map  $f : \mathbb{D} \rightarrow W$  with  $f(0) = x$ , and write  $f(z) = y$ . The growth bound and the quarter theorem recalled in [Section A.3](#) give

$$|x - y| \leq |f'(0)| \frac{|z|}{(1 - |z|)^2} \leq \frac{4d_W(x)}{(1 - |z|)^2}.$$

Under the stated hypothesis this implies  $|z| \geq 1/2$ . The conformal Green identity and  $-\log t \leq 2(1 - t)$  for  $1/2 \leq t \leq 1$  now give [Equation \(6.1\)](#).

For convergence, domain monotonicity gives the common bound  $0 \leq G_{V_n} \leq G_{B_0}$ , with zero understood between components. The upper kernel is bounded on compact sets away from the diagonal and has only an integrable logarithmic singularity on the diagonal. Fix  $y \in V$  and a component  $C$  of  $V$ . On compact subsets of  $C \setminus \{y\}$ , harmonic compactness gives subsequential limits in the first variable. If  $y \in C$ , a fixed disk about  $y$  is eventually contained in one component of  $V_n$ . On that disk subtract  $\log(1/|x - y|)$ . The difference is harmonic and uniformly bounded on a fixed interior circle by the common upper bound; the Poisson formula therefore preserves the unit logarithmic pole in the limit. If  $y \notin C$ , there is no pole.

For  $x \in V_n$ , its distance to the boundary of its component equals  $\text{dist}(x, F_n)$ . Passing [Equation \(6.1\)](#) to a subsequential limit shows that this limit has zero trace on  $\partial C$ , since  $\text{dist}(x, F_n) \rightarrow \text{dist}(x, F)$ . The maximum principle now identifies it as  $G_C(\cdot, y)$  when  $y \in C$ , and as zero when  $y \notin C$ . The removable difference is bounded near its pole and near the boundary, so the maximum principle applies on the whole component. Every subsequential limit is the same. Interior harmonic estimates in both variables give the asserted local uniform convergence. The common logarithmic upper bound also gives convergence of  $G_{V_n}(f, g)$  for fixed smooth tests with supports compactly contained in  $V$ . On the space of connected complements, the local uniform assertion gives continuity of  $G_{B_0 \setminus F}(x, y)$  where  $x, y \notin F$  and  $x \neq y$ . These positive-clearance sets form a countable cover of the interior off-diagonal pairs. Extension by zero off the open set and the usual diagonal value therefore give a Borel kernel, and hence Borel smeared covariances.  $\square$

## 6.2 Moments and boundary traces

Choose once  $R_0 > 2 \text{diam}(B_0)$ .

**Proposition 6.3** (Logarithmic moments). *Let  $z_1, \dots, z_{2m}$  be corner-grid sites in a flat chamber of diameter at most  $R_0/2$ . They may lie on either parity. Then*

$$\left| \mathbb{E} \prod_{i=1}^{2m} K(z_i) \right| \leq C_m \sum_{\mathcal{P}} \prod_{\{i,j\} \in \mathcal{P}} \left( 1 + \log^+ \frac{R_0}{|z_i - z_j| \vee \delta} \right), \quad (6.2)$$

where the sum is over pairings. Odd moments vanish. The same bound holds for independent flat chambers after setting the field to zero off those chambers. For a compactly supported bounded test  $f$ , uniformly over all such chambers,

$$\mathbb{E} |K_\delta(f)|^{2m} \leq C_m \|f\|_\infty^{2m}. \quad (6.3)$$

For any fixed smooth compactly supported profile  $f$ , put  $f_{r,z}(x) = r^{-2} f((x - z)/r)$ . If  $0 < \delta \leq r$ , then

$$\mathbb{E} |K_\delta(f_{r,z})|^{2m} \leq C_{m,f} \left( 1 + \log^+ \frac{R_0}{r} \right)^m. \quad (6.4)$$

This includes a mass-one smooth average of radius  $r$  for a fixed profile.

*Proof.* Condition on the current  $N$  and the  $s$  spins. By [Theorem 3.1](#), the centered height at a wall-parity site is a sum of jumps of size two with independent fair signs, one sign per contributing cluster. A cluster's coefficient at a site is zero or has absolute value two. Its sign is shared by all its odd holes.

The same signed expansion controls opposite-parity sites. For such a site  $x$ , choose an incident wall-parity site  $v$ , and let  $C_x$  be the free  $N$  component containing  $x$ . The corner rule gives

$$K(x) = K(v) + \varepsilon_W s(v) d(C_x),$$

where the fixed sign  $\varepsilon_W$  records which parity carries  $W$ . Absorb this term into the coefficient of  $d(C_x)$ . Conditional on  $(N, s)$ , every height is still a sum of the independent fair component colors, now with coefficients of absolute value at most three. If a component has a nonzero coefficient at a site, that site lies in the component or its filled hull up to distance  $O(\delta)$ : wall-parity contributions come from odd holes, and the extra contribution is incident to the component. Consequently a component shared by two sites has diameter at least their separation minus  $O(\delta)$ .

Expanding an even product and summing the independent signs leaves only terms in which every sign appears an even number of times. Pair the occurrences of each such sign. Overcounting by all pairings gives an upper bound by  $C_m$  times sums of products of shared-cluster counts  $M(z_i, z_j)$ . This argument does not assert independence of the counts.

A contributing cluster surrounds or is incident to the relevant site. The preceding support observation gives the required diameter lower bound for a shared cluster. At a fixed site only boundedly many components of diameter at most  $C\delta$  can contribute, including isolated vertices; put all of these contributions in the lattice cutoff. Above that cutoff, a fixed number of boxes at a cluster's diameter scale detect its passage or enclosure. By [Theorem 3.5](#), the count in each such dyadic shell has uniformly bounded moments of every fixed order. Minkowski's inequality therefore gives

$$\|M(z_i, z_j)\|_{L^m} \leq C_m \left( 1 + \log^+ \frac{R_0}{|z_i - z_j| \vee \delta} \right).$$

Holder's inequality for the product of the  $m$  counts proves [Equation \(6.2\)](#). Height reflection about the flat value proves vanishing of odd moments. The argument is unchanged in several independent chambers; shared-cluster counts between different chambers are zero.

Integrate the estimate against the interpolation weights, or the atomic Riemann weights, of the tests. Each logarithmic factor is uniformly integrable in its two variables, including its lattice diagonal: the latter contributes  $O(\delta^2 \log(1/\delta))$ . Since the pairs have disjoint sets of variables, this proves [Equation \(6.3\)](#). Rescaling a fixed profile by  $x = z + ru$  gives [Equation \(6.4\)](#). Here  $\delta/r \leq 1$  is essential for an atomic smearing: after the rescaling the diagonal part has the additional uniformly bounded factor  $(\delta/r)^2 \log(r/\delta)$ , while the common scale contributes  $1 + \log^+(R_0/r)$ .  $\square$

**Proposition 6.4** (Boundary decay). *Consider deterministic flat chambers in a fixed bounded region, with connected accessible walls, and a Hausdorff limit of their complements. For any limiting open component, moment kernels obtained from [Equation \(6.2\)](#) have zero boundary trace in a distinguished variable when the other variables remain at distinct interior points. The bounds and the trace conclusion are uniform for the discrete chambers. More precisely, a shared cluster joining the height contribution at a site within distance  $r$  of a matching connected wall to one at distance at least  $R$  from that site has probability at most  $C(r/R)^\eta$ , with fixed positive outer clearance and  $\delta \leq r \ll R$ .*

*Proof.* The relevant wall component reaches from the small scale to the outside of the fixed testing window. A shared cluster either is incident to the distinguished site or encloses it. In the incident case its connected passage reaches the outer scale forced by the distant spectator. In both cases it must avoid matching circuits attached to this wall at every scale between  $r$  and  $R$ : such a circuit separates the distinguished contribution from the distant one. The conditional circuit estimate in [Theorem 3.3](#) gives a uniformly positive success probability at a fixed fraction of these separated scales. An attached circuit blocks the cluster, so iteration gives  $C(r/R)^\eta$ . The argument is within the accessible chamber with its wire retained; no regularity of the geometric wall is used.

In every pairing contributing to a mixed even moment, the distinguished site is paired with one spectator. The corresponding shared cluster has a fixed positive minimum diameter. Its count, restricted to surviving the attached-circuit trials, tends to zero in every smaller  $L^p$  norm: apply Holder to the survival indicator and the higher count-moment bound from [Theorem 3.5](#). The remaining counts are bounded by [Equation \(6.2\)](#). This proves vanishing of the mixed moment as the distinguished site approaches the wall. For fixed distinct spectators the pairing bound itself is uniformly bounded near the wall. Smearing first at a radius smaller than the site's wall distance gives the same result for distributional moment limits. Once harmonicity is established below, the mean-value property identifies these smearing limits with the pointwise boundary trace. The proof also applies when the wall develops a pinch: only the actual connected wall crossing the testing scales is used.  $\square$

**Lemma 6.5** (Parity and interpolation). *On compact subsets with positive clearance from the wall, the two parity smearings of  $K$ , and their smooth nodal interpolation, have the same subsequential distributional limits and the same joint test moments. Here each original parity has Riemann weight  $\delta^2$ ; the combined corner grid has weight  $\delta^2/2$  per site. Thus its interpolation is compared to the average of the two parity smearings, not their sum.*

*Proof.* Pair each site of one parity with an incident site of the other parity by a fixed lattice translation. The paired height difference is  $Z_e \in \{-1, 1\}$ . Conditional on  $(N, s)$  it is, up to a deterministic orientation, a fixed  $s$  spin times the fair color of the adjacent  $N$  component. Thus for two well-separated paired edges,

$$|\mathbb{E}[Z_e Z_{e'}]| \leq \mathbb{P}(\text{their adjacent sites belong to the same } N \text{ component}).$$

A matching  $W$  circuit around the first edge separates these components. Bulk circuit trials between the lattice scale and their separation imply that, for separation at least  $r$  and fixed wall clearance,  $|\mathbb{E}[Z_e Z_{e'}]| \leq C(\delta/r)^\eta$ . This is equivalently the conditional fair-color coupling in disjoint pin-free buffers. In particular it does not infer smallness merely from the deterministic bound  $|Z_e| = 1$ .

For a smooth test, the variance of its averaged paired differences is bounded by  $Cr^2 + C(\delta/r)^\eta$ : separate pairs at distance below  $r$  from all other pairs. First let  $\delta$  decrease to zero and then let  $r$  decrease to zero. Changing the test's evaluation point within a paired cell gives an  $O(\delta)$  smooth-test error in  $L^2$  by [Equation \(6.3\)](#). This proves convergence of the difference to zero in  $L^2$ . Uniform higher moments and interpolation give convergence in every finite  $L^p$ . Nodal interpolation is an average of the same local pairings with smooth weight errors, so the argument includes it.  $\square$

**Lemma 6.6** (Tightness and absence of collision measures). *Centered flat heights are tight in  $H_{\text{loc}}^s$ , for every  $s < -1$ , uniformly over the chambers above. Every finite joint test moment converges*

along distributional convergence. Once the limiting moment distributions are harmonic off their collision diagonals, their smooth representatives there represent the full distributions: there are no additional measures or distributions supported on those diagonals.

*Proof.* Put a cutoff  $\chi$  in a fixed square regarded as a torus, with normalized Fourier basis  $e_n$ . The uniform second-moment bound for  $K_\delta(\chi e_n)$  is independent of  $n$ . Hence, for  $q > 1$ ,

$$\sup_{\delta} \mathbb{E} \|\chi K_\delta\|_{H^{-q}}^2 \leq C \sum_{n \in \mathbb{Z}^2} (1 + |n|^2)^{-q} < \infty.$$

For  $1 < q < -s$ , compactness of  $H^{-q} \hookrightarrow H^s$  gives tightness. Use a countable exhaustion of cutoffs and summable exceptional probabilities. Higher versions of Equation (6.3) give uniform integrability of all finite test products, proving convergence of moments. These statements also hold for a finite independent collection of chambers or for fields set to zero on their already exposed complement.

Here is a direct verification of the last assertion, independent of any pointwise choice on diagonals. Mollify each variable of a  $j$ th moment distribution at the same radius  $r$ . For each fixed  $r$  the mesh limit is taken first, so eventually  $\delta \leq r$ . Holder and Equation (6.4) bound the mollified kernel by  $C_j(1 + |\log r|)^{j/2}$  on compact sets, even at collisions. Away from pairwise distances at most  $2r$ , separate harmonicity and radial mollifiers make it exactly the smooth unmollified kernel. Taking the mollification radius to be a fixed fraction of the smallest pair distance also bounds that kernel by a power of its logarithm. The collision neighborhood has volume  $O(r^2)$ , so both integrals over it tend to zero. The mollified kernels therefore converge locally in  $L^1$  to the smooth kernel extended across the diagonals. Their distributional limit is the original moment distribution, proving the assertion.  $\square$

**Lemma 6.7** (Centered remainders on small revealed sets). *Conditional on a stopped exploration using Theorem 3.1, let  $R_\delta$  be its centered flat remainders, set to zero on exposed sites. Let  $E_\delta$  be a set of sites measurable in the stopped data, in a fixed bounded region, and write  $m_\delta$  for the area of their lattice cells. If  $m_\delta \rightarrow 0$  in probability, then, for every bounded deterministic test  $f$ ,*

$$R_\delta(f \mathbf{1}_{E_\delta}) \longrightarrow 0 \quad \text{in } L^2.$$

*The conclusion is uniform over the stopped geometries. An additional revealed mean bounded by a fixed constant has a vanishing contribution on these sets as well.*

*Proof.* Condition on the stopped data and apply the second-moment case of Theorem 6.3. If the ambient diameter is at most  $R_0$ , a rearrangement bound for the logarithmic kernel gives

$$\mathbb{E}[|R_\delta(f \mathbf{1}_{E_\delta})|^2 \mid \mathcal{E}_\delta] \leq C \|f\|_\infty^2 m_\delta^2 \left( 1 + \log^+ \frac{R_0^2}{m_\delta} \right),$$

with value zero at  $m_\delta = 0$ . To verify the discrete bound, for each fixed site sum the decreasing logarithmic kernel first over its nearest  $O(m_\delta/\delta^2)$  cells, allowing the fixed cell-area factor of either parity. A disk of radius comparable to  $\sqrt{m_\delta}$  bounds this sum by  $C m_\delta (1 + \log^+(R_0^2/m_\delta))$ ; its central cell uses the cutoff  $|x - y| \vee \delta$ . Sum once more over  $E_\delta$ . The right side is a bounded function of  $m_\delta$  tending to zero at zero, so averaging proves the claim. A mean bounded by  $M$  contributes at most  $M \|f\|_\infty m_\delta$ . In particular the lemma applies to shrinking neighborhoods of an explored set once their areas have been shown to vanish; it does not presume that area estimate or thinness.  $\square$

### 6.3 Harmonic moment kernels and their collision coefficients

Fix a subsequence along which the flat fields and their moment distributions converge, and extract the plane covariance on the same subsequence. Write  $\kappa > 0$  for the coefficient denoted by  $\beta$  in [Theorem 4.6](#): for smooth compactly supported tests  $\alpha, \beta$  of integral zero,

$$\lim_{\delta \downarrow 0} \mathbb{E}[K_\delta^{\text{pl}}(\alpha) K_\delta^{\text{pl}}(\beta)] = \kappa \iint \alpha(x) \beta(y) \log \frac{1}{|x - y|} \, dx \, dy. \quad (6.5)$$

At this stage  $\kappa$  may depend on the subsequence. In particular, neither Gaussianity nor the final normalization is being assumed.

The complementary open limit  $U$  may have several components even when their discrete precursors belong to one component. Let  $T_j$  denote a subsequential  $j$ th moment distribution on  $U^j$ , with  $T_0 = 1$ . Their existence and convergence against test products follow from [Theorem 6.6](#).

We explain why separated spectator heights are legitimate exterior observations in the first application of screening. Localize the Laplace test to a disk  $B$ , and choose  $B, Q, Q^+$  concentric, with squares  $Q \subseteq Q^+$ , so that the radius of  $B$  is less than one thousandth of the side of  $Q$ . Choose  $Q^+$  compactly inside the limiting component containing  $B$ , disjoint from the spectator disks. An arbitrary local harmonicity test can be partitioned into tests with such small supports. Fix a simple curve surrounding  $Q$  inside  $Q^+ \setminus \overline{Q}$ , at positive distance from both boundaries. For all sufficiently fine meshes, choose a simple corner-grid approximation  $\Gamma_\delta$  of this curve in the discrete chamber and outside  $Q$ .

A spectator in that chamber is read by summing corner increments from its flat wall. Start with any such path. If it enters  $Q$ , replace the portion between its first and last encounters with  $\Gamma_\delta$  by an arc of  $\Gamma_\delta$ . The resulting path stays outside  $Q$ . This also works for spectators in a different limiting component of the same discrete precursor. Spectators in other independently sampled chambers may be adjoined to the exterior sigma-field: independence preserves the screened conditional-mean bound. Thus a bounded function of all these wall-anchored spectators is an allowed exterior observable.

On a product of disjoint interior disks, truncate the spectator product, apply [Theorem 5.1](#), and remove the truncation by Holder's inequality and the higher moment bounds. This gives

$$\Delta_{x_i} T_j = 0 \quad (1 \leq i \leq j).$$

Summing these equations gives an elliptic equation on the product space. Thus  $T_j$  has a smooth, separately harmonic representative off the diagonals. [Theorem 6.6](#) now excludes any additional distribution on the diagonals. Flat height reflection gives  $T_1 = 0$ . The logarithmic moment bound and [Theorem 6.4](#) apply to these representatives, by averaging on disjoint circles. They give, respectively, polylogarithmic growth at a collision and zero boundary trace in each variable away from the other insertions.

**Lemma 6.8** (Uniform logarithmic expansion). *If  $(y, \mathbf{z})$  is a collision-free list of interior points, then, near  $x = y$ ,*

$$T_j(x, y, \mathbf{z}) = L_j(y, \mathbf{z}) \log \frac{1}{|x - y|} + R_j(x; y, \mathbf{z}), \quad (6.6)$$

where  $R_j$  extends harmonically in  $x$  across  $y$ . The coefficient  $L_j$  is smooth in all its arguments. On a compact set of allowed  $(y, \mathbf{z})$ , the expansion holds on a common radius, and the remainder and each fixed order of its  $x$ -derivatives are uniformly bounded on a smaller common radius.

*Proof.* A harmonic function on a punctured disk has constant angular mode  $a + b \log r$  and nonconstant modes  $a_n r^n + b_n r^{-n}$ ,  $n \geq 1$ . Polylogarithmic growth forces every  $b_n$  to vanish. This proves the expansion and removability of its remainder. To obtain parameter uniformity, choose a fixed radius  $q$  avoiding all other insertions. The coefficient is the outward flux

$$L_j(y, \mathbf{z}) = -\frac{1}{2\pi} \int_{|w-y|=q} \partial_{n_w} T_j(w, y, \mathbf{z}) \, ds_w. \quad (6.7)$$

After writing  $w = y + q \exp(i\vartheta)$ , the integrand varies smoothly on a collision-free compact set. This proves smoothness of  $L_j$ . Subtract its logarithm from the boundary values on the circle and apply the Poisson formula. Interior derivative estimates give the claimed uniform bounds on  $R_j$ . This obtains uniformity from a fixed circle, not from a potentially nonuniform singular limit.  $\square$

We next state the precise local independence needed to read this flux. For a smooth test  $\alpha$ , put  $\alpha_r(x) = r^{-2} \alpha((x - y_0)/r)$ . For the neutral insertions below, use the bilinear nodal smearing of [Section 4](#) in both the flat and plane laws. Its coefficient array has exactly zero sum, and this is precisely the discretization in [Equation \(6.5\)](#). In flat chambers [Theorem 6.5](#) identifies this choice with either parity smearing.

**Lemma 6.9** (A neutral pair and distant observations). *Fix an interior disk  $B(y_0, 4R)$  and a finite family of spectator tests supported outside it. Let  $Y_\delta$  be a product of the corresponding flat height averages, or of smoothed plane increments. If  $\alpha, \beta$  are smooth zero-integral tests in disjoint bounded disks, then*

$$\lim_{r \downarrow 0} \limsup_{\delta \downarrow 0} |\mathbb{E}[K_\delta(\alpha_r) K_\delta(\beta_r) Y_\delta] - \mathbb{E}[K_\delta^{\text{pl}}(\alpha_r) K_\delta^{\text{pl}}(\beta_r)] \mathbb{E} Y_\delta| = 0. \quad (6.8)$$

*For flat fields, the assertion is uniform over patched collections in a common bounded region for which the local chamber contains this disk and every spectator buffer lies in its assigned chamber. All boundary constants have been subtracted. Spectators in other independently sampled pieces are allowed.*

*Proof.* Here are the stopping and coloring details of the coupling. Rescale  $R$  to one and take  $\rho = r^{1/2}$ ; all sufficiently small  $r$  have both local supports in  $B(y_0, Cr) \subset B(y_0, \rho)$ . Search inward using only  $(s, W)$  for a monochromatic open  $W$  circuit in the stack of annuli between radii  $\rho$  and  $R$ . The search stops on the first complete circuit and does not inspect its interior. By [Theorems 3.2](#) and [3.3](#), the probability of exhausting the stack without finding a circuit is at most  $C\rho^\eta$ , for some  $\eta > 0$ . The exponent and constants depend on the fixed model parameter and the buffered shapes, not on the flat domain.

On success reveal the entire exterior of the stopped circuit, including its transverse colors. No  $N$ -cluster crosses an open  $W$  circuit, so this revelation determines the outside color identifications without imposing an interior color constraint. By [Theorem 3.1](#), the unexplored interior is a flat law, with its constant  $s$  boundary value and independent fair colors on its free  $N$ -clusters. The sign of the wall causes no change to the law of neutral height increments: reflect a spin sign if necessary.

The spectator heights are measurable in the revealed exterior. For a bounded flat piece, read each of them by summing corner increments from its prescribed wall along paths avoiding  $B(y_0, R)$ . Such paths exist: any path entering this interior disk can be detoured through an annulus contained in the domain. The same detour works in all sufficiently fine approximations,

and averaging uses only nearby exterior sites. For plane increments, choose the integration paths between their two spectator buffers outside the disk. Thus we have not conditioned on absolute heights before obtaining a genuine stopping cut.

Inside the circuit apply the common-cut monotone coupling from [Theorems 3.2 and 3.3](#) to the flat law and the plane law. Between scales  $Cr$  and  $\rho$  there are order  $\log(\rho/r)$  buffered trials. Each successful common signed  $W$  circuit makes the two unexplored interiors identical in law. After this cut, couple their  $(s, W)$  configurations identically and assign the same fair colors to the corresponding interior  $N$ -clusters. Failure has probability at most  $C(r/\rho)^\eta$ . This construction is valid for every exposed exterior history; in each conditional coupling its second marginal is the same unconditioned plane law. Consequently that marginal is independent of the spectator observations. On a failed initial search choose the plane copy independently. If necessary, first perform the construction in finite flat exhaustions and then pass to the plane law. More explicitly, a completed  $W$  cut removes every transverse interaction, so fixing exterior colors leaves a kernel determined only by its signed wall, with no interior color pins. For each history, bracket this kernel and a plane exhaustion by the same ordered extreme boundary laws on an interior buffered region. The common-cut coupling preserves both marginals; after its cut use the identical free interior colors. Its plane marginal does not depend on the history. Thus the circuit trials use only  $(s, W)$  conditioning, never a residual law with arbitrary transverse-color pins.

The coupled local increments therefore agree outside an event  $\mathcal{B}_{\delta,r}$  satisfying

$$\limsup_{\delta \downarrow 0} \mathbb{P}(\mathcal{B}_{\delta,r}) \leq C\{\rho^\eta + (r/\rho)^\eta\} \leq Cr^{\eta/2}. \quad (6.9)$$

Agreement determines local heights up to a constant, which the exactly neutral weights remove. In particular both local smeared variables agree on the successful event.

For completeness, the unbounded factors cost only logarithms. For each fixed  $r$ , the exactly neutral lattice coefficients are discretizations of fixed scaled smooth profiles, with the same bounds once  $\delta/r$  is sufficiently small. Thus [Equation \(6.4\)](#) gives, for every fixed finite  $p$ ,

$$\limsup_{\delta \downarrow 0} \|K_\delta(\alpha_r)\|_{L^p} + \limsup_{\delta \downarrow 0} \|K_\delta(\beta_r)\|_{L^p} \leq C_p(1 + |\log r|)^{1/2}. \quad (6.10)$$

The logarithmic diagonals are integrable, including their lattice cells. The corresponding plane bounds are uniform in  $r$ , by the scale-invariant regularity estimate [[16](#), Theorem 4.5]. The fixed, smoothed spectator product has bounded moments of every finite order. Hölder's inequality with four factors, namely the two insertions, the spectator product, and the failure indicator, now bounds the discrepancy by

$$Cr^{\eta/8}(1 + |\log r|) = o(1).$$

Only original-law and plane-law moments were used. No uniform pointwise height moments or moment bounds under an artificially conditioned law are required. The mesh limit is always first, with the two continuum radii fixed.  $\square$

**Proposition 6.10** (Collision coefficient). *For  $j \geq 2$  and distinct  $(y, \mathbf{z})$  in  $U$ ,*

$$L_j(y, \mathbf{z}) = \kappa T_{j-2}(\mathbf{z}). \quad (6.11)$$

*Proof.* Choose two smooth zero-mass tests  $\alpha, \beta$  with disjoint supports and

$$J = \iint \alpha(u)\beta(v) \log \frac{1}{|u-v|} du dv \neq 0.$$

For an explicit choice let  $\varrho_b$  be a smooth radial unit bump supported within distance  $1/10$  of  $(b, 0)$  and obtained by translating one fixed bump. Take  $\alpha = \varrho_0 - \varrho_1$  and  $\beta = \varrho_3 - \varrho_4$ . Radial harmonic averaging gives  $J = \log(8/9) \neq 0$  exactly.

Take fixed radial unit mollifiers in disjoint disks around the spectator points  $\mathbf{z}$ . By separate harmonicity, averaging  $T_{j-2}$  in those disks gives exactly  $T_{j-2}(\mathbf{z})$ . The same is true of  $T_j$  when its first two points lie in a small disk around  $y_0$  disjoint from the spectator disks. Apply [Theorem 6.9](#) and (6.5). For each fixed  $r$ , the latter plane pairing is  $\kappa J$ : the added constant  $\log(1/r)$  vanishes against neutral tests. Thus

$$\lim_{r \downarrow 0} \iint \alpha(u) \beta(v) T_j(y_0 + ru, y_0 + rv, \mathbf{z}) \, du \, dv = \kappa J T_{j-2}(\mathbf{z}). \quad (6.12)$$

There is no additional large- $L$  limit in this choice of tests.

On the other hand insert [Theorem 6.8](#) in the left side. The term  $L_j(y_0 + rv, \mathbf{z}) \log(1/r)$  integrates to zero in  $u$  *exactly*, before any continuity estimate. The remaining logarithmic term tends to  $L_j(y_0, \mathbf{z})J$ , uniformly for spectator configurations in compact sets. For the remainder, subtract  $R_j(y_0; y_0 + rv, \mathbf{z})$  inside the  $u$  integral. The subtracted term again integrates to zero, and the uniform first-variable derivative bound gives an  $O(r)$  error. Comparison with (6.12) proves the formula. □

## 6.4 Wick's rule in the limiting chambers

The following argument is local at collisions and uses boundary decay, not a parametrization of the rough boundary. It is the collision method of [30, Sections 9–10], with the present cut and coupling estimates supplied above.

**Proposition 6.11** (Gaussian moment recursion). *For every  $j \geq 2$  and distinct points in  $U$ ,*

$$T_j(x_1, \dots, x_j) = \kappa \sum_{i=2}^j G_U(x_1, x_i) T_{j-2}(x_2, \dots, \hat{x}_i, \dots, x_j). \quad (6.13)$$

*Consequently the limiting centered field in  $U$  is a Gaussian field with covariance  $\kappa G_U$ . In particular its restrictions to different components are independent.*

*Proof.* Fix  $x_2, \dots, x_j$  and subtract the right side of (6.13). As a function of  $x_1$  the difference is harmonic away from these points. [Theorem 6.10](#) cancels every logarithmic pole in its component; [Theorem 6.8](#) removes the remaining punctures. Points in other components do not produce a pole. The resulting harmonic function has zero boundary trace by [Theorem 6.4](#). The Dirichlet Green functions have the same zero trace by [Theorem 6.2](#). The difference is bounded, since it is bounded near the boundary and its finitely many interior punctures are removable. The maximum principle therefore makes it zero on each component. This proves the recursion, including mixed moments when discrete components have separated only in the limit.

Starting with  $T_0 = 1$  and  $T_1 = 0$ , the recursion gives zero odd moments and the sum over perfect pairings of products  $\kappa G_U(x_i, x_j)$  for even moments. The products are locally integrable. Since [Theorem 6.6](#) excludes diagonal mass, the identity holds for the full smeared moment distributions. These moments determine the law: if a real linear combination  $Z$  of smeared fields has variance  $s^2$ , then

$$\mathbb{E} Z^{2n} = (2n-1)!! s^{2n}, \quad \mathbb{E} Z^{2n+1} = 0, \quad \mathbb{E} |Z|^n \leq s^n (2n)^{n/2}.$$

The exponential series is integrable for every real argument, so the characteristic function is  $\exp(-s^2 u^2/2)$ . Apply this to all finite linear combinations and then to a countable determining family of tests in the local Sobolev space. The zero cross-component covariance proves independence.  $\square$

## 6.5 Plane increment kernels and their collisions

The free-energy correspondence used below asks for compact-uniform convergence of all plane increment correlations. We now obtain that conclusion from the same screened identity and neutral-pair comparison, keeping the point correlations distinct from the smeared fields.

Use the unit corner-grid coordinates of [Section 4](#). Let  $\widetilde{K}_\delta^{\text{pl}}(x)$  be the piecewise constant extension of the plane unit height used in [\[16, Definition 2.4\]](#). For an increment  $d_i = (x_i, y_i)$ , define

$$\Phi_{m,\delta}(d_1, \dots, d_m) = \mathbb{E} \prod_{i=1}^m (\widetilde{K}_\delta^{\text{pl}}(x_i) - \widetilde{K}_\delta^{\text{pl}}(y_i)).$$

After the fixed rotation and dilation of the grid, this is the source correlation  $\Phi_m^{(\delta)}$  with endpoint convention  $u_i = y_i$ ,  $u'_i = x_i$ , at the corresponding mesh. Its configuration space is

$$\mathcal{D}_m = \{(d_1, \dots, d_m) \in (\mathbb{R}^2 \times \mathbb{R}^2)^m : \{x_i, y_i\} \cap \{x_j, y_j\} = \emptyset \text{ for } i \neq j\}.$$

The two endpoints of one increment may coincide. Put  $Q_0 = 1$ . Height-flip symmetry gives  $\Phi_{m,\delta} = 0$  for odd  $m$ , so their limits  $Q_m$  are zero. For even  $m$ , the regularity estimate and precompactness result [\[16, Theorem 4.5 and Corollary 4.6\]](#) give, after a diagonal extraction on the chosen subsequence,

$$\Phi_{m,\delta} \longrightarrow Q_m \quad \text{uniformly on compact subsets of } \mathcal{D}_m$$

at every order. Each  $Q_m$  is continuous there. These are correlation kernels, not products of point values of a continuum field.

We first relate these kernels to the nodal smooth insertions used above. For each neutral smooth test  $\alpha_i$ , choose a smooth mass-one density  $\rho_i$  in a separate remote disk, with all these disks disjoint from the test supports and from each other. Let  $b_{\delta,v}^{(i)}$  be the nodal interpolation coefficients for  $\alpha_i$ , and let  $r_{\delta,w}^{(i)}$  be discrete weights approximating  $\rho_i$  with sum one. The corresponding signed measures converge to  $\alpha_i(x) dx$  and  $\rho_i(y) dy$ , and their atoms have absolute weights  $O(\delta^2)$ . Exact neutrality gives the finite identity

$$\sum_v b_{\delta,v}^{(i)} K(v) = \sum_{v,w} b_{\delta,v}^{(i)} r_{\delta,w}^{(i)} (K(v) - K(w)).$$

Thus products of nodal insertions test  $\Phi_{m,\delta}$  with independently integrated reference variables, rather than coincident fixed anchors. Away from collisions between endpoints of different increments, use compact-uniform convergence. The regularity estimate bounds the omitted neighborhoods by integrable powers of logarithms, uniformly after the lattice cutoff; their exact coincidence cells contribute at most  $C\delta^2(1 + |\log \delta|)^q$  for a fixed  $q$ . These integrals vanish as the neighborhoods shrink. Coincidence of an increment's own endpoints has the vanishing increment-collapse bound, not a singular term. This identifies the full smeared moment distributions with the integrals of  $Q_m$ , without losing a contribution at an anchor or a collision.

For  $m = 2$ , this identification and Equation (6.5) give the covariance kernel

$$C_\kappa(d_i, d_j) = \kappa \log \frac{|x_i - y_j| |y_i - x_j|}{|x_i - x_j| |y_i - y_j|}. \quad (6.14)$$

as  $Q_2$ . Indeed one may approximate each endpoint by a smooth mass-one bump in a disjoint disk; the finite increment additivity and continuity of  $Q_2$  then recover the displayed point kernel from the neutral pairings.

Screening gives separate harmonicity of  $Q_m$  in every endpoint away from the endpoints of other increments. To see the required exterior measurability, localize a Laplace test in  $B \Subset Q \Subset Q^+$  as before, with  $Q^+$  disjoint from the other endpoint buffers. In the differentiated increment the remote endpoint cancels against the exactly neutral Laplace coefficients. Each remaining increment is read along a path between its two buffers outside  $Q$ , using a fixed annular detour when necessary. Smooth the remote endpoints in their buffers, apply Theorem 5.1 to the truncated spectator product, and remove the truncation by the regularity moment bounds. The smearing identification just proved gives distributional harmonicity of  $Q_m$ . Elliptic regularity gives its smooth separately harmonic representative where endpoints of different increments are distinct. The argument also allows the two endpoints of the differentiated increment to coincide, since its reference term has canceled.

Near a collision between different increments, the source regularity estimate bounds  $Q_m$  by a power of the logarithm of the separation. The analytic proof of Theorem 6.8 therefore applies to each such endpoint collision: there is one logarithmic term, its coefficient is given by a flux on a fixed circle, and the harmonic remainder and its derivatives are uniformly bounded on compact sets of the other endpoints. We can now identify that coefficient.

**Lemma 6.12** (Signed plane collision coefficients). *Fix  $m \geq 2$ , and keep all endpoints other than  $x_1$  distinct. For  $i \geq 2$ , the coefficient of  $\log(1/|x_1 - z|)$  in  $Q_m$  is*

$$\begin{cases} \kappa Q_{m-2}(d_2, \dots, \widehat{d}_i, \dots, d_m), & z = x_i, \\ -\kappa Q_{m-2}(d_2, \dots, \widehat{d}_i, \dots, d_m), & z = y_i. \end{cases}$$

*After subtracting this logarithm, the remainder extends harmonically through the collision. Swapping  $x_1$  and  $y_1$  gives the corresponding rule with the opposite sign for the negative endpoint.*

*Proof.* First consider two positive endpoints. Use the neutral tests  $\alpha, \beta$  from the proof of Theorem 6.10, whose logarithmic pairing is  $J = \log(8/9)$ . Smooth every remote endpoint in a small fixed radial disk. Integrating the two colliding endpoints against  $\alpha_r, \beta_r$  cancels their two remote reference terms exactly. By the smearing identification, the resulting expression is the limit of the two neutral lattice insertions times a product of smoothed plane increments. These spectators have the bounded moments required by Theorem 6.9. That lemma and Equation (6.5) give the limit

$$\kappa J Q_{m-2}(d_2, \dots, \widehat{d}_i, \dots, d_m) \quad (r \downarrow 0).$$

Separate harmonicity removes the spectator mollifiers. In the uniform logarithmic expansion, the term containing  $\log(1/r)$  integrates to zero against  $\alpha$  before any limit; smoothness of the coefficient and the uniform first-variable derivative bound for the remainder leave the coefficient times  $J$ . Comparing the two expressions proves the positive sign. Antisymmetry under reversal of either increment gives the negative signs. This calculation uses only smoothed spectator moments.  $\square$

**Proposition 6.13** (Plane Gaussian correlations). *Along the chosen subsequence, all plane increment correlation functions satisfy Wick's rule with covariance  $C_\kappa$ . More precisely,  $\Phi_{m,\delta} \rightarrow Q_m$  uniformly on compact subsets of  $\mathcal{D}_m$  at every order, and  $Q_m$  is zero for odd  $m$ , while for even  $m$  it is the sum over pairings of products of  $C_\kappa(d_i, d_j)$ .*

*Proof.* The cases  $m = 0, 1, 2$  have been established. For even  $m \geq 4$ , induct on  $m$  and subtract

$$\sum_{i=2}^m C_\kappa(d_1, d_i) Q_{m-2}(d_2, \dots, \hat{d}_i, \dots, d_m) \quad (6.15)$$

from  $Q_m$ . In the variable  $x_1$ , [Theorem 6.12](#) cancels all logarithmic poles at  $x_i$  and  $y_i$ ,  $i \geq 2$ , and its harmonic remainders remove the punctures. The regularity estimate gives continuous extension through  $x_1 = y_1$ , with value zero there; this is the collapse of the first increment, not an additional collision between different factors. The remainder is consequently entire harmonic. With all other endpoints fixed, the same estimate bounds its growth by a fixed power of  $1 + \log(2 + |x_1|)$ , in particular by  $o(|x_1|)$ . An entire harmonic function of sublinear growth is constant: the interior gradient bound on disks of radius  $M$ , followed by  $M \rightarrow \infty$ , makes its gradient zero. At  $x_1 = y_1$  both  $Q_m$  and (6.15) vanish. The constant is therefore zero. This proves the plane Wick recursion.

The argument initially identifies the kernels where all the endpoints are distinct. The quoted equicontinuity and the increment-collapse estimate extend the identity locally uniformly to the full configuration spaces allowing the two endpoints of one increment to coincide. Thus all orders, not only covariance or smooth-test laws, have been identified on one common subsequence.  $\square$

## 6.6 Identifying the coefficient without circularity

**Proposition 6.14** (Plane normalization). *For every subsequence considered above,*

$$\kappa = \frac{1}{2\pi \arcsin(c/2)} = \frac{1}{a^2}. \quad (6.16)$$

*Proof.* The coefficient is strictly positive by [Theorem 4.6](#).

Set  $\sigma^2 = 2\pi\kappa$ . The plane Wick kernels just proved are precisely those of  $\sigma$  times a plane GFF in the normalization with singularity  $-(2\pi)^{-1} \log|x - y|$ . The fixed rotation and dilation used above leave the increment cross-ratio in [Equation \(6.14\)](#) unchanged. Their compact-uniform convergence at every order is the exact hypothesis of [\[16, Theorem 8.1\]](#). That convergence criterion, whose input is the regularity estimate, supplies the full-plane subsequential scaling limit in the sense required by its [Definition 2.7](#). We may therefore apply [\[16, Theorems 4.3 and 4.4\]](#): their free-energy correspondence and explicit computation give

$$\sigma^2 = \frac{1}{\arcsin(c/2)}.$$

The source uses the column slope  $I_{\text{col}}/L$  and per-site free energy, as reconciled in [Section A.2](#); that appendix also records the finite-error factors before their final limits. All three cited inputs apply for  $1 \leq c \leq 2$ ; the present fixed parameter satisfies  $1 < c \leq 2$ . This use does not invoke the narrower-range height convergence theorem of that paper. It follows that  $2\pi\kappa = 1/\arcsin(c/2)$ . Finally  $a^2 = g(\pi/2)^2 = 2\pi \arcsin(c/2)$ , proving (6.16).

The logical order is important. The spectral argument supplied only a subsequential coefficient; the local collision argument identified every Gaussian correlation with that coefficient; the convergence criterion established the requisite plane GFF limit; only now has the free energy fixed the coefficient.  $\square$

## 6.7 Completion of the flat limit and conditioning at a cut

*Proof of Theorem 6.1.* By Theorem 6.2, every compact subset of  $U$  eventually lies inside the discrete open set. On a finite list of interior buffers, Theorem 6.6 gives joint subsequential field and moment convergence. The screened identity applies there with their fixed positive clearances. The boundary estimate uses the connected accessible wall in each chamber, and remains valid at every boundary point of the limiting components, as proved in Theorem 6.4. The Green boundary estimate in Theorem 6.2 supplies the matching zero trace. Thus the proofs of Theorems 6.10 and 6.11 apply to these components without a smooth-boundary assumption. They identify all mixed moment kernels, including those in different components, with the block Dirichlet covariance  $\kappa G_U$ .

The plane coefficient on any simultaneous subsequence is  $\kappa = 1/a^2$  by Theorem 6.14. Hence all subsequential laws agree. Tightness proves full convergence, and the moment bounds give convergence of moments. Theorem 6.5 supplies the asserted parity and interpolation equivalences. For a mesh-dependent number of independent chambers, the moment proof is unchanged: each retained site belongs to at most one chamber, and shared-cluster counts between different chambers are zero. In particular the constants do not accumulate with the number of chambers. The same uniform bounds, with zero extension, prove the final tightness assertion. That assertion does not identify an extra field supported on a possibly positive-area limiting exposed set; subsequent uses of thinness will address the full decomposition across that set.  $\square$

At a stopped cut, let  $A_\delta$  be the revealed closed set with its lattice-scale thickening. On any fixed compact test region  $Q$  separated from the initial domain boundary, the closed complement of the resolved chamber faces satisfies

$$F_\delta \cap Q \subset A_\delta^{C\delta}, \quad A_\delta \cap Q \subset F_\delta^{C\delta},$$

where  $E^\eta = \{x : \text{dist}(x, E) < \eta\}$ . Both descriptions contain the same exposed cells and perimeter edges; only the one-cell medial displacement differs. These inclusions transfer the neighborhood estimates in Section 7 between the revealed cut and the resolved complement.

**Proposition 6.15** (Conditional flat remainders). *Suppose a stopping exploration uses the cuts in Theorem 3.1, so that, conditional on its revealed data  $\mathcal{E}_\delta$ , the unexplored chambers have independent flat laws with pairwise disjoint interiors. Let  $R_\delta$  be their centered unit-height remainders, patched by zero, and let  $U_\delta, F_\delta$  be their open union and closed complement in  $\bar{B}_0$ . Fix real tests  $f_1, \dots, f_\ell \in C_c^\infty(B_0)$ ,  $u_1, \dots, u_\ell \in \mathbb{R}$ , and  $d_0 > 0$ . Put*

$$A_{\delta, d_0} = \left\{ \text{dist} \left( \bigcup_i \text{supp } f_i, F_\delta \right) \geq d_0 \right\}.$$

Then

$$\begin{aligned} & \mathbb{E} \left[ \mathbf{1}_{A_{\delta, d_0}} \left| \mathbb{E} \left[ \exp \left( i \sum_i u_i R_\delta(f_i) \right) \right] \middle| \mathcal{E}_\delta \right] \right. \\ & \quad \left. - \exp \left( -\frac{1}{2a^2} \sum_{i,j} u_i u_j G_{U_\delta}(f_i, f_j) \right) \right] \longrightarrow 0. \end{aligned} \tag{6.17}$$

Here  $G_{U_\delta}$  is the continuum Dirichlet kernel of the resolved open chamber union, with zero covariance between its components.

Multiplying the displayed difference by any uniformly bounded  $\mathcal{E}_\delta$ -measurable variable preserves the same estimate with its clearance indicator. To state the limiting consequence, let  $\mathbf{E}$  be a Polish space and let  $e_\delta$  be an  $\mathcal{E}_\delta$ -measurable  $\mathbf{E}$ -valued variable. Suppose, for one fixed  $s < -1$ , that along a subsequence

$$(e_\delta, F_\delta, R_\delta) \implies (e, F, R) \quad \text{in } \mathbf{E} \times \mathcal{K}(\overline{B}_0) \times H_{\text{loc}}^s(B_0),$$

where  $\mathcal{K}(\overline{B}_0)$  is the space of nonempty compact sets with the Hausdorff metric. The limit  $F$  is almost surely connected and contains  $\partial B_0$ . Put  $U = B_0 \setminus F$  and

$$A_F = \left\{ \text{dist} \left( \bigcup_i \text{supp } f_i, F \right) > 0 \right\}.$$

Then, almost surely,

$$\begin{aligned} \mathbf{1}_{A_F} \mathbb{E} \left[ \exp \left( i \sum_i u_i R(f_i) \right) \middle| \sigma(e, F) \right] \\ = \mathbf{1}_{A_F} \exp \left( -\frac{1}{2a^2} \sum_{i,j} u_i u_j G_U(f_i, f_j) \right). \end{aligned} \tag{6.18}$$

Thus, conditional on  $\sigma(e, F)$ , the restrictions of  $R$  to the component interiors of  $U$  are independent zero-boundary GFFs divided by  $a$ , including components born from a pinched single precursor. This is first tested on deterministic compact buffers with positive limiting clearance and then extended by a countable exhaustion. The retained variable  $e_\delta$  may encode any exact-cut coordinates and flat constants for which this joint convergence holds; the constants are added after the centered assertion. Its  $\mathcal{E}_\delta$ -measurability excludes data from any later revelation.

*Proof.* For a deterministic realized chamber configuration, let its flat product law supply the characteristic function in the first line of Equation (6.17). The error from the displayed Gaussian characteristic function tends to zero uniformly over all such configurations with the stated clearance. Otherwise choose a sequence with error bounded away from zero. Compactness of the closed complements in  $\overline{B}_0$  gives a Hausdorff-convergent further subsequence, and the clearance persists in its limit. The deterministic theorem gives convergence of the flat characteristic function to the Gaussian one for that limit. Theorem 6.2 gives  $G_{U_\delta}(f_i, f_j) \rightarrow G_U(f_i, f_j)$  on the same subsequence. The two characteristic functions therefore have the same limit, a contradiction.

Condition now on the stopped data. By Theorem 3.1, the conditional kernel is exactly the deterministic product kernel just considered, with its constants subtracted. The uniform error proves Equation (6.17). Multiplication by any bounded stopped-measurable variable preserves convergence to zero.

Assume the displayed joint convergence and write  $d(F) = \text{dist}(\bigcup_i \text{supp } f_i, F)$ . Choose a bounded continuous function  $\chi$  of the clearance whose support is contained in  $[d_0, \infty)$  for some  $d_0 > 0$ , and a bounded continuous function  $v$  on  $\mathbf{E} \times \mathcal{K}(\overline{B}_0)$ . Multiply the conditional difference by  $\chi(d(F_\delta))v(e_\delta, F_\delta)$ , and use the tower property for the term involving  $R_\delta$ . The estimate just proved makes the resulting expectation tend to zero. Evaluation of  $R$  on a smooth test and the clearance  $d(F)$  are continuous in the displayed state space. The connected complements containing  $\partial B_0$  form a closed subspace supporting every  $F_\delta$  and  $F$ . On that subspace, Theorem 6.2 makes the

Gaussian expression continuous where  $\chi$  is supported. Joint convergence therefore gives

$$\begin{aligned} & \mathbb{E} \left[ \chi(d(F))v(e, F) \exp \left( i \sum_i u_i R(f_i) \right) \right] \\ &= \mathbb{E} \left[ \chi(d(F))v(e, F) \exp \left( -\frac{1}{2a^2} \sum_{i,j} u_i u_j G_U(f_i, f_j) \right) \right]. \end{aligned}$$

The monotone class theorem on the Polish space  $\mathbb{E} \times \mathcal{K}(\overline{B}_0)$  extends this identity to bounded measurable  $v$ . A countable family of cutoffs  $\chi$  whose positive sets cover  $d(F) > 0$  now gives [Equation \(6.18\)](#). A countable determining family of smooth tests then identifies the regular conditional law on the component interiors. The zero cross-component Green kernel gives their independence, including when one precursor separates into several limiting components. Increasing the list of buffers and decreasing their required clearance gives the full interior statement. Jointly retained constants can then be added.  $\square$

## 7 Stopped local sets and canonical clusters

We now identify the field left by a stopped current exploration. The discrete field in this section is  $h_\delta = aK_\delta$ , whose limit has the normalization of  $h$ . Two kinds of stops are needed. Stopping at a fixed number of odd rims gives the two-valued carpets. A seed exploration instead reveals whole current clusters; its limiting conditional field will be used to decide which odd rims belong to one cluster. For both stops we prove the conditional law on tests that cross the limiting cut. This is stronger than identifying the field only in its open components.

A closed set  $A$ , together with information  $\mathcal{E}$ , is a local set of  $h$  when, conditionally on  $\mathcal{E}$ ,  $h = h_A + h^{D \setminus A}$ , where  $h_A$  is  $\mathcal{E}$ -measurable and harmonic off  $A$ , and  $h^{D \setminus A}$  is the zero-extended GFF with independent zero-boundary restrictions to the components of  $D \setminus A$ . Conditioning down to  $\sigma(A, h_A)$  gives the usual local-set definition. It is *thin* when  $h_A$  is integration against that harmonic function, with no further distribution supported on  $A$ . We include the domain boundary in every closed local set. Thus statements about zero area below concern its intersection with the open domain; a Jordan boundary itself need not have zero area.

For  $p, q > 0$  with  $p + q \geq 2\lambda$ , a boundary-connected thin local set with complementary values in  $\{-p, q\}$  is the canonical  $A_{-p,q}$ , by [6, Proposition 2]. Here boundary-connected means that the closed set, with the boundary adjoined, is connected; it supplies the topological condition in the bounded-type thin-local-set theorem. This uniqueness does not assume that the candidate was already measurable in the field. The resulting canonical set is field-measurable. We use the Jordan-hole property in Jordan parents. When a first odd carpet is constructed in a rough simply connected parent, its two label magnitudes are  $2a > 2\lambda$ ; its loops avoid the parent boundary and hence remain Jordan under the interior conformal map. These geometric facts and the nested constructions are in [5, Sections 3–4]. In particular  $A_{-2\lambda, 2\lambda}$  has  $\text{CLE}_4$  holes with independent fair signs and conditionally independent zero-boundary remainders.

### 7.1 Stops and their cut neighborhoods

We first specify the finite stops. In a free-current chamber of flat value  $b$ , descend through surrounding current clusters, continue through even holes, and stop on the first odd rim along each branch. The closed complement of the unsearched holes is the *first odd carpet*; its hole values

are  $b - 2a$  and  $b + 2a$ . Repeating this operation in the holes defines the odd depth. This counts odd rims, not all clusters inspected: any number of even holes may be passed before the next odd rim. All holes are the sealed component faces of [Theorem 3.1](#), with its medial resolution at pinches.

For a seed stop, use a fixed predictable growth rule: from the current revealed data choose the next vertex and an attachment to the revealed region, reveal the entire current cluster first met there, and stop using the revealed data. Apart from the new vertex, the attachment uses only already revealed vertices or known seed edges, with no unsearched free vertex in between. The rule may follow a preassigned column sweep or choose the lexicographically first unvisited vertex adjacent to the revealed region. It may use discovery indices, observed diameters, or the first observed odd jump around specified points. The seed trace may therefore be adaptive; its union with the revealed clusters remains connected to the exterior. The terminal sigma-field  $\mathcal{E}_\delta$  includes the revealed spins, edges, colors, heights, and the seed. The initial sigma-field  $\mathcal{E}_{0,\delta}$  records only the exact flat cut from which this exploration starts. Conditional on it, the initial unsearched chambers have the independent flat laws of [Theorem 3.1](#).

We also use a composite seed stop for the wired current. Expose its boundary cluster first, and then run one fixed global seed rule through the remaining free chambers, attaching each newly visited seed piece through the already revealed region containing that layer. A half-column sweep can be ordered this way. Seed edges are edges of the underlying graph and need not be current-open. After every added vertex, reveal the whole cluster met there. The revealed union remains connected to the original boundary, and the finite cut law applies simultaneously to all its residual chambers. This is one exploration of the original field, even when one discrete wired hole will later separate into several limiting holes.

Write  $A_\delta$  for the revealed closed set with its one-cell thickening,  $S_\delta$  for the seed at the stop, and  $W_{0,\delta}$  for the initial revealed wall. For a carpet the seed is empty. Let  $U_{0,\delta}$  be the union of the initial open chamber faces and let  $F_{0,\delta}$  be its closed complement in the fixed enclosing disk  $\bar{B}_0$  of [Section 6](#). Define  $U_\delta, F_\delta$  in the same way for the residual faces at the terminal stop. On compact subsets of the initial domain,  $F_\delta$  and  $A_\delta$  differ only by the  $O(\delta)$  medial displacement described immediately before [Theorem 6.15](#).

Use the same atomic smearing for all three finite distributions

$$M_\delta(\phi) = \mathbb{E}[h_\delta(\phi) \mid \mathcal{E}_\delta], \quad R_\delta = h_\delta - M_\delta, \quad h_\delta = M_\delta + R_\delta.$$

Thus  $R_\delta$  is  $a$  times the centered flat remainders of [Theorem 6.15](#), patched by zero on exposed sites. For bounded nonsmooth tests,  $a$  set denotes the union of its lattice cells and the pairing is the corresponding atomic sum. This convention also identifies it with piecewise constant integration up to the smooth-test cell error.

For a discovered cluster  $C$ , let  $\mathcal{O}_\delta(C)$  be the union of its odd holes. A contributor to a compact set  $Q$  is a discovered  $C$  for which  $C$  or  $\mathcal{O}_\delta(C)$  meets  $Q$ ; write  $N_\delta(Q)$  for their number. On the event

$$d_\delta(Q) := \text{dist}(Q, F_{0,\delta} \cup S_\delta) \geq \rho > 0,$$

each contributor either reaches  $Q$  from the seed or surrounds a point of  $Q$ , and has diameter at least a fixed multiple of  $\rho$ . Consequently  $N_\delta(Q)$  is pathwise bounded by the initial whole-box count of free clusters of that diameter. The domination and tail in [Theorem 3.6](#), conditional only on  $\mathcal{E}_{0,\delta}$ , make these counts uniformly tight over the stopping rules. A completed terminal history can already record many contributors, so no count tail conditional on that history is used.

**Lemma 7.1** (Negligible cut neighborhoods). *Let  $Q$  be a compact subset of the initial domain and fix  $\rho > 0$ . Consider a seed stop just described, a carpet of fixed odd depth, or the first wired layer. For the last two cases omit  $S_\delta$  in  $d_\delta(Q)$ . There are  $C, \alpha > 0$ , depending on  $Q, \rho$  and the fixed depth when applicable, such that for  $0 < \eta < \rho/20$ ,*

$$\mathbb{E}[\mathbf{1}_{\{d_\delta(Q) \geq \rho\}} |Q \cap A_\delta^\eta| \mid \mathcal{E}_{0,\delta}] \leq C(\eta + \delta)^\alpha. \quad (7.1)$$

The bound is averaged over the production of the stop and is uniform over the allowed initial flat data and stopping rules.

Let  $\zeta_\rho$  be a continuous function on  $[0, \infty)$ , with values in  $[0, 1]$ , equal to zero on  $[0, \rho]$  and to one on  $[2\rho, \infty)$ . For every bounded  $\phi$  supported in  $Q$ ,

$$\lim_{\eta \downarrow 0} \limsup_{\delta \downarrow 0} \mathbb{E}[\zeta_\rho(d_\delta(Q)) |R_\delta(\phi \mathbf{1}_{A_\delta^\eta})|^2] = 0. \quad (7.2)$$

For a fixed-depth carpet the analogous first-moment statement holds for  $M_\delta$ . For a seed stop it holds after restricting to  $N_\delta(Q) \leq M$  and to a fixed bound on the initial flat labels of the chambers meeting  $Q$ :

$$\lim_{\eta \downarrow 0} \limsup_{\delta \downarrow 0} \mathbb{E}[\zeta_\rho(d_\delta(Q)) \mathbf{1}_{\{N_\delta(Q) \leq M\}} |M_\delta(\phi \mathbf{1}_{A_\delta^\eta})|] = 0. \quad (7.3)$$

The restriction on the initial labels is understood in this display. For the initial free and wired constructions their magnitudes are respectively 0 and  $a$ . The contributor restriction can be removed in probability using the initial count tail.

*Proof.* Fix the initial exact-cut data. Suppose a seed stop satisfies  $d_\delta(Q) \geq \rho$  and its revealed set comes within  $\eta$  of  $z \in Q$ . Up to one lattice cell, a newly revealed cluster then meets  $B(z, O(\eta + \delta))$  and contains a current path from there to its seed vertex at distance at least  $\rho$ . Thus this event is contained in the event of an  $N$ -arm from the small disk to radius  $\rho$ . This inclusion holds for every terminal rule, including rules using the transverse colors.

Test this arm in the  $(s, W)$  marginal of the initial flat kernel. In a stack of separated buffered annuli between these radii, every prescribed-color  $W$  circuit blocks the arm. Reveal only  $(s, W)$  in successive annuli, with endpoints before edges. The conditional bound of [Theorem 3.3](#) gives probability at least  $p > 0$  for each blocking circuit. The arm probability is therefore at most  $(1 - p)^{c \log(\rho/(\eta + \delta))}$ . This calculation never conditions that circuit estimate on the full terminal history. Integrating the event inclusion over  $z \in Q$  proves (7.1) for seed stops, including the product of initial flat chambers in the composite construction.

For a carpet of odd depth  $r$ , use successive pairs of annuli with opposite prescribed  $W$  signs. Between the two circuits in each successful pair, the jump rule of [Theorem 3.1](#) forces an odd rim. Nested successful pairs force distinct odd layers. Reaching the inner disk before depth  $r$  therefore implies fewer than  $r$  successes among logarithmically many one-parity trials. Their conditional lower bounds give a binomial tail; its polynomial factor is absorbed by decreasing the exponent. For the first wired layer, a  $W$  circuit blocks the boundary current's arm, using the bulk pre-removal clause of [Theorem 3.3](#). These arguments take place in the original or fresh exact-cut kernel before any future color revelations.

Condition now on the terminal data, only to estimate the remainder. The residual fields are independent flat fields by [Theorem 3.1](#). For every terminal-measurable union  $E$  of cells in  $Q$ , the logarithmic covariance bound gives

$$\mathbb{E}[|R_\delta(\phi \mathbf{1}_E)|^2 \mid \mathcal{E}_\delta] \leq C \|\phi\|_\infty^2 \iint_{E \times E} \left(1 + \log^+ \frac{R_0}{|z - w| \vee \delta}\right) dz dw \leq C' \|\phi\|_\infty^2 |E|. \quad (7.4)$$

The final bound follows from the uniform integral of the logarithmic kernel on a bounded set; its atomic version includes the diagonal cell cutoff. Multiplying by  $\zeta_\rho(d_\delta(Q))$ , averaging, and using (7.1) proves (7.2).

The revealed values and wall labels at fixed odd depth are bounded by a deterministic depth-dependent constant. At a seed stop, on the stated truncation the exact jump rule instead gives

$$|M_\delta| \leq B + 2aM + a \quad \text{on } Q,$$

where  $B$  bounds the initial labels. Each contributor gives either zero or one jump of size  $2a$ ; on its own sites the extra parity discrepancy is at most  $a$ . The area estimate proves (7.3). The pathwise contributor bound above and the initial whole-box tail remove its truncation in probability.  $\square$

## 7.2 The conditional field across a stopped cut

We specify what is retained when a finite stop has a limit. Let  $\mathbf{e}_\delta$  be a random element of a countable product of Polish spaces, measurable at that same terminal stop. It contains  $A_\delta, F_\delta, F_{0,\delta}, W_{0,\delta}, S_\delta$  as closed-set coordinates. For each rational closed ball  $B$  and each positive rational  $c$ , it also records the terminal chamber label when  $\text{dist}(B, F_\delta) > c$ , and the initial chamber label when  $\text{dist}(B, F_{0,\delta}) > c$ , with a cemetery value for either coordinate when its clearance fails. Rational polygonal paths with positive clearance encode connection between such buffers. One may include specified terminal-measurable contributor coordinates whose chosen encodings are tight, such as the localized finite lists used in Section 9. This retained state excludes future clusters and canonical data that may appear elsewhere in a larger joint representation.

For a fixed  $s < -1$ , pass to a joint subsequence in the product of  $H_{\text{loc}}^s(B_0)^3$  and this retained state space:

$$(h_\delta, M_\delta, R_\delta, \mathbf{e}_\delta) \implies (h, M, R, \mathbf{e}), \quad h = M + R. \quad (7.5)$$

The field tightness and the zero-extension tightness of the remainders give this extraction for the first three coordinates. For a fixed protected ball choose a smooth unit-mass test supported inside it and divide its atomic pairing by the deterministic atomic mass of that test. The mass tends to one, and on the clearance event the normalized  $M_\delta$  pairing is exactly the chamber label. The full pairing is tight and the remainder pairing has uniformly bounded second moment, so the label on that event, with the cemetery value off it, is tight as well. The same argument at the initial cut uses its centered flat remainder and proves tightness of the protected initial-label coordinates. This makes a diagonal extraction possible for the countable state; unprotected labels in chambers collapsing onto the cut are not retained. Write  $\mathcal{E} = \sigma(\mathbf{e})$ , completed under the limiting law. This is the limiting stopped sigma-field used below.

Let  $A, F, F_0, W_0, S$  be the closed-set limits. The one-cell comparison between  $A_\delta$  and  $F_\delta$  gives  $A \cap D = F \cap D$ ; similarly  $W_0 \cap D = F_0 \cap D$ . Put  $U = D \setminus A$ . Its components are defined geometrically from  $F$ : two rational points belong to the same one exactly when some rational polygonal path between small neighborhoods of them has positive clearance. They are not defined by a limiting discrete equivalence relation. At a rational point outside  $F$ , choose a small ball and a rational threshold strictly below its limiting clearance. Its protected coordinate reads the limiting label; the point receives the cemetery value on  $F$ . The labels agree within each component, since a compact connecting path is eventually inside one discrete chamber. Denote this label by  $m_O$  on a component  $O$ .

**Lemma 7.2** (Full conditional field at a stopped cut). *In (7.5), let  $Q \Subset D$  be compact. On the stopped-measurable event  $\Lambda_Q = \{\text{dist}(Q, F_0 \cup S) > 0\}$ , the set  $A \cap Q$  has zero area,  $M$  is the*

distribution given on  $Q$  by the locally integrable function  $m_O$  on each component of  $U$ , and, for every real  $\phi \in C_c^\infty(D)$  supported in  $Q$ ,

$$\mathbf{1}_{\Lambda_Q} \mathbb{E}[\exp(i(h - M)(\phi)) \mid \mathcal{E}] = \mathbf{1}_{\Lambda_Q} \exp\left(-\frac{1}{2}G_U(\phi, \phi)\right). \quad (7.6)$$

The identity holds for every real linear combination of any finite family of such tests. Thus, on tests crossing  $A$  in this region,  $h - M$  is conditionally the zero-extended GFF on  $U$ , with zero covariance between its components. The mean is the limit of the finite jump contributions after the contributor and initial-label truncations of [Theorem 7.1](#), followed by their removal.

For a carpet explored in a single initial parent, take  $D$  to denote that parent; then  $\Lambda_Q$  holds for every  $Q \Subset D$ , and hence  $|A \cap D| = 0$ . For a global wired carpet whose initial wall already contains  $W_0$ , the conclusion is used on compact subsets of each pointed wired hole and combined with the previously identified layer. For seed and composite stops the conclusions hold locally away from  $W_0 \cup S$ . Each stop in a countable family has this conclusion conditional on its own retained state; adjoining the other stops to a joint representation does not enlarge that conditioning sigma-field.

*Proof.* Use a Skorokhod representation of the joint subsequence for this proof. We first derive zero area before using any continuum neighborhood bound. Let  $d_\delta = d_\delta(Q)$  and  $d = \text{dist}(Q, F_0 \cup S)$ . Their convergence follows from Hausdorff convergence. For fixed  $\eta$ , every point of  $A \cap Q$  eventually lies in  $A_\delta^\eta$ . Fatou's lemma and (7.1) therefore give

$$\mathbb{E}[\zeta_\rho(d)|A \cap Q|] \leq \liminf_{\delta \downarrow 0} \mathbb{E}[\zeta_\rho(d_\delta)|Q \cap A_\delta^\eta|] \leq C\eta^\alpha.$$

Letting  $\eta \downarrow 0$  proves zero area where  $d \geq 2\rho$ . Exhausting positive rational  $\rho$  proves it on  $\Lambda_Q$ . The same holds for  $F \cap Q$ , because  $A$  and  $F$  agree in  $D$ .

We next prove the conditional characteristic function across the cut. Fix a small  $\eta > 0$ . Choose a finite nonnegative smooth partition  $(\psi_i)_{i \leq n_\eta}$  equal to one in sum on  $Q$ , with each  $\psi_i$  supported in a rational ball and with a fixed larger rational closed ball  $B_i \Subset D$  of diameter at most  $4\eta$ . Choose a continuous  $\chi : [0, \infty) \rightarrow [0, 1]$  equal to zero on  $[0, 1/4]$  and to one on  $[1, \infty)$ , and put

$$w_{i,\delta} = \chi\left(\frac{\text{dist}(B_i, F_\delta)}{\eta}\right), \quad \phi_\delta^\eta = \sum_i w_{i,\delta} \phi \psi_i. \quad (7.7)$$

Use the corresponding  $w_i, \phi^\eta$  with  $F$  in the limit. These weights are continuous functions of the closed-set coordinate. If  $w_{i,\delta} > 0$ , the whole buffer  $B_i$  has clearance at least  $\eta/4$ . If  $1 - w_{i,\delta} > 0$ , every point in  $\text{supp}(\phi \psi_i)$  is within  $5\eta$  of  $F_\delta$ . Consequently  $\phi - \phi_\delta^\eta$  is bounded by  $\|\phi\|_\infty$  and supported in  $Q \cap F_\delta^{5\eta}$ .

Stratify the finitely many possible active index sets  $\{i : w_{i,\delta} > 0\}$ . On each stratum, [Equation \(6.17\)](#), multiplied by  $a$  in the field normalization, applies to the fixed tests  $\phi \psi_i$ . Its fixed real coefficients may be replaced by the stopped-measurable vector  $(w_{i,\delta})$ : approximate the compact coefficient cube by a finite rational net. The conditional second-moment bound makes the characteristic function uniformly Lipschitz in those coefficients, and  $G_{U_\delta} \leq G_{B_0}$  gives the same uniform continuity for the Gaussian expression. Summing over the active strata yields

$$\mathbb{E}\left|\mathbb{E}[e^{iR_\delta(\phi_\delta^\eta)} \mid \mathcal{E}_\delta] - e^{-\frac{1}{2}G_{U_\delta}(\phi_\delta^\eta, \phi_\delta^\eta)}\right| \longrightarrow 0. \quad (7.8)$$

This used only terminal-measurable coefficients and ancillaries, never information from a later interior revelation.

For indices with positive limiting weight, their buffers retain positive clearance. Green convergence in [Theorem 6.2](#) applies to every pair of their fixed tests. Indices whose weights vanish are harmless by the uniform bound from  $G_{B_0}$ . Hence the Gaussian exponent in (7.8) converges to  $G_U(\phi^\eta, \phi^\eta)$ . The left test converges jointly by (7.5) and the finite continuous weights. Multiplying by a bounded continuous function of  $\mathbf{e}_\delta$  and by  $\zeta_\rho(d_\delta)$  is therefore legitimate.

There are two neighborhood errors to remove. The one-cell relation between  $F_\delta$  and  $A_\delta$ , (7.4), and (7.1) imply

$$\lim_{\eta \downarrow 0} \limsup_{\delta \downarrow 0} \mathbb{E}[\zeta_\rho(d_\delta) |R_\delta(\phi - \phi_\delta^\eta)|^2] = 0.$$

For the limiting kernel, positivity and domain monotonicity give

$$\|\phi - \phi^\eta\|_{G_U}^2 := G_U(\phi - \phi^\eta, \phi - \phi^\eta) \leq C \|\phi\|_\infty^2 |Q \cap F^{5\eta}|. \quad (7.9)$$

Indeed  $0 \leq G_U \leq G_{B_0}$ , whose logarithmic singularity has a uniformly bounded integral. The right side tends to zero on  $\Lambda_Q$  by the zero-area result. Energy Cauchy–Schwarz now shows that  $G_U(\phi^\eta, \phi^\eta) \rightarrow G_U(\phi, \phi)$  there; the other energy factors are bounded by the same outer kernel.

The inequality  $|e^{ix} - e^{iy}| \leq |x - y|$  removes the lattice error in the tested conditional identity. Then (7.9) and bounded convergence remove the continuum error. We obtain, for every bounded continuous  $Z$  of the retained state,

$$\mathbb{E}[\zeta_\rho(d)Z(\mathbf{e})e^{iR(\phi)}] = \mathbb{E}[\zeta_\rho(d)Z(\mathbf{e})e^{-\frac{1}{2}G_U(\phi, \phi)}].$$

Bounded continuous functions determine the law on the retained metric state. A monotone-class argument gives the same equality for all bounded  $\mathcal{E}$ -measurable  $Z$ . Exhausting positive clearances proves (7.6), because  $R = h - M$  in the joint limit. A countable determining family fixes one conditional version; continuity then gives every real linear combination in the finite-vector assertion, including independence across components that shared a discrete precursor.

It remains to identify  $M$ . On an active buffer in (7.7),  $M_\delta$  is exactly its flat chamber label. The protected rational-label coordinates and their agreement along compact connecting paths show that  $M_\delta(\phi_\delta^\eta)$  converges to  $\int_U m_O \phi^\eta$ . Terms of zero limiting weight vanish by tightness of the fixed  $M_\delta(\phi\psi_i)$  pairings. The joint extraction also identifies this limit with  $M(\phi^\eta)$ . We spell out the bound on the limiting labels. A finite deterministic cover of  $Q$  by rational balls of radius at most  $\rho/8$  can be chosen so that every point lies in one of them. On  $d_\delta(Q) \geq \rho$ , any initial chamber meeting  $Q$  contains such a ball with clearance at least  $\rho/2$  from  $F_{0,\delta}$ . Thus the maximum of its initial labels is bounded by the maximum of finitely many protected initial-label coordinates and is tight. On this clearance event, a contributor bound  $N_\delta(Q) \leq M$  and an initial-label bound  $B$  bound every terminal chamber label meeting  $Q$  by  $B + 2aM + a$ . Transfer this bound first to any finite family of protected limiting buffers, using their joint convergence and a slightly smaller clearance. The probability of its failure is at most the initial count or label tail. Exhaust the countable rational buffers; the same tail bound holds for the supremum of the limiting labels on  $Q \setminus A$ . Sending  $M, B$  to infinity proves that this supremum is finite almost surely on each positive-clearance localization.

On these truncations the omitted finite contribution vanishes by (7.3); the omitted limiting integral vanishes by the local label bound and  $|Q \cap F^{5\eta}| \rightarrow 0$ . Joint convergence of the two  $M_\delta$  pairings transfers the finite omission bound to  $M(\phi) - M(\phi^\eta)$  in probability. Letting  $\eta \downarrow 0$  therefore gives

$$M(\phi) = \int_U m_O(z)\phi(z) \, dz \quad \text{on } \Lambda_Q.$$

The initial whole-box tail of [Theorem 3.6](#) and label tightness remove the truncations in probability. This also proves local  $\mathcal{E}$ -measurability of  $M$ . No mean or centered distribution remains on the cut in the tested region. Countable rational localizations and a separate retained state for each stop give the simultaneous assertion.  $\square$

### 7.3 The stopped two-valued sets

**Proposition 7.3** (Convergence of stopped carpets). *The wired boundary cluster converges jointly with  $h_\delta$ , with its labels, to  $A_{-a,a}(h)$ . In a free flat parent, after subtracting its label and writing  $f$  for the zero-boundary limiting remainder, the first odd carpet converges to  $A_{-2a,2a}(f)$ . At depth  $r$ , define the global closed carpet as the closed complement of the holes still unsearched after every branch has met  $r$  odd rims; it includes all earlier layers and, in the wired case, the first wired layer. For each fixed finite  $r$ , this global carpet converges in ordinary Hausdorff distance to the corresponding canonical nested carpet. Its labels and restarted zero-boundary fields converge on every finite family of pointed compact buffers in the limiting holes. These pointed conclusions include distinct limiting components with the same discrete precursor; they do not assign one whole precursor carpet to several separate Hausdorff limits.*

*Proof.* Take a joint subsequence as in (7.5), also retaining the raw closed carpets. Compactness of closed subsets, field tightness, and the finite label sets permit this extraction. The exact cuts give residual labels  $\pm a$  for the wired layer and  $\pm 2a$ , relative to the parent, for the first free odd carpet. Their closed limits are boundary-connected. Every open component is represented by rational interior buffers: if a disk lay outside the closed limit but in no pointed component, its compact containment in the discrete complement would give a contradiction.

[Theorem 7.2](#) gives the full local-set decomposition and its bounded label mean on tests throughout the parent. It also gives zero area inside that parent. Hence the limit is a boundary-connected bounded thin local set, including across its cut. The uniqueness theorem of [6, Proposition 2] identifies the two limits as  $A_{-a,a}$  and  $A_{-2a,2a}$ , since  $2a > 2\lambda$  and  $4a > 2\lambda$ .

The original wired domain is Jordan. Its limiting holes are Jordan by the two-valued-set geometry recalled above. The first odd holes have label magnitudes  $2a > 2\lambda$ , so their loops avoid even a rough parent boundary and are Jordan as well. Restart in such a hole after subtracting its observed label. The conditional characteristic-function estimate is uniform on deterministic interior buffers; a countable exhaustion handles the random hole. Induction gives every fixed finite odd depth, with a single global closed carpet at that depth and pointed compact restrictions inside its holes. In the wired case, the local applications are on compact subsets of pointed wired holes, and the already identified  $W_0$  is supplied by the preceding layer. Write  $K_r$  for the connected global limit at depth  $r$ . If  $V$  is a pointed Jordan hole of  $K_{r-1}$ , then  $\partial V \subset K_{r-1} \subset K_r$ , and  $K_r \cap \bar{V}$  is connected. Indeed, a separation of this compact intersection would put the connected  $\partial V$  on one side and leave the other side compactly inside  $V$ . Adjoining  $K_r \setminus V$  to the first side would then separate  $K_r$ . Thus the child restriction is boundary-connected. Inside each pointed parent, uniqueness identifies that restriction; outside the union of those parents the global closed set is the preceding layer. This identifies the whole global closed limit even when one precursor splits. The sets are canonical functions of the same limiting field, so every subsequential law is the stated one.  $\square$

## 7.4 Fields at seed stops

**Proposition 7.4** (Conditional fields at a cluster stop). *Consider a seed stop in an initial free parent of label  $b_0$ , or the composite wired stop defined above. Let  $A, S, W_0, \mathcal{E}$  be its limiting explored set, seed trace, initial wall, and retained stopped sigma-field. The set  $A$  is connected to the original boundary, and every component  $O$  of  $D \setminus A$  is simply connected. Conditionally on  $\mathcal{E}$ ,*

$$h|_O = m_O + h^O, \quad m_O - b_0(O) \in 2a\mathbb{Z}, \quad (7.10)$$

where the  $h^O$  are independent zero-boundary GFFs. Here  $b_0(O)$  is the label of its initial parent chamber; it is 0 in the initial free domain and  $\pm a$  in a wired hole. The labels are tight on every finite family of pointed components.

On every compact subset of  $D$  away from  $W_0 \cup S$ ,  $A$  has zero area and the full conditional mean  $M$  is the locally integrable piecewise constant function  $m_O$ . On the same region, Equation (7.6) gives the full zero-extended conditional GFF law of  $h - M$ , also for tests whose supports meet  $A$ . Locally  $M$  is the limit of the finite signed odd-hole contributions of the discovered clusters, with the initial parent label added. These statements hold for every member of a specified countable family of sweeps and observable indexed stops, each conditional only on its own retained state.

For a composite wired stop, every  $O$  is contained in one pointed limiting wired hole. The conclusions use the global exploration of the original  $h$ ; they do not require a separate lattice sweep selected from that limiting hole.

*Proof.* The initial wall, the attached seed, and all clusters met by it form one boundary-connected revealed union. Each residual face has one sealed monochromatic  $W$  perimeter with its actual connections, by Theorem 3.1. Its flat label differs from that of its initial parent by an element of  $2a\mathbb{Z}$ . Hausdorff limits preserve the connected revealed union. The planar connected-complement criterion then makes every open component simply connected. In the wired composite case  $A$  contains  $W_0$ , so each open component lies in one component of  $D \setminus W_0$ .

Before the limit, reflection of a centered flat law makes its mean zero. Thus on an unsearched site the exact jump rule gives

$$M_\delta(z) = b_{0,\delta}(z) + 2a \sum_{C \text{ discovered}} \epsilon_C \mathbf{1}_{O_\delta(C)}(z), \quad z \notin A_\delta, \quad (7.11)$$

where  $b_{0,\delta}(z)$  is its initial chamber label and  $\epsilon_C \in \{-1, 1\}$  is the cluster's relative jump sign. The contributor bound and Theorem 7.1 give the finite local truncation of this formula. On a connected compact ball away from  $W_0$ , the initial label is one constant. On exposed sites the parity discrepancy from an incident wall is bounded and is removed with the cut neighborhood.

Theorem 7.2 now proves both the full mean identity and the full residual law. Its interior restriction is (7.10); the block Green kernel gives independence even when one discrete chamber pinches into several limiting components. The label tightness was proved in the joint extraction. For the composite wired stop, all inputs are those of the single global finite exploration, so the same argument covers such a pinch without a future choice of lattice chamber.  $\square$

## 7.5 Geometry of the candidate clusters

The incoming label must be kept when a canonical cluster is restarted. If its parent field is  $b + f$ , the construction in Theorem 2.1 has the following values:

| incoming label | CLE sign | field in the CLE hole | labels after the split |
|----------------|----------|-----------------------|------------------------|
| $b$            | +1       | $b + 2\lambda + f_V$  | $\{b, b + 2a\}$        |
| $b$            | -1       | $b - 2\lambda + f_V$  | $\{b - 2a, b\}$ .      |

Equivalently,  $b$  becomes  $b + \xi 2\lambda$  and then one of  $\{b, b + \xi 2a\}$ . A hole of a retained cluster is *odd* when its label is  $b + \xi 2a$ , and is even when its label is  $b$ . Thus every descendant baseline lies in  $b_0 + 2a\mathbb{Z}$ . Here  $b_0 = 0$  in the free domain and  $b_0 = \pm a$  behind the wired layer.

We first justify the nested continuation that will be used twice.

**Lemma 7.5** (Completion between fixed labels). *Let  $f$  be a zero-boundary GFF in a bounded simply connected parent  $P$ . Let  $E$  be a field-measurable boundary-connected bounded thin local set whose value  $c_H$  is constant on each component  $H$  of  $P \setminus E$ . Suppose*

$$-p \leq c_H \leq q, \quad p, q > 0, \quad p + q \geq 2\lambda.$$

*For these fixed  $p, q$ , suppose also that all  $c_H$  lie in one fixed deterministic countable subset  $\mathcal{L} \subset [-p, q]$ . In each component with  $-p < c_H < q$ , insert  $A_{-p+c_H, q-c_H}(f^H)$  in its zero-boundary remainder. Leave components already labelled  $-p$  or  $q$  untouched. The closed union is  $A_{-p, q}(f)$ . In particular every such partial continuation is contained in that same canonical target.*

*Proof.* Point each component at its first rational point and use the normalized Riemann map from the disk. The kernel theorem makes this map Borel in the pointed domain. For each  $c \in \mathcal{L} \cap (-p, q)$ , choose one Borel version of the canonical  $A_{-p+c, q-c}$  map in the disk, apply it to the pulled-back remainder, transport its interior set, and adjoin the component boundary. This gives jointly measurable insertions in the random pointed components for the countably many fixed pairs. Enumerate the selected components by rational interior points and let  $E_N$  insert only the first  $N$  complete two-valued sets. Conditional local-set gluing in these distinct components makes  $E_N$  local. Its mean distribution is the old mean plus the zero-extended relative means of those insertions. On a new hole the value is  $c_H$  plus its relative value, hence is  $-p$  or  $q$ . The old density on the inserted set contributes nothing, because a two-valued set has zero area inside its open parent [5, Proposition 3.1]. This says nothing about the area of the parent boundary. Thus  $E_N$  is thin, field-measurable, boundary-connected, and a  $K$ -BTLS for  $K = \max(p, q)$ .

Here is the needed justification for passing through countably many holes. Choose an integer  $M$  with  $K \leq 2M\lambda$ . By [6, Propositions 1 and 3], all  $E_N$  lie in the same larger canonical set  $Z = A_{-2(M+1)\lambda, 2(M+1)\lambda}(f)$ . The compact-interior dimension bound of [5, Proposition 3.1] gives  $|Z \cap P| = 0$  by exhaustion. Let  $E_\infty = \bigcup_N E_N$ , with closure in  $\bar{P}$ . The increasing-local-set result [7, Lemma 2.3(3)] applies: the sets are field-measurable, and after adjoining  $\partial P$  each has one nontrivial connected component. It makes  $E_\infty$  local and gives convergence of the conditional mean distributions.

For every smooth compactly supported  $\varphi$ , thinness gives  $|f_{E_N}(\varphi)| \leq K\|\varphi\|_{L^1(P)}$ . Take an almost surely convergent further subsequence on a countable dense family of tests, pass this bound to the limiting distribution, and then use continuity. By  $L^1$  duality that distribution has an  $L^\infty(P)$  density bounded by  $K$ . Locality identifies it with the harmonic mean off  $E_\infty$ . Since  $E_\infty \cap P \subset Z \cap P$  has zero area, this is the full mean distribution, so  $E_\infty$  is thin. Every complementary component is an untouched terminal hole or a hole of one inserted set: a compact subset of one original  $H$  cannot be affected by insertions in the other components. Its value is therefore  $-p$  or  $q$ . Uniqueness in [6, Proposition 2] identifies the union with  $A_{-p, q}(f)$ .

This also proves the nested-interval rule for each fixed target interval and a partial local set whose labels range over a fixed deterministic countable set: after centering the incoming field, in  $[L, U]$  with  $L < 0 < U$  and  $U - L \geq 2\lambda$ , a hole of value  $c \in (L, U)$  is continued by the relative set  $A_{-c-L, U-c}$ , while a hole already at an endpoint is left alone. The uses below have finite label sets. This is a same-field statement, consistent with the monotonicity and nested construction in [5, Section 3.2].  $\square$

The next lemma records the facts needed to identify how several odd rims belong to one cluster. It does not assume local finiteness over all generations.

**Lemma 7.6** (Canonical cluster geometry). *In the free procedure of Theorem 2.1 the retained sets form a countable family of pairwise disjoint connected compact sets, each strictly inside its parent domain. They and all their labelled holes are measurable functions of the GFF. Each retained cluster is the closure of the union of its odd-hole boundaries. Conditional on the unlabelled recursive geometry, including which odd holes belong to the same retained cluster, their relative jump signs are independent and fair.*

*In a parent field  $b + f$ , a positive first-generation retained cluster is contained both in the positive excursion of  $f$  and in the boundary-connected upper first-passage set of  $f$  at level  $2a$ . The containing cable sets have paths respectively above  $b$  and below  $b + 2a$  in total field units. A negative retained cluster satisfies the reflected containments, giving paths below  $b$  and above  $b - 2a$ . The carpet formed by continuing through even holes and stopping at first odd holes is  $A_{-2a, 2a}(f)$ . These statements remain valid at  $a = 2\lambda$ .*

*Proof.* Write  $P$  for the current parent. Each retained set is a two-valued set inside a Jordan  $\text{CLE}_4$  hole, with that hole's boundary included. Its closed set is connected and compact. Distinct outer  $\text{CLE}_4$  loops are disjoint, and every subsequent restart is strictly inside a complementary hole of the previous retained set. Induction gives disjointness at all finite generations; their union is countable because open components each contain a rational point. Canonical measurability follows at each restart from local-set measurability. None of these observations asserts all-generation local finiteness.

Subtract the incoming  $b$  and condition on the outer CLE data. Consider a positive  $\text{CLE}_4$  hole  $V$ . The field there is  $2\lambda + f_V$ , and the retained split is  $A = A_{-2\lambda, 2a-2\lambda}(f_V)$ , with total values 0 and  $2a$ . The first-passage set  $F = A_{-2\lambda}(f_V)$  is the increasing limit obtained by sending the upper two-valued barrier to infinity. Thus it contains  $A$ , and  $F$  together with the outer CLE rim is the positive excursion inside that loop [7, 9].

The FPS decomposition on  $V$  reads  $2\lambda + f_V = \nu + f^F$ , where  $\nu = (f_V)_F + 2\lambda \, dz$  is a positive measure whose relative support is all of  $F \cap V$  [9, Definition 8 and Theorem 9]. The tower property for the nested sets  $A \subset F$  gives, for a nonnegative smooth test  $q$  compactly supported in  $V$ ,

$$\mathbb{E}[\nu(q) \mid V, A \text{ and their labels}] = 2a \int_{\mathcal{O}(A)} q(z) \, dz, \quad (7.12)$$

where  $\mathcal{O}(A)$  is the union of its odd holes. Indeed the centered remainder after  $F$  has conditional mean zero, and thinness of  $A$  makes the conditional mean of  $2\lambda + f_V$  exactly the density  $2a \mathbf{1}_{\mathcal{O}(A)}$ . A countable family of nonnegative tests, multiplied by the event that their supports miss the odd holes, shows from positivity that  $\text{supp}_V \nu$  is contained in their closure. Since that support contains  $A \cap V$ , every interior neighborhood of a point of  $A \cap V$  meets an odd hole. The point itself lies outside those interiors, so their boundaries accumulate there.

The relative support statement does not include the outer rim. Condition on  $V$  and fix a conformal map from the disk onto this Jordan domain. On a closed angular subarc take a smooth nonnegative probability density  $w$  relative to  $d\vartheta/(2\pi)$ , and push its averages on circles of radii  $r \uparrow 1$  into  $V$ . Smooth radial approximations give compactly supported smooth tests  $q_r$  with the same limiting estimates. The disk Green kernel is nonnegative, and its average against uniform angular measure on both circles of radius  $r$  is  $-\log r$ . Hence the Green energy of these weighted averages is at most  $\|w\|_\infty^2 (-\log r)$ , which tends to zero. In particular  $f_V(q_r) \rightarrow 0$  in  $L^2$ .

The centered remainder  $f^F$  after the FPS continuation has conditional covariance bounded above by  $G_V$ . Therefore  $f^F(q_r) \rightarrow 0$  in  $L^2$  as well. If an ambient neighborhood of the chosen boundary arc missed the interior FPS, its positive measure would vanish on  $q_r$  for all sufficiently large  $r$ . On this event the full FPS decomposition would give  $2\lambda + f_V(q_r) = f^F(q_r)$ . Both centered terms tend to zero in probability, whereas their difference on that event is the nonzero constant  $-2\lambda$ . The event consequently has probability zero. A countable basis of angular arcs and ambient neighborhoods, using the Jordan boundary extension [22, Chapter I, Theorem 3.1], proves  $\partial V \subset \overline{F} \cap \overline{V}$ . This is the boundary-average argument behind FPS nontriviality [7, Proposition 4.6], localized at the outer rim; the closed excursion and its ambient support are also described in [9, Theorem 1 and Section 3].

By Equation (7.12), the interior FPS lies in the closure of the odd holes. The preceding inclusion thus puts the outer rim in that closure too. Since a point of  $A$  is outside every odd-hole interior, approaching it from an odd hole also approaches its boundary. This proves the claimed closure identity on the whole retained cluster.

For the upper FPS containment, expose the outer  $\text{CLE}_4$  set and this positive split. Finite conditional gluing makes it a field-measurable boundary-connected BTLS with relative labels among  $\{-2\lambda, 0, 2\lambda, 2a\}$ . Fix  $R > 2\lambda$ . All but  $2a$  lie strictly between  $-R$  and  $2a$ , since  $a > \lambda$ . By Theorem 7.5, completing these holes to  $-R, 2a$  yields  $A_{-R, 2a}(f)$ , containing the split. As  $R \uparrow \infty$  through a deterministic sequence, reflection of the FPS construction [7, Remark 4.2 and Proposition 4.4] identifies the closed increasing union as the upper FPS of  $f$  at  $2a$ . This last set is generally nonthin; the preceding bounded-density argument is not used as the barrier grows. Its metric approximants connect points to the parent boundary by sublevel paths, with coupled FPS convergence given by [8, Proposition 4.7]. For excursions, Theorem 8.5 below proves directly the one-sided same-field approximation needed for connected paths between strict interior hits. In the total field the two path levels are  $b$  and  $b + 2a$ . Reflection gives levels  $b$  and  $b - 2a$  for a negative cluster.

For the sign assertion, let  $\xi$  be the fair outer  $\text{CLE}_4$  sign. The zero-boundary remainder  $f_V$  is independent of it. The field  $\xi f_V$  has the same law and is still independent of  $\xi$ . Reflection carries the negative split into the positive split, preserving its unlabelled geometry and the distinction between odd and even holes. Conditional remainders in the holes can be reflected in the same way before restarting. This proves the independent-fair-sign assertion against every finite recursive geometry cylinder. A monotone-class argument gives the assertion conditional on the entire countable unlabelled geometry. It does not assert independent labels on arbitrary two-valued-set loops.

Finally, fix the incoming  $b$ , subtract it, and put  $T = A_{-2a, 2a}(f)$ . At a completed first-odd stage, holes labelled  $\pm 2a$  are terminal and holes labelled zero are active. In every active hole insert its complete outer CLE set, then the appropriate complete split in each CLE hole. Define each layer by enumerating its holes and taking the closed limit of finite insertions. Every finite partial layer is a field-measurable boundary-connected BTLS by finite gluing. Its labels lie among  $0, \pm 2\lambda, \pm 2a$ , hence inside  $[-2a, 2a]$ . Its completion under Theorem 7.5 is the same  $T$ , so the partial layer lies in  $T$ . For a full layer, the increasing-local-set and bounded-density proof of that lemma applies with  $Z = T$  and  $K = 2a$ . It remains a BTLS contained in  $T$ .

Let  $B_0 = \partial P$ , let  $B_n$  be the completed stage after  $n$  rounds of even continuation, and let  $B_\infty = \overline{\bigcup_n B_n}$ . The same increasing-local-set result makes  $B_\infty$  local and gives convergence of the mean distributions. Their absolute values on a test  $\varphi$  are bounded by  $2a\|\varphi\|_1$ , so the limiting mean has an  $L^\infty$  density bounded by  $2a$ . Since  $B_\infty \subset T$  and  $|T \cap P| = 0$ , that density agrees with the harmonic mean as a full distribution. This proves thinness of the increasing limit.

The active region has vanishing area. Conditional on a positive CLE hole, thinness and mean zero for  $f_V$  give, for Lebesgue-a.e.  $z$ ,

$$0 = -2\lambda(1 - p(z)) + (2a - 2\lambda)p(z), \quad p(z) = \lambda/a,$$

where  $p(z)$  is the conditional probability that the split assigns the odd value. Reflection gives the same probability in a negative hole. Fubini therefore bounds the expected area still active after  $n$  rounds by  $(1 - \lambda/a)^n |P|$ . Monotone convergence makes the area surviving all rounds zero almost surely. Terminal labels persist, so the limiting harmonic mean takes values  $\pm 2a$  almost everywhere off  $B_\infty$ . Harmonic continuity and connectedness make it one of these constants throughout each open component. BTLS uniqueness gives  $B_\infty = T$ . At the next odd depth the observed  $b \pm 2a$  becomes the new incoming label before the argument is restarted.

At  $a = 2\lambda$  the split has width  $4\lambda$  and both relative barriers remain positive. The FPS support proof (7.12), the sign argument, and the containments still apply, and the odd probability is  $1/2$ . Thus the endpoint is included.  $\square$

## 8 Local cable tests and the no-crossing principle

A canonical cluster is a function of the limiting field, whereas the stopped law in Theorem 7.4 is conditioned on retained exploration data. To compare them, we turn a local cable path test into a bounded function of the continuum field on a slightly larger buffer. Calibration in the canonical parent makes this function equal to one on a crossing event. Its expectation in the stopped component tends to zero under that component's conditional GFF law.

### 8.1 Auxiliary cables and the tests

All cable fields in this section are auxiliary Gaussian approximations; their index  $n$  and dyadic mesh  $\varepsilon_n = 2^{-n}$  are separate from the current mesh  $\delta$  in Section 7. They have the normalization of  $h$ , with Cameron–Martin norm

$$\|u\|_{\nabla}^2 = \frac{1}{2\pi} \int |\nabla u|^2 \, dz, \quad u \in H_0^1. \quad (8.1)$$

Fix the corresponding square cable grids  $\Gamma_n$  in one bounded box  $\mathcal{B}$ . All later subsequences are drawn from this dyadic sequence. A cable GFF  $\tilde{\Phi}_n$  has discrete GFF values at vertices and independent Brownian bridges on the metric edges conditional on those values. Edge resistances and the field scaling are fixed once for this normalization. For the metric-domain approximations below, planar pairings use the ALS harmonic extension of the *whole cable function* to the box, with zero data at the box boundary [8, Sections 2.1 and 4.1]; denote it by  $\hat{\Phi}_n$ . Thus  $\hat{\Phi}_n(f) = \int f(z) \hat{\Phi}_n(z) \, dz$ . It is not a vertex-only pairing. Full-field convergence and full-field functionals below use  $H^s(\mathcal{B})$  for one fixed  $s < -1$ , the distribution topology of the cited cable convergence.

For every test polygon  $B$ , fix once a connected closed union  $B_n^\square$  of complete primal mesh cells approximating  $\bar{B}$  from inside and exhausting its compact subsets. Put  $C_{B,n} = \Gamma_n \cap B_n^\square$ . It includes the contained and perimeter edges, but not the open part of an outward edge. The same convention defines  $C_{K,n}$  for a smaller test polygon.

The reference polygon  $P$  uses a distinct approximation so that it is literally a metric domain of [8, Section 4.1]. Let  $P_n^\circ$  be the interior of a fixed simply connected union of dual mesh cells centered

at grid vertices, exhausting compact subsets of  $P$  with complement-Hausdorff convergence. Its boundary crosses primal cables at their midpoints. Subdivide there and write  $P_n = \Gamma_n \cap P_n^\diamond$ , including the cut points as Dirichlet boundary vertices. The inherited half-edge resistances preserve the cable law. In every compared ambient approximation, this same subdivided  $P_n$  must occur unchanged for fine mesh, with no missing cable, altered resistance, or extra pin there. The cut points are unpinned in the ambient field; they become Dirichlet vertices only in the reference residual after the exterior is conditioned. The reference is the zero-boundary cable GFF in  $P_n$ . An admissible ambient graph is the source metric domain  $\Gamma_n \cap D_n$  for a Euclidean open  $D_n \subset \mathcal{B}$ , with its boundary intersection points adjoined as Dirichlet endpoints. The regular inner approximations used below exhaust compact subsets and contain the prescribed  $P_n$  for all sufficiently large  $n$ . Their complements converge in Hausdorff distance, and their numbers of complementary components are uniformly bounded. These are the primary admissible approximations in this section. [Theorem 8.4](#) adds a second auxiliary class, with an explicit whole-cable embedding and the same local kernel.

Here are the tests used later. A collar choice consists of a compact rational polygon  $Q$  and regular rational polygons

$$Q \Subset V \Subset K \Subset B \Subset P.$$

For  $t \in a\mathbb{Z}$ , let  $T_{n,t}^+$  be the indicator that, on the cables in  $C_{K,n}$ , a path in  $\{\tilde{\Phi}_n \geq t\}$  joins  $Q$  to  $K \setminus \bar{V}$ . Define  $T_{n,t}^-$  with  $\tilde{\Phi}_n \leq t$ . A canonical inner hit will be in  $\text{int } Q$ , and its outer hit will have positive distance from  $\bar{V}$  and from  $\partial K$ . These strict hits persist under Hausdorff convergence. The tests are measurable functions of the same cables in every domain. More generally, the next two lemmas allow any  $[0, 1]$ -valued measurable test supported in  $C_{K,n}$ ; no continuity of that test as a function of a limiting distribution is assumed.

## 8.2 The reverse buffer estimate

For a fixed polygon  $B$ , put

$$\mathcal{G}_n(B) = \sigma(\tilde{\Phi}_n(x) : x \in C_{B,n}), \quad \mathcal{G}(B) = \sigma(\Phi(f) : f \in C_c^\infty(B)).$$

The first sigma-field contains the full cable restriction, including the perimeter bridges. By the cable Markov property, the residual after conditioning is zero on  $C_{B,n}$  and is an independent killed cable GFF on its complement. Extend this residual by zero on  $C_{B,n}$  before applying the original ambient harmonic extension operator. Write  $\mathsf{K}_n^B(f, g)$  for the covariance of the resulting planar pairings. In particular, the conditioned perimeter cables remain zero traces in this extension. We do not replace this form by the extension from a graph in which those cables were deleted.

**Lemma 8.1** (Reverse buffer estimate). *Let  $K \Subset B \Subset D$ , with  $B$  a regular polygon, and let  $\tilde{\Phi}_n$  be zero-boundary cable GFFs in the admissible approximations of  $D$ . Suppose the full extended Green forms converge to  $G_D$  on smooth tests. Then*

$$\mathsf{K}_n^B(f, g) \longrightarrow G_{D \setminus \bar{B}}(f, g).$$

*If  $T_n \in [0, 1]$  is supported in  $C_{K,n}$  and  $(\hat{\Phi}_n, T_n)$  converges jointly along a subsequence to  $(\Phi, T)$ , then*

$$\mathbb{E}[T \mid \Phi] = \mathbb{E}[T \mid \mathcal{G}(B)]. \tag{8.2}$$

*The same conclusion holds for a bounded vector of local tests and after adding deterministic harmonic means whose extended functions converge in the full-field topology of the joint convergence*

and converge smoothly on the buffer neighborhoods, in the interior finite-difference sense used below.

*Proof.* The covariance limit is for the precise clamped form just defined. The deterministic set  $C_{B,n}$  is optional for the cable field, connected and nontrivial, and converges in Hausdorff distance to  $\overline{B}$ . The metric-local-set convergence theorem [8, Lemma 4.9] therefore applies under the stated domain hypotheses. It uses the ambient harmonic extensions of the revealed and residual fields. Every limit is the deterministic local-set decomposition at  $\overline{B}$ , so the residual limit is the zero-extended GFF on  $D \setminus \overline{B}$ . Its finite pairing vectors and their prelimit vectors are centered Gaussian. Convergence of their characteristic functions forces convergence of their covariance matrices, proving the displayed limit. This argument retains the whole zero cable trace; it makes no assertion that deleting perimeter edges commutes with harmonic extension.

Let  $\mathbf{f} = (f_i)_{i \leq \ell}$  be a tuple of smooth tests, put  $F_n = (\hat{\Phi}_n(f_i))_{i \leq \ell}$  and  $F = (\Phi(f_i))_{i \leq \ell}$ , and let  $Q_n = \mathbb{E}[F_n \mid \mathcal{G}_n(B)]$ . Linearity of the ambient extension and the cable Markov decomposition give the exact identities

$$\text{Cov}(F_n - Q_n) = \mathsf{K}_n^B(\mathbf{f}, \mathbf{f}), \quad \text{Cov}(Q_n) = \hat{G}_n^D(\mathbf{f}, \mathbf{f}) - \mathsf{K}_n^B(\mathbf{f}, \mathbf{f}), \quad (8.3)$$

where  $\hat{G}_n^D$  is the covariance form of  $\hat{\Phi}_n$ . The residual is independent of  $\mathcal{G}_n(B)$ , hence of  $T_n$ . Both matrices in (8.3) converge. In the continuum, conditioning on  $\mathcal{G}(B)$  has the corresponding killed covariance: the functions in  $H_0^1(D)$  that vanish on  $B$  are the zero extensions of  $H_0^1(D \setminus \overline{B})$ . Polygonal regularity makes this the usual domain Markov decomposition.

Choose a countable dense family  $(f_i^B)$  of interior smooth pairings for the Gaussian Hilbert space  $\mathcal{G}(B)$ , discarding dependent coordinates in each finite list. Every cell meeting the support of a fixed  $f_i^B$  belongs to  $B_n^\square$  for all sufficiently large  $n$ . Thus its ALS face pairing is exactly  $\mathcal{G}_n(B)$ -measurable. Let  $Q_n^{(r)}$  be the Gaussian projection of  $F_n$  onto the first  $r$  such pairings. For fixed  $r$ , convergence of the covariance matrices gives joint convergence with the field to the corresponding finite projection  $Q^{(r)}$ . If  $Q = \mathbb{E}[F \mid \mathcal{G}(B)]$ , density gives  $Q^{(r)} \rightarrow Q$  in  $L^2$ . Orthogonality and (8.3) yield

$$\lim_{r \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{E}|Q_n - Q_n^{(r)}|^2 = \lim_{r \rightarrow \infty} \mathbb{E}|Q - Q^{(r)}|^2 = 0. \quad (8.4)$$

This convergence of projection norms is the information needed for an arbitrary local test.

For a bounded Lipschitz function  $v$  of  $F_n$ , Gaussian conditioning writes  $\mathbb{E}[v(F_n) \mid \mathcal{G}_n(B)] = \Psi_n(Q_n)$ , where  $\Psi_n$  convolves  $v$  with the residual Gaussian law. Its covariance limit and (8.4) approximate this expression in  $L^1$ , uniformly for fine mesh, by bounded continuous functions of finitely many interior pairings. The limit is  $\mathbb{E}[v(F) \mid \mathcal{G}(B)]$ . Multiply this identity by  $T_n$  and by a bounded continuous function of finitely many interior pairings. The  $L^1$  error remains valid because  $T_n$  is bounded. Passing to the two limits proves the conditional factorization with  $T$  for every full-field Gaussian cylinder given  $\mathcal{G}(B)$ . A monotone-class argument proves (8.2). The vector and harmonic mean versions follow by the same argument.  $\square$

### 8.3 A domain-independent local kernel

**Lemma 8.2** (Common local kernel). *Fix  $K \in B \in P$  as above and a sequence of  $[0, 1]$ -valued cable tests  $T_n$  supported in  $C_{K,n}$ . There are a subsequence and a measurable  $k$ , with  $0 \leq k \leq 1$ , on continuum field restrictions to  $B$ , such that:*

1. *In the zero-boundary reference field in  $P$ , every joint limit along this subsequence satisfies  $\mathbb{E}[T \mid \Phi] = k(\Phi|_B)$ .*

2. The same  $k$  is the conditional kernel in every deterministic domain  $D$  with  $\overline{P} \subset D$ , for the same cable tests in admissible approximations containing the identical subdivided  $P_n$ . For a prescribed harmonic mean, including a constant boundary value, require its ALS extended means to converge in  $H^s(\mathcal{B})$  to the chosen full continuum ALS extension, and to converge smoothly on buffer neighborhoods in the interior finite-difference sense used below. Equivalently, for every bounded continuous field functional  $v$ ,

$$\mathbb{E}[T_n v(\hat{\Phi}_n)] \longrightarrow \mathbb{E}[k(\Phi|_B) v(\Phi)]$$

along the selected subsequence. This convergence of mixed expectations does not require convergence of the tests as field functions.

3. Let a fixed random parameter  $\Theta$  select  $D$  and its mean, and suppose that conditionally on  $\Theta$  the original total parent field  $\Phi_D$  has that GFF law. Sample auxiliary cable fields conditionally on  $\Theta$  in admissible approximations for almost every realization, with the mean convergence required in item 2 for almost every realization. For every joint limit

$$(\Theta, \hat{\Phi}_n, T_n) \Longrightarrow (\Theta, \Phi_D^{\text{aux}}, T)$$

along the selected subsequence, the auxiliary field marginal satisfies  $(\Theta, \Phi_D^{\text{aux}}) \stackrel{d}{=} (\Theta, \Phi_D)$ , and

$$\mathbf{1}_{\{\overline{P} \subset D\}} (\mathbb{E}[T \mid \Theta, \Phi_D^{\text{aux}}] - k(\Phi_D^{\text{aux}}|_B)) = 0.$$

One subsequence works for any countable collection of tests, polygons, and thresholds. For the collar tests at  $t \in a\mathbb{Z}$ , write the selected kernels as  $k_t^\pm$ . In the restriction to  $B$  of every admissible zero-boundary field  $g^D$ ,

$$k_t^\pm(t + g^D|_B) = k_0^\pm(g^D|_B) \quad \text{almost surely.} \quad (8.5)$$

The same identity holds under the conditional zero-boundary law of a random selected domain. It is an identity under these Gaussian laws, with the countable thresholds fixed in advance.

*Proof.* For the reference field, take a joint subsequential limit of  $(\hat{g}_n^P, T_n)$ . The reverse buffer lemma makes its conditional mean a function  $k(g^P|_B)$ . We show that this selected function also governs the mixed expectations in other domains.

First translate the reference cable field by the linear cable interpolation  $u_n$  of a fixed  $u \in C_c^\infty(P)$ . Write  $X_n = (\tilde{g}_n^P, u_n)_{\nabla, n}$  for the Gaussian Cameron–Martin coordinate; this is the isonormal pairing, not a Dirichlet integral of a rough cable sample. The likelihood is

$$L_{n, u_n} = \exp \left\{ X_n - \frac{1}{2} \|u_n\|_{\nabla, n}^2 \right\}. \quad (8.6)$$

We need strong identification of  $X_n$ , since  $T_n$  is arbitrary. Let  $f = -\Delta u / (2\pi)$ , and let  $\hat{u}_n$  be the same harmonic face extension of  $u_n$ . The Gaussian reproducing identity and linearity of the extension give

$$\text{Var } X_n = \|u_n\|_{\nabla, n}^2, \quad \text{Cov}(X_n, \hat{g}_n^P(f)) = \int \hat{u}_n f, \quad \text{Var}(\hat{g}_n^P(f)) = \hat{G}_n^P(f, f).$$

The three expressions tend respectively to  $\|u\|_{\nabla}^2$ ,  $\int u f = \|u\|_{\nabla}^2$ , and  $G_P(f, f) = \|u\|_{\nabla}^2$ . Here the energy is a Riemann sum,  $\hat{u}_n \rightarrow u$  uniformly on compact subsets, and the last limit is Green convergence. Hence

$$\mathbb{E}|X_n - \hat{g}_n^P(-\Delta u / (2\pi))|^2 \longrightarrow 0. \quad (8.7)$$

The likelihood coordinate therefore converges jointly with every bounded  $T_n$ , not only marginally. For  $r > 1$ ,

$$\mathbb{E}[L_{n,u_n}^r] = \exp \left\{ \frac{1}{2} r(r-1) \|u_n\|_{\nabla,n}^2 \right\}. \quad (8.8)$$

These bounds give uniform integrability on bounded energy sets. Tilting the reference joint law by (8.6) is exactly the law of the translated total field and the test evaluated on that translated cable field. The limiting likelihood is a function of the continuum field. Such a tilt preserves its conditional kernel  $k$ , evaluated at the total restriction.

For two shifts in one reference law, the Gaussian relative entropy bound and Pinsker's inequality give

$$\|L_u - L_v\|_{L^1} \leq \min\{2, \|u - v\|_{\nabla}\}. \quad (8.9)$$

The same inequality holds on the cables. Finite smooth nets in the Dirichlet norm, together with (8.8), therefore make the translated mixed-expectation limit uniform on compact families of smooth shifts. This is equality of limiting measures tested first on bounded continuous field functions. It does not require evaluating a possibly discontinuous  $k$  on the discrete fields.

Now take a compared ambient graph containing the identical subdivided  $P_n$ . Split at its boundary cut points and condition on the outside cables and those boundary vertex values. The Markov decomposition on  $P_n$  is exactly

$$\tilde{\Phi}_n = \tilde{g}_n^P + H_n,$$

where  $\tilde{g}_n^P$  has the reference cable law and is independent of  $H_n$ . The mean  $H_n$  is discrete harmonic at the interior vertices and linear on inside cables. Choose a smooth cutoff  $\chi$  supported in  $P$  and equal to one near  $\overline{B}$ . At vertices set  $u_n = \chi H_n$  and interpolate linearly on cables. It equals  $H_n$  on all cables needed for the pairings and tests in  $B$ . The reference and ambient harmonic extensions also agree on those complete cells.

Interior discrete harmonic estimates and Green convergence make the values and scaled finite differences of  $H_n$  converge in law on the support of  $\chi$  to those of a smooth harmonic limit. They also give convergence of the cutoff energies. This is a statement about interior finite differences; the piecewise harmonic face extension need not have a global classical smooth norm. Uniformly over the domains, the covariance of  $H_n$  on a smaller buffer is bounded in quadratic-form order by the harmonic covariance obtained from the fixed outer box. Interior estimates consequently bound fixed smooth norms of the limiting harmonic parts and their discrete analogues in moments. Truncate such a bound, approximate the resulting family by a finite smooth net in a weaker norm controlling the cutoff energy, and apply (8.9). Removing the truncation proves the mixed-expectation limit on the restriction to  $B$ . A prescribed harmonic mean is treated by the same cutoff.

For completeness, the mean premise is automatic for the constant labels used below in the regular inner class. For a constant  $t$ , the full continuum ALS extension  $u_t$  equals  $t$  on  $\overline{D} \cap \mathcal{B}$ , is harmonic on  $\mathcal{B} \setminus \overline{D}$ , and has value zero on  $\partial\mathcal{B}$ . The prelimit mean equals  $t$  on  $\Gamma_n \cap D_n$ ; its ambient extension is  $t$  times the planar probability of hitting that cable domain before  $\partial\mathcal{B}$ . The maximum principle bounds its absolute value by  $|t|$ . For  $z \in \partial D \cap \mathcal{B}$ , fix a radius  $r > 0$  below its distance to the box boundary and to a fixed interior point. For every small  $\eta > 0$ , connectedness of  $D$  gives a compact path in  $D$  from  $B(z, \eta)$  to outside  $B(z, r)$ . Inner exhaustion supplies a connected cable path with the same property, up to a vanishing mesh error, for all sufficiently large  $n$ . The planar Beurling estimate [8, Lemma 4.1] bounds the probability of avoiding that path until leaving  $B(z, r)$  by  $C(\eta/r)^\beta$ . First let  $n$  tend to infinity and then  $\eta$  to zero. The extended mean

thus tends to  $t$  at every such boundary point. Away from  $\partial D$ , convergence is [8, Lemma 4.2(1)]. Bounded convergence now gives convergence to  $u_t$  in  $L^2(\mathcal{B})$ , hence in  $H^s(\mathcal{B})$ , even when  $\partial D$  has positive area. On the buffer cells the mean is exactly constant, so the interior finite-difference requirement also holds.

Thus the conditional kernel on that restriction is  $k$ . Applying [Theorem 8.1](#) in the actual ambient domain promotes it to conditioning on the full ambient field. The restricted Gaussian laws are mutually absolutely continuous: conditional on  $H$ , each is a Cameron–Martin translate of the reference restriction with a strictly positive likelihood, and integrating such translates preserves equivalence. The representative of  $k$  is therefore unambiguous under all these restrictions. The subsequence principle gives the mixed-expectation convergence in item 2 even if a general bounded test has several joint subsequential laws with the same conditional mean.

For item 3, apply that deterministic mixed-expectation conclusion for almost every fixed  $\Theta = \theta$ . The full-field convergence of the centered fields and the extended means identifies the limiting conditional field marginal with the selected parent GFF law. Integrating against bounded functions of  $\Theta$  gives  $(\Theta, \Phi_D^{\text{aux}}) \stackrel{d}{=} (\Theta, \Phi_D)$ . The cutoff comparison identifies the conditional mean given  $(\Theta, \Phi_D^{\text{aux}}|_B)$ ; then the reverse buffer argument, applied conditionally after centering the selected mean, identifies it given  $(\Theta, \Phi_D^{\text{aux}})$ . Integrate the bounded mixed expectations against bounded functions of  $\Theta$ , and use equality of measures and a monotone class. This proves the displayed conditional identity on the auxiliary limit; the displayed equality of marginals relates it to the separately given original field. A countable choice of polygons is handled by partitioning according to that choice. The reference subsequence is never reselected for a realized domain.

Finally use a diagonal extraction for the countable tests and thresholds. On  $C_{K,n}$  the exact identity  $T_{n,t}^\pm(\tilde{g}_n + t) = T_{n,0}^\pm(\tilde{g}_n)$  holds. The corresponding extensions differ by  $t$  on the complete cells near  $B$ , though they need not do so outside the selected domain. Apply the two selected kernel statements to these identical joint restricted laws. Their conditional means give (8.5). The same mixed-expectation argument conditional on  $\Theta$  gives its random-domain version.  $\square$

The cutoff estimate also gives the following fixed-buffer fact.

**Lemma 8.3** (Interior exhaustion). *Let  $D_r \uparrow D$  be regular inner domains and  $B \Subset D_1$ . The laws of the zero-boundary continuum GFFs restricted to  $B$  converge in total variation to the restriction of the GFF in  $D$ . This holds for simply connected  $D$  without a Jordan-boundary assumption.*

*Proof.* Choose a fixed polygon  $P$  with  $\overline{B} \subset P \Subset D_r$  for all large  $r$ . In the Markov coupling, the field in  $D$  restricted to  $D_r$  is an independent zero-boundary field in  $D_r$  plus a harmonic Gaussian  $H_r$ , with covariance  $G_D - G_{D_r}$ . Interior Green convergence and harmonic estimates make  $H_r \rightarrow 0$  in mean square in each fixed smooth norm near  $\overline{B}$ . Cut it off inside  $P$ . The cutoff Dirichlet norm tends to zero in mean, and (8.9), conditional on  $H_r$ , bounds the total variation distance by  $\frac{1}{2}\mathbb{E}[\min\{2, \|\chi H_r\|_\nabla\}]$ . This tends to zero.  $\square$

## 8.4 Calibration on killed lattice graphs

**Lemma 8.4** (The source killed graphs and the common kernel). *The conclusions of [Theorems 8.1](#) and [8.2](#), with the same reference kernel  $k$ , also hold for the following auxiliary approximations of a bounded simply connected domain  $D$ . Choose a fixed dyadic point  $r \in D$  and put*

$$V_n = \{v \in \varepsilon_n \mathbb{Z}^2 \cap D : \text{a lattice path from } v \text{ to } r \text{ stays at distance at least } \varepsilon_n \text{ from } D^c\}.$$

For sufficiently large  $n$  this is the approximation of [25, (2.6)], translated to  $r$ . Use the cable realization of the vertex GFF killed on leaving  $V_n$ : every edge incident to  $V_n$ , including a full exit edge ending at an outside Dirichlet vertex, has its original resistance and its usual conditional Brownian bridge. Complete this cable function to the full grid in  $\mathcal{B}$  by setting missing vertices and missing edges identically to zero, and harmonically extend it into each grid face. Denote this extension by  $\hat{X}_n$ . The missing edges have deterministic zero traces, not additional Brownian bridges.

For a simply connected regular polygon  $B \Subset D$ , use the connected, hole-free complete-cell approximations  $B_n^\square$  and reveal all of  $C_{B,n}$ , including its perimeter cables. Extend the residual, zero on  $C_{B,n}$ , by the same completed-grid operator. Its covariance form  $\mathsf{K}_n^B$  and the full covariance form satisfy, for smooth tests  $f, g$  on the box,

$$\mathrm{Cov}(\hat{X}_n(f), \hat{X}_n(g)) \longrightarrow G_D(f, g), \quad \mathsf{K}_n^B(f, g) \longrightarrow G_{D \setminus \bar{B}}(f, g), \quad (8.10)$$

where the limiting kernels are zero-extended to  $\mathcal{B}$ . The corresponding fields are tight in the fixed  $H^s(\mathcal{B})$ ,  $s < -1$ . For a constant parent label  $t$ , add directly to  $\hat{X}_n$  the chosen continuum ALS extension  $u_t$ : it equals  $t$  on  $\bar{D}$ , is harmonic on  $\mathcal{B} \setminus \bar{D}$ , and is zero on  $\partial\mathcal{B}$ .

*Proof.* This is a second, explicit planar embedding convention for auxiliary fields. It preserves the entire source cable field, including its exit-edge bridges. On every fixed compact subset of  $D$ , its face extension agrees with the earlier whole-cable convention for fine mesh. In particular every test on  $C_{K,n}$  is unchanged.

Let  $F_n$  be the set of vertices in  $C_{B,n}$ . The full vertex covariance is the killed random-walk Green kernel on  $V_n$ ; the residual vertex covariance after the revelation is the killed kernel on  $V_n \setminus F_n$ . Conditional on endpoint values, revealing the bridges in  $C_{B,n}$  supplies no further information about vertices outside it. All unrevealed edge bridges remain present. In particular an unrevealed edge with two pinned endpoints still contributes its independent zero-endpoint bridge.

The two killed vertex kernels converge off the diagonal to  $G_D$  and  $G_{D \setminus \bar{B}}$ , respectively. We recall the boundary argument to specify its applicability to these full exit stubs. The active vertex sets exhaust compact subsets of the indicated open sets. Stop the walk in a compact inner domain; the invariance principle identifies its limiting interior walk and harmonic functions. The discrete Beurling estimate then bounds the probability that a walk which has entered a thin neighborhood of a killing boundary returns a fixed distance into the domain before being killed. This bound tends to zero uniformly in fine mesh. The outer complement is connected and nondegenerate; clamping  $B_n^\square$  adds the disjoint nondegenerate polygonal obstacle. Apply the estimate at the component approached by the walk. This identifies killing at the limiting boundary, and the potential-kernel representation gives Green convergence. This is the Green-convergence proof of [8, Lemmas 4.1–4.2], applied to the killed vertex walk. Realizing its killing transition by a full cable stub does not change its vertex transition matrix.

All these kernels are bounded by the killed Green kernel of one fixed containing square. The square-lattice potential-kernel estimate gives

$$0 \leq G_n(v, w) \leq C \left( 1 + \log_+ \frac{R}{|v - w| \vee \varepsilon_n} \right), \quad (8.11)$$

with fixed  $R, C$ ; see also the potential-kernel references in [2, Appendix B]. If  $v_n$  tends to a boundary point and  $w_n$  to a distinct point, Beurling applied before reaching a fixed neighborhood of  $w_n$ , followed by (8.11), gives  $G_n(v_n, w_n) \rightarrow 0$ . Thus the piecewise constant vertex kernels converge off the diagonal to the zero-extended continuum kernel even when the outer boundary has

positive area. The logarithmic bound is integrable, and the integrated contribution of a shrinking neighborhood of the diagonal vanishes uniformly. Dominated convergence proves convergence of the paired vertex Green forms for smooth box tests.

We next retain, rather than discard, the cable bridges. Split the completed cable field into the piecewise linear interpolation of its vertex values and its centered edge bridges, and extend both parts harmonically into the square faces. The harmonic basis function of a vertex is nonnegative, supported on its four incident faces, and has integral  $\varepsilon_n^2$ . Hence its coefficient against  $f$  differs from  $\varepsilon_n^2 f(v)$  by  $O_f(\varepsilon_n^3)$ , and the total absolute coefficient error is  $O_f(\varepsilon_n)$ . The containing-square diagonal Green bound therefore bounds the variance of this error by  $O_f(\varepsilon_n^2 \log(1/\varepsilon_n))$ . A centered edge bridge contributes through at most two faces. Its paired variance is  $O_f(\varepsilon_n^4)$ , since its covariance is bounded at the fixed resistance normalization and those faces have area  $O(\varepsilon_n^2)$ . There are  $O(\varepsilon_n^{-2})$  independent retained bridges, independent also of the vertex field. Consequently, for the full field and for the clamped residual,

$$\mathbb{E} \left| \widehat{X}_n(f) - \varepsilon_n^2 \sum_v f(v) X_{n,v} \right|^2 \leq C_f \varepsilon_n^2 (1 + \log(1/\varepsilon_n)) \rightarrow 0. \quad (8.12)$$

For the residual use its pinned vertex values in this display. Unrevealed bridges between two pinned endpoints are included in the bridge estimate. Thus no perimeter cable has been deleted or replaced. The use of the completed grid makes this estimate valid for global box tests, including boundary-adjacent faces. Together with the vertex Green convergence it proves (8.10).

The nonnegative face basis also carries (8.11) to the planar covariance, with the logarithm truncated at mesh distance. The bridge covariance is bounded and supported on pairs of adjacent faces. Integrating against the logarithmic kernel for the  $H^{-1}$  norm gives a uniform bound on the expected squared  $H^{-1}$  norm. Compact inclusion into  $H^s$ ,  $s < -1$ , proves the asserted tightness.

Finally  $\overline{P} \subset D$  implies that a whole neighborhood of  $\overline{P}$  lies in  $V_n$  for all sufficiently fine mesh: its compact pieces have lattice connections to  $r$  with a fixed positive clearance. Subdivide the prescribed cables at  $\partial P_n^\diamond$ . The reference  $P_n$  is then literally the same graph, with the same inherited half-edge resistances and unpinned cut points in the ambient field. Conditioning on its exterior gives exactly the original reference residual.

After covariance convergence, the proof of [Theorem 8.1](#) uses Gaussian projection, the cable Markov decomposition, and measurability of compact interior face pairings. All these statements hold for the present extension, so that proof applies unchanged. The comparison proof of [Theorem 8.2](#) uses the same reference  $P_n$  and harmonic estimates on fixed interior buffers, where the operators and cables agree exactly. It therefore gives the same  $k$ , including the conditional-parent conclusion by integrating the bounded mixed expectations. For a constant label the added deterministic function  $u_t$  is bounded, agrees exactly at every mesh with the prescribed full continuum mean, and is identically  $t$  near the buffers. Thus it has the required full-field and interior mean convergence, without altering any local threshold test.  $\square$

## 8.5 Forcing in the canonical parent

We first establish the one-sided excursion approximation needed for calibration. All fields in this argument have the normalization (8.1). Fix a bounded simply connected domain  $D$  and a dyadic rational point  $z_0 \in D$ . For all sufficiently large  $n$ , let  $D_n^{\text{tr}}$  be the component of  $z_0$  in the  $\varepsilon_n$ -grid whose connecting paths stay at distance at least  $\varepsilon_n$  from  $\partial D$ , as in [25, (2.6)]. Use its killed cable graph, retaining full-resistance edges to outside Dirichlet vertices. These auxiliary graphs contain every fixed interior reference  $P_n$  for large  $n$ ; [Theorem 8.4](#) makes [Theorem 8.2](#) applicable to them.

**Lemma 8.5** (One-sided forcing by an outermost excursion). *Let  $\widehat{\Phi}_n$  be the completed-grid whole-cable extensions of zero-boundary GFFs on these graphs. Fix a collar  $Q \Subset V \Subset K \Subset B \Subset P \Subset D$ . In every joint subsequential limit*

$$(\widehat{\Phi}_n, T_{n,0}^+, T_{n,0}^-) \longrightarrow (\Phi, T^+, T^-),$$

*the following holds almost surely. If an outermost canonical positive excursion of  $\Phi$  meets both  $\text{int } Q$  and  $\text{int } K \setminus \overline{V}$ , then  $T^+ = 1$ . The analogous negative excursion forces  $T^- = 1$ .*

*Proof.* Fix one  $\gamma \in (0, 2)$ . Couple the cable GFF to a critical cable loop soup  $\widetilde{\mathcal{L}}_n$  by the signed isomorphism [28]. Write  $\mathcal{L}_n$  for the random-walk soup obtained by taking its vertex trace. Vertex occupation times are preserved by this trace construction; see also [8, Section 2.2]. Write  $\mathcal{M}_{\gamma,n}$  for the normalized unsigned thick-point measure at vertices and  $\mathcal{M}_{\gamma,n}^\pm$  for its restrictions to vertices with the indicated cable-field sign. Thus  $\mathcal{M}_{\gamma,n} = \mathcal{M}_{\gamma,n}^+ + \mathcal{M}_{\gamma,n}^-$ . We use exactly the source normalization [25, (2.5), (2.10), Theorem 2.1], whose limiting expectations are  $\mathbb{E}\mathcal{M}_\gamma(dz) = 2dz$  and  $\mathbb{E}\mathcal{M}_\gamma^+(dz) = dz$ . All measure limits below may be read in the local weak topology on  $D$ . The deterministic conformal-radius factor in this normalization is continuous, positive and bounded with bounded inverse on each compact subset of  $D$ , so that changing the GMC convention there causes no boundary reweighting issue.

The cable version of [25, Corollary 8.4 and the paragraph following it], in the vertex-trace coupling, gives

$$(\widetilde{\mathcal{L}}_n, \mathcal{M}_{\gamma,n}^+) \longrightarrow (\widetilde{\mathcal{L}}, \widetilde{\mathcal{M}}_\gamma^+), \quad (8.13)$$

where  $\widetilde{\mathcal{L}}$  is a Brownian loop soup and  $\widetilde{\mathcal{M}}_\gamma^+$  is its chaos restricted to independently positively marked clusters. Retain separately the projected random-walk soup coordinate  $\mathcal{L}_n$ , whose joint limits with unsigned and cluster-restricted chaos are given by [25, Theorem 8.3]. Call its limiting soup  $\mathcal{L}$ . We will not need to identify the two limiting time parametrizations.

Their multisets of individual loop ranges do agree. Indeed, drawing each vertex-trace jump along its edge changes the range of a cable loop visiting vertices by at most  $2\varepsilon_n$  in Hausdorff distance. Erased excursions stay on edges incident to visited vertices, and loops with no vertex have diameter at most  $\varepsilon_n$ . At every positive diameter cutoff which is not the diameter of a limiting loop, the limiting soups have finitely many loops. The finite point-process matching in the loop-soup topology, followed by the range map, therefore passes this trace matching to the limit; compare [8, Section 4.1.2]. Exhausting such cutoffs gives equal multisets of individual ranges, and hence identical cluster closures. This uses equality of the limiting range families, not continuity of a cluster-formation map.

The positive and negative thick-point calculations in [2, Section 9, proof of Theorem 1.5] also give convergence of each signed measure jointly with its underlying vertex GFF to the corresponding GMC of that same GFF. Since both limiting GMCs are measurable functions of the field, this also identifies their joint limit. In the source occupation-time normalization the vertex field is  $X_n(v) = \sigma_v \sqrt{4\pi\ell_v}$ , so its logarithmic covariance coefficient is one, as required by (8.1). The whole-cable estimate (8.12) identifies this vertex-field limit with the completed-grid extension in Theorem 8.4.

In particular, every joint limit has  $\mathcal{M}_\gamma^\pm = \text{GMC}_{\pm\gamma}(\Phi)$ , where the GMC notation includes the stated deterministic normalization. The reconstruction theorem [12, Theorem 1.1] supplies a measurable map  $R_\gamma$  such that

$$\Phi = R_\gamma(\mathcal{M}_\gamma^+) \quad \text{almost surely.}$$

The deterministic conformal-radius factor between the GMC conventions is included in  $R_\gamma$ . Thus (8.13) and the same-field GMC limit identify the entire joint law  $(\Phi, \tilde{\mathcal{L}}, \mathcal{M}_\gamma^+)$ . This step uses measurability after passage to the limit, not continuity of the reconstruction map.

There is also one common unsigned measure in the two soup coordinates. The exact prelimit identity and the two same-field limits give

$$\mathcal{M}_\gamma = \text{GMC}_\gamma(\Phi) + \text{GMC}_{-\gamma}(\Phi).$$

In the just identified cable-soup coupling this is its unsigned chaos  $\tilde{\mathcal{M}}_\gamma$ , by [25, Theorem 1.4]; in the projected-soup coordinate it is the unsigned chaos supplied by [25, Theorem 8.3]. Thus the two limiting soups have both the same cluster geometry and the same unsigned measure. Equality of the latter follows from their common vertex observable and the two GMC identities, without assuming that chaos is determined by unparametrized ranges.

We next identify the cluster marks in this joint law as the canonical excursion signs. The continuum results [9, Theorems 1 and 2] describe the joint law of the closed cluster supports, their Minkowski measures, their independent symmetric signs, and their signed sum. The Minkowski construction in [25, Theorems 1.9 and 1.10] normalizes the same Euclidean neighborhood measures. Since both normalizations have finite nonzero limits, their deterministic ratio converges to one positive constant, common to every cluster. For example, compare the total masses for one cluster: the ratio can have neither zero nor infinite subsequential limits, and two distinct finite limits would give two different multiples of a nonzero limiting measure. The signed sums therefore differ by that same constant; their GFF covariance normalizations fix it. The order of summation can be changed to a common sign-independent enumeration: Hilbert-space  $L^2$  convergence and independent-sign orthogonality give  $\mathbb{E} \sum_i \|\mu_i\|_{H^{-q}}^2 < \infty$  for a sufficiently large  $q > 1$ , hence conditional square summability and unconditional convergence of the signed series. Consequently the joint law just identified has precisely the canonical excursion supports and signs of  $\Phi$ . Only the continuum parts of [9] are used here.

For the fixed  $\gamma$ , every soup cluster  $C$  selected by a rational-point label satisfies

$$\mathcal{M}_\gamma(C) > 0 \quad \text{almost surely.} \tag{8.14}$$

Here is a useful argument that does not presume positivity of a restriction to a random thin set. Flip only the independent sign of  $C$ , keeping the soup and all other signs fixed. This transformation preserves the signed-soup law. Hence the original and flipped samples both satisfy the same reconstruction identity with  $R_\gamma$ , on the intersection of two full-probability events. On  $\{\mathcal{M}_\gamma(C) = 0\}$  the positive-chaos measure is unchanged, so the reconstructed field is unchanged. Its signed-Minkowski representation, however, changes by a nonzero multiple of the nonzero positive measure of  $C$ , by [25, Theorems 1.4, 1.9 and 1.10]. This contradiction proves (8.14).

Fix a rational  $x \in D$ , and let  $C_n(x)$  be the outermost ordinary random-walk loop cluster surrounding  $x$ , with projected loops drawn along the grid edges. Its trace is contained in a cable sign cluster; call that sign  $\sigma_n(x)$ . If no such cluster exists, use the empty set and assign sign  $+1$ ; its restricted measure below is then zero. When nonempty, it is a connected one-sign path witness. Retain also

$$\mu_n(x) = \mathcal{M}_{\gamma,n}|_{C_n(x)}.$$

The printed random-walk statement [25, Theorem 8.3] gives, jointly with the soup,

$$C_n(x) \longrightarrow C(x), \quad \mu_n(x) \longrightarrow \mathcal{M}_\gamma|_{C(x)},$$

in the Hausdorff and weak-measure topologies, respectively. Take an arbitrary further joint limit including  $\sigma_n(x)$ . At every mesh,

$$\begin{aligned}\sigma_n(x) = +1 &\implies \mathcal{M}_{\gamma,n}^+ \geq \mu_n(x), \\ \sigma_n(x) = -1 &\implies \mathcal{M}_{\gamma,n} - \mathcal{M}_{\gamma,n}^+ \geq \mu_n(x).\end{aligned}$$

These inequalities pass to the limit because the positive-measure cone is closed. In the identified continuum coupling, the positive chaos restricted to  $C(x)$  is either its entire unsigned chaos or zero, according to the canonical sign  $\tau(x)$ . By (8.14), the limiting inherited sign must therefore equal  $\tau(x)$ . No independence of the signs of different random-walk clusters is asserted: they may share a cable cluster.

The rational labels cover every outermost continuum cluster. For each fixed label, strict hits in the two open target sets persist under Hausdorff convergence, and the inherited sign has just been identified. A path in the projected cluster from its inner hit to its outer hit is a cable sign path. Since  $\bar{V} \subset \text{int } K$ , an initial portion reaches a point outside  $\bar{V}$  before any exit from  $K$ . This portion forces the corresponding test. Applying Skorokhod representation to any further joint subsequence proves the assertion, first for each rational label and then simultaneously for all labels by countability.  $\square$

We now apply this deterministic-domain calibration to the canonical parent before revealing its remainder field. Let  $\Theta$  be the local-set data of the canonical ancestors, let  $\Phi_D$  be the total field in the selected parent, and fix a collar with  $\bar{P} \subset D$ . Conditionally on  $\Theta$ , choose the killed auxiliary graphs above for excursion calibration and subtract the constant parent label. A retained cluster is treated in its own recursive parent, where its containing excursion is outermost. Theorems 8.4 and 8.5 and the threshold identity in Theorem 8.2 apply along the already selected mesh subsequence. For an FPS, use instead the admissible metric-domain approximations and [8, Proposition 4.7]. Its inner hit is connected to the artificial parent boundary by the prescribed one-sign path. Since the parent contains  $P_n$ , this path exits  $V$  while still inside  $K$ . Thus in either case the canonical strict-hit event forces  $T = 1$  in every auxiliary joint limit.

This conditional calibration does not require a measurable family of Skorokhod representations over all possible parents. For each fixed parent, augment an arbitrary subsequence as in the proof above. The continuous hit-depth function

$$J_U(C) = \sup_{x \in C} \min\{1, \text{dist}(x, U^c)\}, \quad J_U(\emptyset) = 0,$$

shows that the product of the two required hit depths with  $1 - T$  vanishes in every limit. The resulting bounded mixed-expectation conclusion holds along the selected subsequence. Apply it pointwise to the conditional auxiliary kernels for  $\Theta = \theta$ , and integrate by dominated convergence. Together with item 3 of Theorem 8.2, this gives

$$\mathbb{E}[T \mid \Theta, \Phi_D^{\text{aux}}] = k(\Phi_D^{\text{aux}}|_B).$$

If  $E^{\text{aux}}$  is the corresponding canonical strict-hit event, then it belongs to  $\sigma(\Theta, \Phi_D^{\text{aux}})$ , and  $T = 1$  on that event. Positivity gives  $\mathbf{1}_{E^{\text{aux}}}(1 - k(\Phi_D^{\text{aux}}|_B)) = 0$ . Equality of the  $(\Theta, \Phi_D^{\text{aux}})$  and  $(\Theta, \Phi_D)$  marginals transfers this identity to the original parent:

$$\mathbf{1}_E(1 - k(\Phi_D|_B)) = 0 \quad \text{almost surely.} \quad (8.15)$$

Canonical ancestor data are functions of the original field, so this identity remains true in any extension carrying the stopped exploration. The later upper bound conditions only on that stop; it does not assert a formula for  $\mathbb{E}[T \mid \Phi_D, \text{stop}]$ .

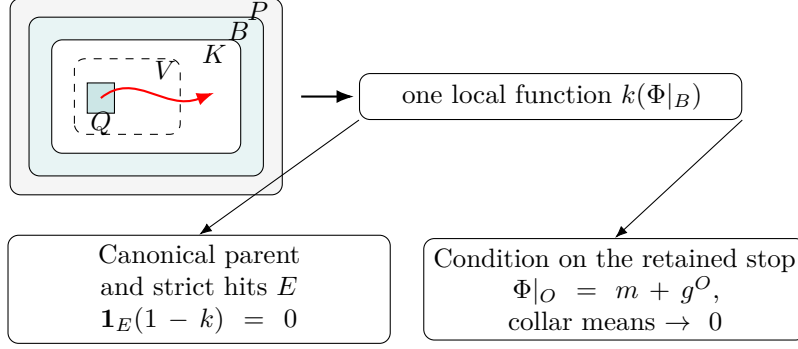


Figure 4: The path test crosses  $V$  inside  $K$ , and the separated buffers  $B$  and  $P$  make its kernel common to different Gaussian domains. Calibration in the canonical parent gives an identity on the field space. The upper bound uses the conditional GFF law of the stopped component.

## 8.6 The zero-boundary collar bound

Fix a bounded simply connected  $O$ , a compact rational polygon  $Q \Subset O$ , and a regular inner exhaustion  $Q \Subset O_1 \Subset O_2 \Subset \dots \uparrow O$ . For the  $j$ -th collar take

$$V = O_j, \quad K = O_{j+2}, \quad B = O_{j+3}, \quad P = O_{j+4}.$$

Use regular inner cable approximations of  $O$  on the common grid. They have complement-Hausdorff convergence, uniformly bounded connectivity, and contain every fixed  $P_n$  for fine mesh. Write  $T_{j,n}^\pm$  for the zero-threshold tests. We claim

$$\lim_{j \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(T_{j,n}^\pm = 1) = 0. \quad (8.16)$$

We prove this estimate directly on the auxiliary metric graphs. Choose the inner approximating domains  $D_n \subset O$  to be simply connected unions of dual mesh cells, with their boundary cable intersection points adjoined as Dirichlet endpoints, as for the reference polygons above. The approximations exhaust compact subsets of  $O$  and eventually contain every prescribed reference subgraph. Such a choice follows by first exhausting the simply connected  $O$  by regular simply connected polygons and then taking sufficiently fine inner cell approximations, along a diagonal sequence. Their complements converge in Hausdorff distance, their connectivity is one, and they converge in the pointed Carathéodory sense. Write  $O_n = \Gamma_n \cap D_n$  for the resulting zero-boundary metric domains. All distances below are ambient Euclidean distances. In the metric-graph coupling [8, Section 2.3 and Proposition 2.4, with zero boundary data], the nonzero sign components of the cable GFF are the clusters of a metric-graph Brownian loop soup of intensity  $1/2$ . The constant field rescaling used for our normalization does not change these components.

Set

$$d = \frac{1}{2} \text{dist}(Q, \partial O) > 0, \quad \eta_j = \sup_{z \in O \setminus O_j} \text{dist}(z, \partial O).$$

Because  $O$  is bounded and the nested  $O_j$  exhaust every compact subset of  $O$ , we have  $\eta_j \rightarrow 0$ . A zero-threshold crossing has a supporting sign component whose closure meets  $Q$  and  $O \setminus O_j$ . Indeed, after erasing loops, the path uses finitely many cable intervals. Interior vertex values are nonzero almost surely, and a Brownian bridge almost surely has no interior local minimum or

maximum at the deterministic level zero. Thus allowing zero endpoints does not concatenate distinct sign components. In the loop-soup coupling, it supplies a cluster  $C$  with

$$\text{dist}(C, \partial O_n) \leq 3\eta_j, \quad \sup_{z \in C} \text{dist}(z, \partial O) \geq d.$$

Here distances to a cluster may equivalently be taken to its closure. For each fixed  $j$ , these bounds hold for all sufficiently fine meshes. The first uses the inner approximation property: distance to the Euclidean boundary of  $D_n$  is at most distance to  $\partial O$ , and every point of  $\partial D_n$  is within  $\varepsilon_n/2$  of a metric boundary cut point. The extra margin also allows replacing an endpoint in the closure by a point of the cluster. The second bound follows from the hit on  $Q$ .

The boundary-avoidance estimate [8, Lemma 4.13], at intensity  $1/2$ , applies to the chosen complement-Hausdorff approximations, which also converge in the pointed Carathéodory sense and have the required bounded connectivity. For the fixed  $d$ , it bounds the probability of the last displayed event by a quantity tending to zero as  $\eta_j \rightarrow 0$ , uniformly in the mesh. Therefore

$$\lim_{j \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(T_{j,n}^\pm = 1) = 0.$$

This proof uses metric loop-soup boundary avoidance, without requiring convergence of diameter-ranked excursion clusters.

Let  $k_{j,0}^\pm$  be the common kernels selected for these countably many collar choices. For fixed  $j$ , the mixed-expectation conclusion of [Theorem 8.2](#) in  $O$  identifies its integral with the mesh limit of the bounded test. Thus

$$\lim_{j \rightarrow \infty} \mathbb{E}[k_{j,0}^\pm(g^O|_{B_j})] = 0, \quad g^O \text{ a zero-boundary GFF in } O. \quad (8.17)$$

This proof applies directly to rough  $O$ . The total variation statement in [Theorem 8.3](#) concerns a fixed buffer and supplies no estimate with a collar moving toward the boundary.

## 8.7 No crossing of a stopped component

**Proposition 8.6** (No crossing). *Let  $O$  be a component of a stopped exploration from [Theorem 7.4](#), with retained sigma-field  $\mathcal{E}$ . Suppose  $O$  lies in an initial free parent  $P_0$  of label  $b_0 \in a\mathbb{Z}$ , or in one pointed wired parent  $P_0$  of label  $b_0 = \pm a$ . As provided there,  $O$  is simply connected and, conditionally on  $\mathcal{E}$ ,*

$$h|_O = m + g^O, \quad m - b_0 \in 2a\mathbb{Z}.$$

*No canonical free cluster descended from that parent meets both  $O$  and  $\partial O$ . The conclusion holds simultaneously for all pointed components and all recursive generations, and after a restart in a captured odd layer with its new incoming label.*

*In the wired case  $O$  may be a residual component of the global composite exploration of the original  $h$ . The proof uses  $O \subset P_0$ , simple connectedness, and the displayed conditional law. It requires no connectedness of the explored set relative to that parent and no separately selected lattice sweep inside a limiting wired hole.*

*Proof.* First work in an indexed canonical parent  $D_{\text{par}}$  whose total field is  $b + f$ . For a positive first-generation cluster  $C$ , [Theorem 7.6](#) supplies a containing positive excursion at total level  $b$  and a containing upper FPS with sublevel paths at total level  $b + 2a$ . For a negative cluster the containing paths are below  $b$  and above  $b - 2a$ . The parent is the initial domain or a canonical

Jordan hole; its auxiliary approximations are the source killed graphs for the excursion test and the regular metric domains for the FPS test.

Suppose  $C$  meets both  $O$  and  $\partial O$ , and choose a rational  $Q \Subset O$  whose interior it meets. In a regular inner exhaustion as above, connectedness of  $C$  makes it meet  $\partial O_{j+1}$  for every large  $j$ . This gives the two strict neighborhoods for the excursion test: the second lies in  $O_{j+2} \setminus \overline{O}_j$ . The FPS test needs only the strict inner hit, because its boundary-connected path must exit the fixed collar. On the relevant-parent event  $O \subset D_{\text{par}}$ , all buffers are contained in this canonical parent.

The stopped label and this parent label belong to the same affine lattice  $b_0 + 2a\mathbb{Z}$ . For a positive cluster, either  $m \leq b$ , so the excursion forces the above- $m$  test, or  $m \geq b + 2a$ , so the upper FPS forces the below- $m$  test. For a negative cluster, either  $m \geq b$ , using the below- $m$  excursion, or  $m \leq b - 2a$ , using the above- $m$  FPS. These alternatives exhaust the lattice. The absolute threshold  $m$  belongs to  $a\mathbb{Z}$ .

Calibrate both tests at every fixed threshold in  $a\mathbb{Z}$  in the canonical parent, conditional on its ancestor data. For the excursion, use [Theorem 8.5](#) for each rational-point label and take the countable union; for the FPS use the whole-set identity. The path implications and (8.15) then hold simultaneously for the countable choices of parent index, rational-point label, threshold, and collar. They are identities on the original field space and survive adjoining the stop. To state the bound globally, fix an indexed candidate  $C$  and a rational  $Q$ , and select the collar from the stopped state by the rule described below. Let  $H_j \in \mathcal{E}$  record that the selected pointed component and its  $j$ th collar are defined with the required clearances inside  $O$ . Let  $L_j$  record that the indexed candidate and its parent  $D_{\text{par}}$  exist and that  $\overline{P}_j \subset D_{\text{par}}$ . This latter event may depend on canonical data. Declare it false when  $H_j$  fails, use the empty set for an absent candidate or component, and put

$$C_Q = \{C \cap \text{int } Q \neq \emptyset, C \cap \partial O \neq \emptyset\}.$$

The right side below is defined to be zero outside  $H_j$ . Substituting the stopped-measurable  $m$  in the simultaneous forcing identities gives the global pointwise bound

$$\mathbf{1}_{H_j \cap L_j \cap C_Q} \leq \mathbf{1}_{H_j} (k_{j,m}^+(h|_{B_j}) + k_{j,m}^-(h|_{B_j})). \quad (8.18)$$

The direction selected by the unknown parent label  $b$  has disappeared from this inequality. In particular  $b$  need not be measurable in  $\mathcal{E}$ .

Condition now only on  $\mathcal{E}$ . The event  $H_j$ , the collar, and  $m$  are stopped-measurable. The component law gives

$$\begin{aligned} \Pr(H_j \cap L_j \cap C_Q \mid \mathcal{E}) &\leq \mathbf{1}_{H_j} \mathbb{E} \left[ k_{j,m}^+(m + g^O|_{B_j}) + k_{j,m}^-(m + g^O|_{B_j}) \mid \mathcal{E} \right] \\ &= \mathbf{1}_{H_j} \mathbb{E} \left[ k_{j,0}^+(g^O|_{B_j}) + k_{j,0}^-(g^O|_{B_j}) \mid \mathcal{E} \right], \end{aligned}$$

where (8.5) is applied under that conditional Gaussian law. For each realized  $O$ , (8.17) makes the right side tend to zero as  $j \rightarrow \infty$ . On the branch where this parent is relevant, a crossing with witness  $Q$  belongs to  $H_j \cap L_j$  for every sufficiently large  $j$ . Conditional Fatou applied to the left indicators therefore gives zero conditional probability for that relevant crossing. No conditioning on  $L_j$  is used.

All random choices can be made within the countable families already selected. An open hit has a rational polygon  $Q$  as witness. The closed-set coordinate of the stop determines whether a rational polygon has positive clearance inside  $O$ . Choose successively the first regular rational polygon containing the preceding one and the next finite list of rational compact buffers. Simple

connectedness ensures such a nested exhaustion exists, and the clearance tests make the choice  $\mathcal{E}$ -measurable. Partition by each finite collar and threshold choice before conditioning. The random-domain clause of [Theorem 8.2](#) and bounded convergence justify the displayed conditional integrals. There are countably many components and canonical clusters.

It remains to justify that the same argument reaches every recursive parent. Assume it for the ancestors of a candidate cluster. If an ancestor lies wholly in  $O$ , its filled disk lies in  $O$ : its outer Jordan loop is inside the simply connected  $O$ , so the connected complement of  $O$  cannot enter that loop without crossing it. All descendants then stay in  $O$ . Otherwise, an ancestor that does not cross  $O$  and does not lie inside it is disjoint from  $O$ . Connectedness puts  $O$  in one complementary canonical hole, unless the candidate descendant is outside  $O$  already. In the relevant hole the incoming label is some  $b \in b_0 + 2a\mathbb{Z}$ . The same absolute threshold argument applies there. On this branch  $O \subset D_{\text{par}}$ , so  $L_j$  holds for every sufficiently large collar and the global bound applies. Calibration conditions on that parent's canonical data, while the upper bound continues to condition only on the original retained stop. Induction proves the claim.

For a global composite wired stop,  $A$  contains the wired layer, so  $O$  lies in one pointed wired hole. Its label satisfies  $m - b_0 \in 2a\mathbb{Z}$  with  $b_0 = \pm a$ , and all canonical descendants have baselines in the same affine lattice. The preceding proof applies in the original field using those absolute labels. No further exploration is introduced inside the selected limiting hole.  $\square$

## 9 Identification of the complete cluster collections

The stopped carpets determine the odd holes and their relative height jumps. They do not determine which holes belong to one current cluster. We identify that partition first. Finite families of visible holes then transfer the lattice count bounds to all generations of the canonical construction. The remaining task is to recover every point of a cluster, including points that are not detected by an odd hole.

### 9.1 Marked odd holes and their supplying clusters

Fix a countable dense set  $\mathcal{Q} \subset D$ . For each  $q \in \mathcal{Q}$  choose a nearest lattice representative on the relevant  $W$  lattice, using one deterministic rule to break ties. We record odd holes by a family label  $\iota \in \{f, w\}$ , a mark  $q$ , and an odd depth  $r \geq 1$ . The label  $f$  refers to the original free dual current;  $w$  refers only to the free primal descendants after the wired boundary cluster has been exposed.

At depth zero,  $V_{\delta,0}^f(q)$  is the original free chamber, with wall value  $b_{\delta,0}^f(q) = 0$ . For  $\iota = w$ , it is the component face left by the wired layer that contains the representative of  $q$ , with its observed value  $b_{\delta,0}^w(q) \in \{-a, a\}$ . Given a valid  $V_{\delta,r-1}^\iota(q)$ , expose its first odd carpet and let  $V_{\delta,r}^\iota(q)$  be the unsearched hole containing the representative, with observed wall value  $b_{\delta,r}^\iota(q)$ . Write  $C_{\delta,r}^\iota(q)$  for the current cluster supplying that odd rim, and put

$$\epsilon_{\delta,r}^\iota(q) = \frac{b_{\delta,r}^\iota(q) - b_{\delta,r-1}^\iota(q)}{2a} \in \{-1, 1\}.$$

If the representative is outside the chamber, lies on the revealed carpet, or no such hole remains, give this observation and all its descendants the cemetery value  $\dagger$ , and set its sign to zero. Validity, odd depth, and the supplying cluster depend only on the unsigned data  $(N, s)$ , not on the free transverse colors  $d$ .

The continuum observations  $V_r^\iota(q)$ ,  $b_r^\iota(q)$ , and  $c_r^\iota(q)$  are defined in the same way from the canonical odd carpets of [Theorem 7.6](#). At depth zero the free parent is  $D$ , while the wired parent is the component of  $D \setminus C_\partial$  containing  $q$ , with its label  $\pm a$ . Use the same cemetery convention if a mark lies on a continuum carpet. For a valid observation,  $C_r^\iota(q)$  denotes the retained canonical cluster whose odd hole is  $V_r^\iota(q)$ . The one-arm estimates in the proof of [Theorem 7.1](#) make every fixed mark miss the limiting carpets almost surely. Every canonical odd hole contains a mark, so these countably many observations see all odd holes at every finite depth.

Here and below convergence of a marked hole is *pointed*: the closed carpet and its complement determine the limiting component containing the mark, and every compact subset of that component is eventually inside the discrete component. We do not assert Hausdorff convergence of the closures of individual holes. In particular, one discrete component may split into several pointed limiting components.

Abbreviate an address  $(\iota, q, r)$  by  $i$ , and write  $V_\delta(i)$ ,  $C_\delta(i)$ , and  $\epsilon_\delta(i)$ , with the analogous notation in the limit. For two addresses in the same family, define  $i \sim_\delta j$  when both are valid and  $C_\delta(i) = C_\delta(j)$ ; otherwise the relation is false. Define  $i \sim j$  by the same rule for the supplying canonical clusters, with a false relation whenever either address is invalid.

Fix any mesh subsequence. The whole-domain bound [Theorem 3.6](#) and [Theorem 2.2](#) give compact containment of the two encoded collection factors  $F(\mathcal{C}_\delta^f)$  and  $F(\mathcal{C}_\delta^w)$  in the locally finite part of  $\mathfrak{X}$ , the compact hyperspace over the fixed enclosing square  $X$ . The height is tight by [Theorem 6.1](#). We pass to a joint further subsequence retaining the closed carpets, their marked pointed components and labels, the marked supplier supports in  $\mathcal{H} \cup \{\dagger\}$  with an isolated cemetery value, and the bits  $i \sim_\delta j$  for every fixed pair of addresses. These supplier coordinates are compact; they are separate from the contributor lists introduced later. Write  $F_*^\iota$  for the locally finite outer Hausdorff limits of the encoded collections. The relation coordinates are finite-valued; on a joint representation they are eventually constant, with limiting relation denoted by  $\sim_*$ . This retains the finite identity relations needed below, not numerical cluster identifiers across meshes. By [Theorem 7.3](#) and the one-arm estimate, for every finite family of addresses the validity indicators and the signs in  $\{-1, 0, 1\}$  converge jointly to their canonical counterparts. In particular all bounded products of these signs converge in expectation.

The use of  $X$  is necessary because an admissible polygon may protrude outside  $D$ . For each compact set disjoint from  $\partial D$ , uniform convergence of the boundary parametrizations preserves its winding number for all sufficiently small mesh. Hence the polygonal closures eventually lie in every fixed neighborhood of  $\overline{D}$ , and every Hausdorff limit of lattice supports lies in  $\overline{D}$ .

We will use one elementary property of pointed convergence. If a sealed discrete hole has pointed limit  $V$ , every point of  $\partial V$  is approached by its sealing boundary. Otherwise a ball around such a point would avoid the discrete boundaries along a subsequence. Since it meets a compact subset of  $V$ , that connected ball would then lie in the discrete hole, enlarging its pointed limit across the boundary. The sealing boundary lies within  $O(\delta)$  of its supplying current cluster. Consequently any Hausdorff limit of those suppliers contains  $\partial V$ ; if the supplier has been revealed at a stop, the limiting explored set contains  $\partial V$  as well.

**Lemma 9.1** (The restricted field determines its flat label). *Let  $O$  be a random bounded simply connected domain with a marked point  $q \in O$ , and let  $f$  be a distribution on  $O$ . Suppose  $O, q, m$  are measurable in a sigma-field  $\mathcal{E}$  and, conditionally on  $\mathcal{E}$ , one has  $f = m + h^O$ , where  $m \in \mathbb{R}$  and  $h^O$  is a zero-boundary GFF. There is a measurable function of  $(O, q, f)$  alone that equals  $m$  almost surely. Thus two such conditional descriptions of the same pointed domain and restricted field have the same label.*

*Proof.* Let  $\phi_O : \mathbb{D} \rightarrow O$  be normalized by  $\phi_O(0) = q$  and  $\phi'_O(0) > 0$ . Rational polygonal inner exhaustions, normal-family compactness, and uniqueness of the normalized limit make  $\phi_O$  measurable in  $(O, q)$ . Push uniform angular probability measure  $d\vartheta/(2\pi)$  on  $|z| = r$  forward by  $\phi_O$ . Conformal invariance of the Green kernel gives covariance  $-\log \max(r, s)$  for the zero-boundary averages at radii  $r, s$ . The radius- $r$  average of  $f$  thus has mean  $m$  and variance  $-\log r$ .

For  $r_j = 1 - 2^{-j}$ , approximate the angular measure by a smooth radial probability test  $\nu_j$  compactly supported in the disk, with squared Green-energy error at most  $2^{-j}$ . Such choices are deterministic: the covariance  $-\log \max(r, s)$  is continuous near each positive  $r_j$ , so radial smoothing converges in energy. Push  $\nu_j$  forward by  $\phi_O$  to obtain a smooth compactly supported test  $\psi_j(O, q)$ . This test is measurable in the pointed domain, and evaluation  $f(\psi_j(O, q))$  is measurable in  $(O, q, f)$ . Its conditional mean is  $m$ , and its conditional variance is at most

$$2(-\log r_j) + 2^{1-j}.$$

The sum of these variances is finite. Chebyshev's inequality and Borel–Cantelli therefore give  $f(\psi_j(O, q)) \rightarrow m$  almost surely. Define the function in the statement by this limit when it exists, and by zero otherwise. The construction uses conformal maps only on compact subsets of the disk, so no boundary extension of  $O$  is needed. The conformal identities are recalled in [1, Chapter 6, Sections 1.1 and 5.2].  $\square$

**Proposition 9.2** (The visible partition). *Almost surely,  $i \sim_* j$  if and only if  $i \sim j$ , simultaneously for every pair of marked addresses in either current family. Their relative signs also agree. Each retained canonical cluster  $C$  has a marked address  $i$  such that every Hausdorff limit of  $C_\delta(i)$  contains  $C$ . For every finite family of distinct canonical clusters, their chosen supplying lattice clusters are eventually distinct.*

*Proof.* We first show  $i \sim j \Rightarrow i \sim_* j$ . All odd holes of one retained canonical cluster have the same odd depth  $r$  and the same parent at depth  $r - 1$ : entering any odd hole increases the depth, and later clusters lie inside it. Let  $V_1, V_2$  be two marked odd holes of such a cluster  $C$ , and suppose that their lattice suppliers are distinct along the retained subsequence. If  $V_1 = V_2$ , a compact path inside that hole already puts both marks in the same discrete hole for all sufficiently small mesh. We may therefore assume that  $V_1$  and  $V_2$  are distinct.

Use the following fixed connected search. Starting from the exterior, repeatedly add the lexicographically first unvisited current-lattice vertex adjacent in the underlying graph to the explored region, and reveal its entire current cluster. For the wired family start from the completely exposed wired layer; the same rule interleaves its residual chambers. The explored union remains connected to the original boundary. For the two fixed marks count the surrounding odd rims already revealed, and stop when either counter first reaches  $r$ . This rule uses only the revealed history; it never asks for the identity of a future hole. If neither counter reaches  $r$ , give the stop a cemetery value; on the event under consideration it is eventually active. Adjoin the retained stopped states for all these countably many fixed rules to the joint subsequence. By Theorem 7.4, each has the stated component laws conditional on its own limiting stopped state. The future supplier relation is not included in that state.

The two marks lie in one pointed parent at depth  $r - 1$ . A compact path inside that parent joining them is eventually inside the same discrete parent, by pointed convergence. This implication does not assert the converse through a pinch. The exterior seed must cross all of their shared enclosing odd rims before it can reach either  $r$ th rim. Suppose the stop discovers the rim of  $V_1$  first. The other mark has then seen exactly the common  $r - 1$  odd jumps; any intervening even holes contribute zero. Its stopped value is the common parent label  $b$ .

The newly revealed supplier and its sealing boundary put  $\partial V_1$  in the limiting explored set  $A$ . Hence  $C$  meets  $A$ . If a connected cluster meeting  $A$  entered any component of the complement, it would also meet that component's boundary. [Theorem 8.6](#), applied at the appropriate canonical generation and after subtracting the parent label, forbids this. Therefore  $C \subset A$ , in particular  $\partial V_2 \subset A$ .

The entire discrete interior of the second odd hole is still unsearched. A seed vertex could enter it only after meeting its enclosing supplier, which would have raised the second counter to  $r$ . A different current cluster inside that hole is trapped there by the supplier and its sealing rim, so it could have been discovered only from a seed vertex in the same interior. Thus the whole explored set, not only the seed, misses the second discrete hole. Every compact subset of  $V_2$ , with a slightly larger compact neighborhood, eventually lies in that untouched hole. It follows that  $A$  misses  $V_2$ . Since  $\partial V_2 \subset A$ , the domain  $V_2$  is itself a stopped complementary component.

Its stopped conditional law has flat label  $b$ . The labelled odd carpet gives the same  $V_2$  the label  $b + 2a\epsilon(i)$ , which differs from  $b$ . Both are conditional descriptions of the same restricted field on the same pointed domain, so [Theorem 9.1](#) makes this impossible. The conditional laws were used at the fixed observable stop, not after conditioning on the hypothesis of distinct suppliers. This proves the inclusion at every finite odd depth, including clusters reached after any number of even restarts.

To exclude extra identifications, use the relative signs. Conditional on the unsigned lattice data  $(N, s)$ , the signs of different free current clusters are independent and fair, and one cluster has the same sign at all its odd holes. The validity of a marked address is also unsigned. With our zero sign and false relation at  $\dagger$ , [Theorem 3.1](#) therefore gives, for every fixed pair of addresses in the same current family,

$$\mathbb{E}[\epsilon_\delta(i)\epsilon_\delta(j)] = \mathbb{P}(i \sim_\delta j). \quad (9.1)$$

The canonical fair-sign statement in [Theorem 7.6](#) gives the same identity with  $\epsilon$  and  $\sim$  for each fixed pair in that same family. Joint labelled-carpet convergence passes the bounded sign product to the limit, while the retained relation bits pass the right side to  $\mathbb{P}(i \sim_* j)$ . Hence

$$\mathbb{P}(i \sim_* j) = \mathbb{E}[\epsilon(i)\epsilon(j)] = \mathbb{P}(i \sim j).$$

The inclusion already proved makes  $\mathbf{1}_{\{i \sim_* j\}} - \mathbf{1}_{\{i \sim j\}}$  nonnegative. Its expectation is zero, so it vanishes almost surely. This argument uses global fixed addresses and does not require equality of discrete parent components to be continuous through a pinch. The countable intersection over all pairs proves the asserted partition equality; the relative signs are already the limits of the marked labels.

Finally choose, for each canonical cluster, one address  $i$  supplying one of its odd holes. Every other marked odd hole of that cluster has the same lattice supplier as  $i$  for all sufficiently small mesh. The pointed-boundary observation above therefore puts all its odd-hole boundaries in every Hausdorff limit of  $C_\delta(i)$ . Their union is dense in the canonical cluster by [Theorem 7.6](#). This proves containment. For finitely many distinct canonical clusters, the corresponding relation bits are eventually false, so their suppliers are eventually distinct.  $\square$

Fix an enumeration of the marked addresses. For each canonical cluster, choose its first supplying address in this enumeration and call the associated sequence  $C_\delta(i)$  its *visible precursor*. This is auxiliary information retained for the proof; the matching topology itself forgets multiplicity.

**Corollary 9.3** (All-generation local finiteness). *Almost surely, the entire canonical free cluster family has only finitely many members of diameter greater than any given  $\varepsilon > 0$ . The same holds*

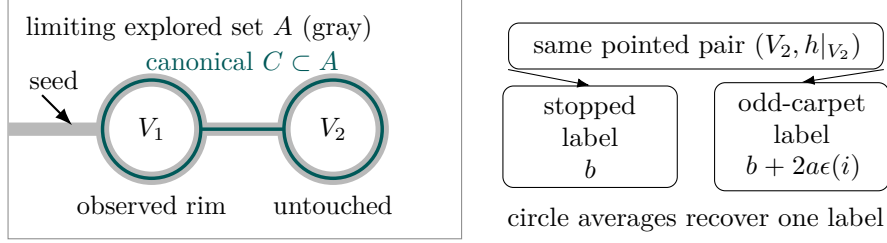


Figure 5: The hypothetical distinct-supplier case at the fixed marked stop. The first rim puts  $\partial V_1$  in  $A$ ; no crossing then puts the whole canonical cluster  $C$ , including  $\partial V_2$ , in  $A$ , while the discrete interior of  $V_2$  remains untouched. Its two proposed labels contradict [Theorem 9.1](#). The strokes depict inclusion and connectedness, not area or distances.

for the wired family, including all descendants. These bounds are compatible with the uniform tight macroscopic lattice counts.

*Proof.* Consider any finite family of  $M$  canonical clusters of diameter greater than  $\varepsilon$ . In each cluster choose two points more than  $\varepsilon$  apart. Density of odd rims gives two nearly as separated rim points, witnessed by finitely many nearby odd holes and dense interior test points. The individual holes need not have diameter comparable to  $\varepsilon$ . By [Theorem 9.2](#), the two witnesses in each cluster have one precursor, and the  $M$  precursors are distinct. For sufficiently small mesh each has diameter greater than  $\varepsilon/2$ . This finite choice involves only finitely many odd depths and marks.

Thus, on the joint representation, the event of at least  $M$  canonical clusters of diameter greater than  $\varepsilon$  implies at least  $M$  lattice clusters of diameter greater than  $\varepsilon/2$  eventually. Finite choices from the countable canonical family and the dense test list suffice; no measurable choice from an uncountable family is needed. Fatou's lemma and [Theorem 3.6](#) make the probability of this event tend to zero as  $M \rightarrow \infty$ . Countably many rational  $\varepsilon$  give the result. The same finite-witness argument can be performed in the wired descendants, with the boundary cluster counted once. This proof does not assume nested all-generation local finiteness in advance.  $\square$

## 9.2 A flat region cannot contain a revealed crossing

We give the analytic observation that detects cluster pieces with no visible jump. It concerns a distributional conditional mean, not merely a harmonic function on the open complement.

**Lemma 9.4** (Flat patches). *Let  $h$  be a zero-boundary GFF in a bounded simply connected domain  $O$ . Suppose an exploration with information  $\mathcal{F}_A$  describes a boundary-connected closed set  $A$  and a  $\mathcal{F}_A$ -measurable distribution  $h_A$  such that, conditionally on  $\mathcal{F}_A$ , the residual  $h^A = h - h_A$  is the GFF on  $O \setminus A$  extended by zero through  $A$ , as a distribution on  $O$ . The complementary components are assumed simply connected. Extra randomness may be included in the exploration. For every deterministic ball  $B \Subset O$ , almost surely on the event that  $h_A$  is one constant distribution throughout  $B$ , one has  $A \cap B = \emptyset$ . If the conditional decomposition is given only for tests supported in a neighborhood of  $B$  and on a  $\mathcal{F}_A$ -measurable event  $\Lambda$ , the conclusion holds on  $\Lambda$ .*

*Proof.* Fix a nonnegative smooth unit-mass mollifier  $\rho$  supported in the unit disk, and put  $h_u = h * \rho_u$ . For a nonnegative  $b \in C_c^\infty(B)$  set

$$X_u = u^{-1} \int b(x) \exp(i\beta h_u(x)) \, dx, \quad \beta^2 > 2. \quad (9.2)$$

On the compact support of  $b$ , the normalized Green kernel gives

$$\mathbb{E} \exp(i\beta(h_u(x) - h_u(y))) \leq C \left( \frac{u}{|x - y| \vee u} \right)^{\beta^2}.$$

Consequently

$$\mathbb{E}|X_u|^2 \leq Cu^{-2} \int_0^{O(1)} \left( \frac{u}{r \vee u} \right)^{\beta^2} r \, dr \leq C. \quad (9.3)$$

For a smooth Gaussian exponential cylinder  $Y = \exp(ih(\phi))$ , the Gaussian characteristic function and the bounds  $\text{Var}(h_u(x)) = \log(1/u) + O(1)$  and  $|\text{Cov}(h_u(x), h(\phi))| \leq C_\phi$  imply

$$|\mathbb{E}[X_u Y]| \leq C_\phi u^{\beta^2/2-1} \longrightarrow 0.$$

The linear span of these cylinders is dense in  $L^2$  of the field. Together with (9.3), this proves  $X_u \rightharpoonup 0$  weakly in  $L^2$ . It also proves weak convergence on the full probability space with extra randomness: replace any  $L^2$  test variable by its conditional expectation given  $h$ .

We next bound the conditional *mollified* variance near  $A$ . Let  $U$  be a simply connected component of  $O \setminus A$  and write  $r = |f^{-1}(w)|$ , where  $f : \mathbb{D} \rightarrow U$  maps zero to  $z$ . The conformal Green formula and the growth and quarter estimates proved or cited in [Section A.3](#) give

$$G_U(z, w) = -\log r, \quad |z - w| \leq \text{cr}_U(z) \frac{r}{(1-r)^2}.$$

For  $r \geq 1/2$  the Green function is at most  $\log 2$ ; for  $r < 1/2$  the second inequality gives

$$G_U(z, w) \leq \log 2 + \log^+ \frac{4 \text{cr}_U(z)}{|z - w|}.$$

Koebe's quarter theorem gives  $\text{cr}_U(z) \leq 4 \text{dist}(z, \partial U)$ . If  $\text{dist}(x, A) \leq u$  and  $z$  belongs to the mollifier at  $x$ , then  $\text{dist}(z, \partial U) \leq 2u$ . Different components have zero covariance, so the killed Green kernel satisfies on the mollifier supports

$$G_{O \setminus A}(z, w) \leq \log 2 + \log^+ \frac{32u}{|z - w|}. \quad (9.4)$$

Rescaling  $z = x + u\zeta$ ,  $w = x + u\omega$  and integrating the locally integrable logarithm against the two mollifiers yields

$$\text{Var}(h_u^A(x) \mid \mathcal{F}_A) \leq C_\rho \quad \text{if } \text{dist}(x, A) \leq u. \quad (9.5)$$

This is uniform in the exploration. It does not assert a finite pointwise variance for the GFF itself.

Choose  $B_0 \Subset B_1 \Subset B$  and take  $b = 1$  on  $B_1$ . Let  $E$  be the event that  $h_A = m$  throughout  $B$  and  $A \cap \overline{B_0} \neq \emptyset$ . This is an exploration-measurable event: obtain  $m$  by testing against one fixed smooth unit-mass function, and check equality of distributions on a countable dense test family. In the localized version replace  $E$  by  $E \cap \Lambda$ . On  $E$ , for small  $u$ ,

$$e^{-i\beta m} \mathbb{E}[X_u \mid \mathcal{F}_A] = u^{-1} \int b(x) \exp \left( -\frac{\beta^2}{2} \text{Var}(h_u^A(x) \mid \mathcal{F}_A) \right) dx. \quad (9.6)$$

All integrands are nonnegative. Since  $A$  is connected to  $\partial O$  and meets  $B_0$ , its  $u$ -neighborhood in  $B_1$  has area at least  $cu$ , with deterministic  $c > 0$ . To verify this without assuming path

connectedness, use the distance from a point in  $A \cap \overline{B_0}$ : its image on the connected set  $A \cup \partial O$  is an interval. Choose points at radial spacings  $3u$  up to a fixed positive radius small enough that they lie in  $B_1$ . Since  $B_1 \subseteq O$ , these points cannot lie on  $\partial O$  and therefore belong to  $A$ . Their disks of radius  $u/2$  are disjoint, lie in  $B_1$ , and are within distance  $u$  of  $A$ . There are at least  $c/u$  of them.

Using (9.5) in (9.6) therefore gives

$$\mathbb{E}[\mathbf{1}_E e^{-i\beta m} X_u] \geq c' \mathbb{P}(E).$$

The left side tends to zero by weak  $L^2$  convergence, since  $\mathbf{1}_E e^{-i\beta m}$  is a fixed bounded random variable. Thus  $\mathbb{P}(E) = 0$ . A countable exhaustion of  $B$  proves the lemma. For a random initial component one first conditions on it and subtracts its initial constant, then uses rational balls with positive clearance. The conclusion is still a probability-zero statement, so no bound uniform over all such domains is needed.  $\square$

### 9.3 Open hole witnesses and the discovered mean

A cluster with no limiting odd hole might still have many small discrete holes. The next observation shows that those holes cannot retain positive area away from the limiting cluster.

**Lemma 9.5** (Open witnesses for hole contributions). *Let  $O$  be open. Suppose compact sets  $C_n \subset X$  converge in Hausdorff distance to  $C$ , and  $C \cap O$  has zero area. Let  $U_n \subset O$  be open, with actual relative topological boundary contained in  $C_n^{\eta_n}$ , where  $\eta_n \rightarrow 0$ . If the indicators of  $U_n$  converge on a dense subset of  $O \setminus C$  to those of a union  $U$  of components of  $O \setminus C$ , then*

$$\mathbf{1}_{U_n} \longrightarrow \mathbf{1}_U \quad \text{in } L^1_{\text{loc}}(O).$$

*In particular, if every nonempty pointed open witness is absent, the area of  $U_n$  on each compact subset of  $O$  tends to zero.*

*Proof.* For  $x \notin C$ , choose a small ball  $B_x \subseteq O \setminus C$ . Its closure misses  $C_n^{\eta_n}$  for all large  $n$ . Membership in  $U_n$  is then constant throughout  $B_x$ ; otherwise the connected ball would meet its actual relative boundary. A dense test point in  $B_x$  determines this constant eventually. The indicator of  $U$  is also constant there, since  $B_x$  lies in one component of  $O \setminus C$ . Thus the indicators converge at every point outside  $C$ . Dominated convergence and the zero area of  $C$  prove the claim.  $\square$

We now specify the sweeps used for complete support identification. They are performed for each current family; the family label is suppressed in the sweep notation. For  $e \in \{\pm e_1, \pm e_2\}$ , visit current-lattice vertices in increasing order of  $e \cdot v$ , completing every earlier lattice column before beginning the next one. Within a column use a fixed lexicographic order. The virtual exterior of the free graph is the initial connected attachment. In the wired case first reveal the boundary cluster completely and use it, together with the exterior, as the attachment for all residual free chambers. When a vertex is visited, add the edge to its neighbor in the preceding column and reveal its entire current cluster. That neighbor has already been visited or belongs to the initial filled region; it is never an unsearched vertex farther ahead. The seed edges need not be current edges. Thus the initial region, the added seed, and all revealed clusters stay connected to the original boundary, and the exact cuts in Theorem 3.1 apply throughout.

We use two countable families of observable stops. A *column stop* ends after all vertices with  $e \cdot v \leq t$  have been visited, for a rational  $t$ . For rational  $b > 0$ , a *recorded stop* ends just after

the  $k$ th newly discovered free cluster of diameter at least  $b$ , for a fixed  $k \geq 1$ . If that discovery never occurs, give the stop a cemetery value. The rank  $k$  counts only these recorded discoveries; arbitrarily many smaller clusters may precede them and remain part of the explored set.

Write  $W_{0,\delta}$  for the initial revealed region: the boundary perimeter in the free case and the wired current support in the wired case. Its limit is respectively  $W_0 = \partial D$  or  $W_0 = C_\partial$ . Write  $S_\delta$  for the added moving seed and  $A_{\delta,\tau}$  for the full explored set at a fixed stop  $\tau$ . For the wired family this is one composite exploration of the original field, across all initial chambers. It does not choose a new lattice exploration from a limiting hole that might have separated from a shared discrete precursor.

Fix rational compact rectangles  $Q \subset \text{int } Q^+ \Subset D$  and a rational  $r > 0$ . On the event

$$\text{dist}(Q^+, W_{0,\delta} \cup S_\delta) \geq r,$$

call a discovered free cluster a *contributor to  $Q^+$*  if its support meets  $Q^+$  or its union  $\mathcal{O}_\delta(C)$  of odd component faces meets  $Q^+$ . List these clusters in their relative discovery order, as  $C_{\delta,1}, \dots, C_{\delta,L_\delta}$ . A listed cluster contains a seed contact. If its support meets  $Q^+$ , it has diameter at least  $r - O(\delta)$ . If an odd hole meets  $Q^+$ , choose a point of that hole there. The cluster surrounds that point and contains a seed contact at distance at least  $r$ . A point in a bounded complementary component lies in the convex hull of the surrounding compact set, so the latter's diameter is at least that distance, again up to the lattice boundary convention. Therefore, for all sufficiently fine meshes,

$$L_\delta \leq N_{\delta,0}(r/3), \tag{9.7}$$

where  $N_{\delta,0}(b) = \mathcal{N}_{\delta,\text{initial}}(b)$  counts all free clusters of diameter at least  $b$  in the initial chamber family, as in Equation (3.15). Theorem 3.6 gives a uniform tail for this right side conditional on the initial cut data. This is a pathwise bound valid at every observable stop; it is not a tail bound after fixing the completed history at that stop.

For these countably many fixed stops, retain the stopped states of Theorem 7.4. For each fixed choice of sweep, stop,  $Q, Q^+$ , and  $r$ , retain also the active clearance flag and, when active, the integer  $L_\delta$ , the ordered supports, their relative signs, and the membership bits of their odd-hole unions at all dense marks. Give the list a cemetery value when inactive. The integer is tight by Equation (9.7), and a fixed-length list has compact support coordinates in  $\mathcal{H}$ . A further diagonal extraction therefore makes  $L_\delta$  eventually equal to a finite  $L$  on the joint representation, with limits  $C_{*,1}, \dots, C_{*,L}$ . The order is reindexed by  $1, \dots, L_\delta$ ; raw discovery indices are not asserted to be tight. Retain also, on the global representation, the equality bits between each list entry and each marked supplier. A missing list entry or invalid address has equality bit zero. A list entry is *visible* if it eventually equals a supplier at some fixed valid marked address, and *invisible* otherwise. By Theorem 9.2, a visible entry has one associated canonical cluster, denoted  $C_j^{\text{can}}$ .

We retain one further kind of auxiliary list on the global representation. For each fixed  $(e, b)$ , let  $D_\delta(e, b)$  be the number of recorded discoveries during the complete sweep, and write their ordered supports as  $\hat{C}_{\delta,1}^{e,b}, \dots, \hat{C}_{\delta,D_\delta(e,b)}^{e,b}$ . The bound  $D_\delta(e, b) \leq N_{\delta,0}(b)$  makes the integer length tight. A further diagonal extraction makes it eventually equal to a finite  $D(e, b)$ , and makes each recorded support converge in  $\mathcal{H}$ ; missing ranks have the cemetery value. Retain also the equality bits between every record and every marked supplier, set to zero for a missing rank or invalid address. They are eventually constant on the joint representation. Call a stabilized record visible if one such bit is eventually one for a fixed valid address, and invisible otherwise. These lists keep their order and multiplicity even if several supports have the same limit. Their lengths

are auxiliary coordinates and are not identified with counts obtained by cutting off the limiting encoded family at diameter  $b$ .

Only coordinates measurable at the fixed stop generate its limiting sigma-field  $\mathcal{E}_\tau$ : the explored set, seed, observed labels, and discovered list data. The global representation also carries future marked suppliers, complete recorded lists, and the canonical field. They are not added to  $\mathcal{E}_\tau$  when applying the stopped conditional laws. In later uses  $r$  is chosen below a positive limiting clearance, so the discrete clearance flag is eventually active.

**Proposition 9.6** (The discovered mean). *Let  $\tau$  be one of the fixed stopped sweeps above, with limiting explored set  $A_\tau$ , and write  $M_\tau = \mathbb{E}[h \mid \mathcal{E}_\tau]$  as a distribution on tests. On each connected compact localization  $Q \subset \text{int } Q^+$  above at positive distance from  $W_0$  and the moving seed,  $M_\tau$  is the finite signed sum*

$$M_\tau = b_0 + 2a \sum_{\substack{j \text{ visible} \\ \text{in the localized list}}} \epsilon_j \mathbf{1}_{\mathcal{O}(C_j^{\text{can}})}. \quad (9.8)$$

Here  $b_0 = 0$  for the original free family. For the wired composite exploration,  $b_0 \in \{-a, a\}$  is the label of the initial pointed wired hole containing the region. The sum is finite on each such localization because its contributor list is finite. An invisible entry has zero contribution. Consequently  $M_\tau$  is a constant distribution on every ball missing  $W_0$ , the moving seed, and the canonical supports associated with its local contributors.

*Proof.* Work first with  $Q \subset \text{int } Q^+$  and a positive clearance as above, on an outcome where the list length has stabilized. Every  $C_{*,j}$  is contained in  $A_\tau$ . By [Theorem 7.4](#), it has zero area in  $\text{int } Q^+$ , and the actual relative boundary of  $\mathcal{O}_\delta(C_{\delta,j})$  lies within  $O(\delta)$  of  $C_{\delta,j}$ .

We identify membership at a dense mark  $q \in \mathcal{Q} \cap (\text{int } Q^+ \setminus C_{*,j})$ . If  $q$  belongs to the odd-hole union along infinitely many meshes, its positive distance from  $C_{*,j}$  gives a fixed disk about  $q$  inside that union on those meshes. The relevant odd depth is bounded by  $L$ . Indeed every enclosing odd ancestor is a distinct cluster: after entering its odd hole, later clusters lie strictly inside that hole. The exterior-connected seed must discover each such ancestor before  $C_{\delta,j}$ , and its odd-hole union contains  $q$ . It is therefore another entry of the same list for  $Q^+$ . After passing to a further subsequence, the depth is one fixed  $r \leq L$ , and the representative of  $q$  lies in the fixed disk. Thus  $C_{\delta,j}$  supplies the marked address  $(\iota, q, r)$ . The stabilized equality bit makes the entry visible, and [Theorem 9.2](#) identifies its canonical cluster and odd hole.

Conversely, suppose the entry is visible and  $q \notin C_{*,j}$  lies in an odd hole of  $C_j^{\text{can}}$ . That hole has the marked address  $(\iota, q, r)$  at a finite depth. The partition equality makes its supplier equal to the list entry for all sufficiently small mesh, so  $q$  eventually belongs to  $\mathcal{O}_\delta(C_{\delta,j})$ . These two implications show that the membership indicators converge at every dense mark off  $C_{*,j}$ . In the visible case their limit is

$$\mathbf{1}_{\mathcal{O}(C_j^{\text{can}}) \setminus C_{*,j}},$$

and in the invisible case it is zero. This also rules out oscillation between membership and nonmembership: any infinite membership subsequence supplies a fixed visible address, after which the converse gives eventual membership.

The displayed visible set is a union of components of  $\text{int } Q^+ \setminus C_{*,j}$ . In fact [Theorem 9.2](#) gives  $C_j^{\text{can}} \subset C_{*,j}$ , and the boundary of every canonical odd hole is contained in  $C_j^{\text{can}}$ ; a component of the smaller complement cannot cross such a boundary. Removing  $C_{*,j}$  changes no area on this region. We may therefore apply [Theorem 9.5](#) to each list entry. Its odd-hole indicators converge in  $L^1(Q)$  to the canonical odd-hole indicator, or to zero for an invisible entry. Thus even a collection of long or numerous collapsing holes has vanishing area unless it supplies an open marked witness.

The signs have stabilized and the list is finite, so the discrete signed sum in Equation (7.11) converges in  $L^1(Q)$  to the sum in the statement. For a visible entry its sign is the canonical sign because it eventually supplies a fixed marked address. In the wired case the connected rectangle  $Q^+$ , being disjoint from  $W_0$ , lies in one pointed initial wired hole. Compact containment puts it in one discrete initial hole for fine mesh, even if that hole pinches elsewhere. Its initial label is therefore one constant  $b_0 \in \{-a, a\}$  on  $Q^+$ . This supplies the additional baseline term. Remove a deterministic truncation  $L \leq M$  using Equation (9.7) and the initial count tail. The uniform higher logarithmic moments in Theorem 6.3 give uniform integrability when passing tested conditional means.

Finally, Theorem 7.4 identifies the full conditional mean on these tests, including across  $A_\tau$ , by its neighborhood removal argument. Hence the ordinary function just obtained is  $M_\tau$ , with no extra distribution on the explored set. On a connected ball avoiding  $C_j^{\text{can}}$ , its odd-hole indicator is constant. The baseline is also constant on a ball avoiding  $W_0$ . The localized list contains only finitely many entries, regardless of the diameters of their associated canonical clusters. This proves the last assertion.  $\square$

## 9.4 Complete matching, including invisible clusters

**Proposition 9.7** (Matching of all positive-diameter clusters). *Every positive-diameter Hausdorff limit of lattice free-current clusters is one retained canonical cluster, and every retained canonical cluster is the limit of its visible precursor. No distinct invisible precursor has a positive-diameter limit. The same assertions hold for the free primal descendants after the wired layer, jointly with that layer and the height. Consequently*

$$F_*^{\text{f}} = F(\mathcal{C}^{\text{f}}), \quad F_*^{\text{w}} = F(\mathcal{C}^{\text{w}}),$$

and both collections converge in the matching topology of Theorem 2.3.

*Proof.* Every point is on a canonical support or the initial wall. Let  $C_\delta$  be lattice free clusters with a positive-diameter Hausdorff limit  $C_*$ . The limit is compact and connected. Take  $x \in C_* \setminus W_0$  and a distinct point  $y \in C_*$ . Choose an axial direction  $e$  and a rational  $t$  with  $e \cdot y < t < e \cdot x$ . At the fixed column stop  $\tau(e, t)$ , the cluster  $C_\delta$  has been discovered for all sufficiently small mesh, because it has a point near  $y$  in an earlier column. A small rational ball about  $x$  lies ahead of the seed and has positive clearance from  $W_0$ . The same clearances hold at fine mesh. Choose  $Q \subset \text{int } Q^+$  around that ball and a rational lower clearance  $r$ . The target cluster is an entry in the finite localized list at this already specified stop.

Let  $\mathcal{U}_\tau$  be the finite union of the canonical supports associated with that list's visible entries. If  $x \notin \mathcal{U}_\tau$ , a smaller rational ball  $B$  around  $x$  misses  $\mathcal{U}_\tau$ ,  $W_0$ , and the moving seed. By Theorem 9.6,  $M_\tau$  is a constant distribution on  $B$ . Let  $\Lambda_{Q^+} = \{\text{dist}(Q^+, W_0 \cup S) > 0\}$ , the stopped-measurable clearance event for this localization. In the present initial domains it is the event in Equation (7.6). That identity gives, for every real smooth test  $\varphi$  supported in  $Q^+$ ,

$$\mathbf{1}_{\Lambda_{Q^+}} \mathbb{E} \left[ e^{i(h - M_\tau)(\varphi)} \mid \mathcal{E}_\tau \right] = \mathbf{1}_{\Lambda_{Q^+}} \exp \left\{ -\frac{1}{2} G_{D \setminus A_\tau}(\varphi, \varphi) \right\}. \quad (9.9)$$

It holds for all linear combinations of finitely many tests, with the killed GFF extended by zero across  $A_\tau$ . It therefore supplies the local residual law required by Theorem 9.4, including the mollifiers whose supports cross the explored set. Since  $A_\tau$  is boundary-connected and the complementary components are simply connected, that lemma says  $A_\tau \cap B = \emptyset$ . But  $C_\delta$  was revealed at this stop, so  $C_* \subset A_\tau$ , a contradiction.

The choice of  $B$  may depend on the later canonical data. Its existence implies one of the countably many  $\mathcal{E}_\tau$ -measurable zero events that  $M_\tau$  is constant on a fixed rational ball and  $A_\tau$  meets it. The conditional law used in each of these events is the one at the fixed stopped state. All directions, column thresholds, balls, and localizations come from countable families, and each localized list has finitely many entries. The preceding conclusion therefore holds simultaneously for every possible subsequential  $C_*$  and every  $x \in C_* \setminus W_0$ . We have proved

$$C_* \subset W_0 \cup \bigcup_{C \in \mathcal{C}_{\text{can}}} C, \quad (9.10)$$

where  $\mathcal{C}_{\text{can}}$  is the complete canonical free family under consideration. For wired descendants the argument used the single composite exploration of the original  $h$ . A ball away from  $W_0$  lies in one pointed wired hole and has a constant initial label, even when its discrete initial chamber pinches into several limiting holes.

The sets in the cover are compact and pairwise disjoint, including  $W_0$ : this follows from [Theorem 7.6](#) and the fact that each retained cluster lies strictly inside its parent. They form a countable family. Their intersections with  $C_*$  are therefore a disjoint closed cover of the compact connected set  $C_*$ . Sierpiński's continuum partition theorem [32], with a complete proof in [18, Chapter V, Theorem 5.10], says that at most one member of such a cover is nonempty. Consequently  $C_*$  is contained in one canonical cluster or entirely in  $W_0$ .

We exclude the latter case using the actual chamber wall. Let  $\Gamma_\delta$  be the component-side open  $W$  perimeter of the initial chamber containing  $C_\delta$ , with the medial resolution at pinches. In the wired case  $W_{0,\delta}$  is a current support, whereas the spatial estimate uses the attached matching wall. [Theorem 3.7](#) gives

$$\sup_{z \in C_\delta} \text{dist}(z, \Gamma_\delta) \leq \sup_{z \in C_\delta} \text{dist}(z, W_{0,\delta}) + O(\delta).$$

Its first-exit proof uses the resolved chamber face: a segment from an adjacent chamber point toward the revealed support first meets this chamber's own medial rim, including at a pinch. The free case uses the polygonal boundary. If  $C_* \subset W_0$ , Hausdorff convergence of  $C_\delta$  and  $W_{0,\delta}$  makes the right side tend to zero.

Choose one fixed buffered square  $Q_{\text{all}}$  containing  $X$  inside a larger regular ambient box, and use the event  $\Xi_\delta(\tau_0; b, \eta)$  of [Theorem 3.8](#) in this square, at the initial stop  $\tau_0$ . Its connected cell region contains the initial revealed support up to  $O(\delta)$ . If the limit above has diameter greater than  $2b$ , then for every fixed  $\eta > 0$ , at all sufficiently fine meshes  $C_\delta$  has a connected passage of diameter at least  $b$  in  $Q_{\text{all}}$  inside that  $\eta$ -collar. It is still unsearched at  $\tau_0$ . Thus the existence of such a limit forces membership in  $\liminf_n \Xi_{\delta_n}(\tau_0; b, \eta)$  for every fixed  $\eta$ . For any deterministic  $\eta_m \downarrow 0$ , Fatou's inequality and the bound in [Equation \(3.17\)](#), integrated over the initial cut data, give

$$\mathbb{P} \left[ \bigcap_m \liminf_n \Xi_{\delta_n}(\tau_0; b, \eta_m) \right] \leq \inf_m \limsup_{\delta \downarrow 0} \omega_{Q_{\text{all}}, b}(\eta_m, \delta) = 0.$$

The collar width is fixed before each mesh limit; its possible dependence on the future Hausdorff limit is used only for the eventual inclusion. Countably many rational  $b > 0$  exclude every positive-diameter limit contained in  $W_0$ . We conclude that  $C_*$  is contained in one canonical cluster.

*Visible precursors.* For the visible precursor of a canonical cluster  $C$ , [Theorem 9.2](#) puts  $C$  in every Hausdorff limit. The preceding containment puts that limit in a single canonical cluster.

Pairwise disjointness forces it to be  $C$  itself. Compactness of  $\mathcal{H}$  then shows that the entire visible precursor sequence converges to  $C$ . In particular every canonical cluster is obtained.

*Invisible precursors.* Return to any lattice sequence with positive-diameter limit  $C_* \subset C$ . Choose an axial direction  $e$  in which

$$w = \max_{z \in C_*} e \cdot z - \min_{z \in C_*} e \cdot z > 0,$$

and fix a rational  $b \in (0, w/4)$ . This sequence is eventually among the  $(e, b)$  records. Their length has stabilized to a finite number, so one fixed rank  $k$  supplies it along infinitely many meshes. The full sequence  $\hat{C}_{\delta, k}^{e, b}$  already has a Hausdorff limit, which must equal  $C_*$  on that subsequence. Replace the possibly alternating original sequence by this fixed recorded sequence. This replacement classifies its support limit; it makes no assertion that visibility transfers from the original choices.

If this record is visible, [Theorem 9.2](#) puts its associated canonical cluster inside  $C_* \subset C$ . Pairwise disjointness makes that cluster  $C$ , and the stabilized supplier bits identify the record with the visible precursor of  $C$  for all fine meshes. It was already treated. Suppose therefore that the record is invisible. Write  $C_\delta^{\text{vis}}$  for the visible precursor of  $C$ . It too is eventually a record because it converges to  $C$  and  $\text{diam } C \geq w > 4b$ . Among the finitely many stabilized ranks, exactly one has its supplier bit equal to one for the fixed chosen address of  $C$ . Thus  $C_\delta^{\text{vis}}$  has one fixed rank  $j$ , and  $j \neq k$ . The two ranks give one eventual discovery order. These complete-list coordinates remain outside the conditioning sigma-field of every fixed recorded stop. Smaller discoveries remain in each explored set; their local contributions are controlled by [Equation \(9.7\)](#).

Suppose first that the visible precursor is discovered earlier, at the fixed recorded stop  $\tau$ . Its limit is  $C$ . For each fixed  $\eta > 0$ , the convergences  $\hat{C}_{\delta, k}^{e, b} \rightarrow C_* \subset C$  and  $C_\delta^{\text{vis}} \rightarrow C$  imply that eventually the later cluster lies inside the  $\eta$ -neighborhood of the frozen cell region at  $\tau$ , using the  $O(\delta)$  displacement from  $A_{\delta, \tau}$ . It is still unsearched at  $\tau$  and has a passage of diameter at least  $b$  inside  $Q_{\text{all}}$ . Let

$$\mathcal{X}_\delta(b, \eta) = \{\exists 0 \leq i \leq D_\delta(e, b) : \Xi_\delta(\tau_i; b, \eta)\}$$

for the fixed  $(e, b)$  sweep in that square. The event under consideration is contained in  $\liminf_n \mathcal{X}_{\delta_n}(b, \eta)$  for every fixed  $\eta$ . For every integer  $K$ , [Equation \(3.18\)](#) and Fatou's inequality yield

$$\mathbb{P} \left[ \bigcap_m \liminf_n \mathcal{X}_{\delta_n}(b, \eta_m) \right] \leq C e^{-cK} + (K+1) \inf_m \limsup_{\delta \downarrow 0} \omega_{Q_{\text{all}}, b}(\eta_m, \delta) = C e^{-cK}.$$

Letting  $K \rightarrow \infty$  gives probability zero. The spatial bound is applied to the existential future passage under its allowed history  $\mathcal{H}_\tau$ , for finitely many fixed indices before the count tail is removed. Later visibility and limiting containment only give the eventual inclusion in the fixed- $\eta$  events; they are not added to that conditioning.

Suppose instead that the invisible cluster is discovered first, and stop immediately after that recorded discovery. Let  $t_\delta$  be the projection of its discovery column. Every earlier column has already been visited, so the first occupied column of this cluster satisfies

$$\left| t_\delta - \min_{z \in \hat{C}_{\delta, k}^{e, b}} e \cdot z \right| = O(\delta).$$

Hausdorff convergence preserves the minimum and maximum of the projection. Choose a point  $x \in C_*$  whose projection is more than  $w/2$  above the minimum. A rational ball about  $x$  lies a

fixed positive distance ahead of the moving seed for all fine meshes. It also has positive clearance from  $W_0$ , since  $C_* \subset C$  and  $C$  is compactly inside its initial parent. The stopped explored set meets that ball.

Use its localized contributor list at this fixed recorded stop. The newly discovered record meets the ball, so it is the last entry of that list and has the fixed index  $L$  after the length stabilizes. That entry is invisible: a stabilized marked supplier bit equal to one would also make the record visible. It has zero mean contribution by [Theorem 9.6](#). Every earlier relevant contributor, whether its support meets the ball or its odd hole surrounds a point there, is included in the same finite list. Its associated canonical support cannot be  $C$ : by [Theorem 9.2](#) that would make it the already chosen visible precursor of  $C$ , contrary to the discovery order. The earlier visible supports are therefore a finite family of compact sets disjoint from  $C$ . Shrink to a rational ball around  $x$  missing this union. Its baseline and all remaining odd-hole indicators are constant, so  $M_\tau$  is constant there. The full residual law [Equation \(9.9\)](#) applies at this stop, and [Theorem 9.4](#) contradicts the hit of the explored set. As before, the later choice of a rational ball only implies a stopped-data zero event and adds no conditioning. This excludes the second order and hence every distinct invisible record with a positive-diameter limit. Since every positive limit of an arbitrary lattice sequence was also the limit of a recurrent fixed record, the classification is exhaustive. There is only one visible rank for a given canonical cluster. A sequence distinct from that precursor along infinitely many meshes has a recurrent rank  $k \neq j$  on those meshes, by finiteness of the record list. If that rank were visible, its limiting support inside  $C$  would make it the same chosen supplier and hence rank  $j$  eventually. Thus it is an invisible recurrent rank, which has just been excluded.

*The collection topology.* For clarity, outer Hausdorff convergence has the following direct link with the individual limits just classified. If  $C \in F_*^\ell$  has positive diameter, then  $\text{dist}_{\mathcal{H}}(C, Z) > 0$ . A nearest member of  $F(\mathcal{C}_\delta^\ell)$  approaching  $C$  therefore cannot be a singleton for fine mesh; it is an actual lattice cluster. Conversely, every positive-diameter Hausdorff limit of lattice members belongs to  $F_*^\ell$ , because these encoded families converge and  $F_*^\ell$  is closed. The fixed sweeps and their finite localized lists made the preceding classification simultaneous, so it applies to these members even when they are selected from the outer limit.

It follows that the positive members of  $F_*^f$  are exactly  $\mathcal{C}^f$ . For  $F_*^w$ , a selected lattice member has a subsequence of one recurrent type: either the distinguished boundary cluster, whose limit is  $W_0 = C_\partial$ , or a free descendant, which has just been classified. The visible precursors give the reverse inclusion in both cases. Thus the two encoded limits equal those in the statement. [Theorem 2.2](#) converts this outer Hausdorff convergence to the stated matching topology. The exclusion of duplicate invisible precursors is a stronger conclusion retained in the auxiliary lists; the set-valued metric itself forgets multiplicity.  $\square$

## 9.5 Completion of the joint limit

*Proof of Theorem 2.3.* By [Theorem 6.1](#),  $K_\delta$  converges to  $h/a$  in every local Sobolev space  $H_{\text{loc}}^s$ ,  $s < -1$ . The arguments above can be performed on any simultaneous subsequence of the field, the wired current, and the free dual current. By [Theorem 7.3](#), the distinguished wired boundary cluster converges in ordinary Hausdorff distance to  $A_{-a,a}(h)$ . Its remaining pointed domains have the conditional zero-boundary GFF remainders of that same field.

Apply [Theorem 9.7](#) to the free dual family in the original domain and to the free primal descendants in the composite wired exploration. The positive outer  $\text{CLE}_4$  label is  $2\lambda$ , and its retained split has absolute values  $0, 2a$ ; the negative one is its reflection. The identified recursion is therefore exactly [Theorem 2.1](#), with its sign-sensitive choice of split, and with every generation

retained. Both families have been identified as canonical functions of the same limiting  $h$ .

For completeness, these are measurable collection-valued functions. Enumerate the recursive clusters by generation and by the first mark of  $\mathcal{Q}$  in each generating hole, using a cemetery value for missing entries. The cluster coordinates in this countable enumeration are measurable by [Theorem 7.6](#). Append  $C_{\partial}$  as one additional slot for the wired family. [Theorem 9.3](#) makes their singleton-augmented families closed. For every open set  $\mathcal{V} \subset \mathcal{H}$ , the event that  $F(C^i)$  meets  $\mathcal{V}$  is the union of the corresponding countably many cluster hit events, together with the deterministic alternative  $Z \cap \mathcal{V} \neq \emptyset$ . These hyperspace hit events prove measurability of both encoded factors. The boundary marker  $A_{-a,a}(h)$  is measurable in the same field.

The height and collection tightness give simultaneous joint subsequences, and every one has the pushforward of the GFF law by this single measurable construction. Hence the full mesh limit exists in the product topology, including the distinguished boundary set. Finally  $H_{\delta} = K_{\delta}/2$  and  $2a = \pi\sqrt{g}$ , giving precisely  $h/(\pi\sqrt{g})$ . The parameter inequalities used throughout include  $a = 2\lambda$ , so the Potts endpoint is part of the conclusion.  $\square$

## A Conventions and elementary corrections in the inputs

The proof uses the versioned statements of [\[16\]](#) specified in the text. We record elementary corrections needed when reading their proofs literally, and give the conformal estimate used in [Theorem 9.4](#). These are our reconciliations of the indicated formulas, not an author-issued erratum. They do not replace the substantive regularity, convergence, or free-energy theorems.

### A.1 The path decomposition in the increment estimate

In [\[16, Lemma 21.3\]](#), the printed path-length factor  $S'$  must be  $\max\{1, -S'\}$ , as used in its subsequent Equation (367). Indeed  $S'$  may be negative, whereas a nonconstant path has positive length. Here is a direct construction of the required path.

Order the increments so that  $|a - a'| \leq |b - b'|$ . If  $a = a'$ , use the empty path. Otherwise put  $R = |a - a'| > 0$ ,  $B = \{b, b'\}$ ,  $d = \text{dist}(\{a, a'\}, B)$ , and  $D = \max\{1, d\}$ . Fix an order  $k$  and write  $A = \exp(20k^2)$  and  $M = 1600A$ . We construct a path from  $a$  to  $a'$  with at most

$$20000A \max\{1, \log(R/D)\} \tag{A.1}$$

steps. Every step is either a nearest-neighbor step or an increment whose endpoint-set distance from  $B$ , divided by its length, is at least  $A$ . This is the alternative needed for the source's increment decomposition.

In the disk of radius  $R$  centered at  $(a + a')/2$ , choose  $z$  outside the at most four cones of half-angle 0.1 starting at  $x \in \{a, a'\}$  and pointing toward  $y \in B \setminus \{x\}$ . The total area of those cones within the disk is at most  $4(0.1)(3R/2)^2 = 0.9R^2$ , so such a point exists. The segment from  $x$  to  $z$  has length  $L \leq 3R/2$ . At arclength  $s$  along it, elementary triangle geometry gives distance at least  $(d_x + s)/100$  from  $B$ , where  $d_x = \text{dist}(x, B)$ : the excluded angle bounds the distance to each  $y \neq x$  by a fixed fraction of  $|x - y| + s$ . If  $y = x$ , the distance is exactly  $s$ , so the same bound holds. Round all segment points by one fixed nearest-lattice rule, with error at most one, using the same rounded endpoint  $z$  on both segments.

If  $d_x < M$ , start with a shortest Manhattan path to the rounded point at  $s_0 = \min\{L, M\}$ ; it takes at most  $2M + 2$  nearest-neighbor steps. If  $d_x \geq M$ , put  $s_0 = 0$ . On the remaining segment, increase  $s$  at each step by  $(d_x + s)/(800A)$ , or stop at its endpoint if that comes first. With

$\rho = d_x + s \geq M$ , the rounded step has length at most  $\rho/(400A)$  and both rounded endpoints have distance at least  $\rho/200$  from  $B$ . Its separation ratio is therefore at least  $2A$ . Delete repeated consecutive lattice points.

Geometric growth of  $d_x + s$  bounds the number of these sampled steps on a leg by

$$1 + 1600A \log \frac{d_x + L}{d_x + s_0}.$$

Whenever that leg is nonempty its denominator is at least  $D$ . Concatenating the first leg with the reversed second leg gives at most

$$6400A + 6 + 3200A \log \left( 1 + \frac{3R}{2D} \right) \leq 20000A \max\{1, \log(R/D)\}$$

steps, proving Equation (A.1). The case  $a = a'$  uses the empty path. The source orders the two increment lengths; if the other increment collapses, that ordering reduces to this same case. All constructed paths are finite and thus lie in sufficiently large exhausting volumes. Since  $S' = \log(D/R)$ , this supplies exactly the positive factor used in Equation (367), with a constant depending only on  $k$ .

## A.2 Slope, free energy, and the Gaussian multiplier

Let  $I_{\text{col}}$  be the left-minus-right arrow count in one vertical column of an  $M$  by  $L$  torus. Conservation makes it independent of the column. The unit-height slope is

$$s = I_{\text{col}}/L = I_{\text{total}}/(ML), \quad I_{\text{total}} = MI_{\text{col}}.$$

The free energy is still normalized per site, by  $(ML)^{-1} \log Z$ . Thus the sector in [16, Definition 4.2] is  $I_{\text{col}}/L = 2\lfloor Ls/2 \rfloor / L$ . Its printed denominator  $ML$  cannot apply to a *single-column* count:  $|I_{\text{col}}| \leq L$ , so a nonzero fixed slope would then disappear as  $M \rightarrow \infty$ . The corrected convention agrees with the filling fraction  $(1 - s)/2$  in its Equation (413) and with the per-site sector convention of [14, Section 1.2 and Theorem 2].

For completeness, the derivative calculation in [16, Equations (412) and (416)] then reads

$$f''(0) = -2 \left( \frac{\pi - \zeta}{\pi} \right)^2 \frac{\pi^2}{4(\pi - \zeta)} = -\frac{\pi - \zeta}{2} = -\arcsin(c/2).$$

The endpoint  $c = 2$ , or  $\zeta = 0$ , uses the endpoint calculation specified there. No factor of the torus length is part of this second derivative.

The conditional Gaussian comparison uses Dirichlet energy  $\text{Dir}(H) = \int |\nabla H|^2$ , without a factor  $1/2$ . For the neutral capacitary test  $\phi_A$  in [16, Section 25], the Gaussian variance is  $\sigma^2 / \text{Dir}(H_A)$ ; hence its logarithmic tail coefficient is  $-\text{Dir}(H_A)/(2\sigma^2)$ . The source's Equation (484) gives  $\text{Dir}(H_A) \leq 4\rho + C$ , with  $C$  independent of  $\rho$ . In its Equation (482) this yields  $f''(0) \geq -(1 + C/(4\rho))/\sigma^2$ ; letting  $\rho \rightarrow \infty$  gives  $f''(0) \geq -1/\sigma^2$ .

In the reverse inequality, retain the factor  $(1 - \varepsilon)^2$  in its Equation (485), because the circuit threshold in Equation (441) is  $(1 - \varepsilon)k$ . Combined with Equation (486), this gives  $(1 - \varepsilon)^4$  in the bound for  $f''(0)$ . Only after the prescribed large-scale limits let  $\varepsilon \downarrow 0$ ; the resulting inequality remains  $f''(0) \leq -1/\sigma^2$ . The geometric normalization in the two splitting estimates is consistent: Equations (503)–(504) use height  $2n(k - 24)$  across scale  $2nN$ , whereas Equations (523)–(524) use

height  $4n(k+4)+8$  across the same scale. These give slopes  $(k-24)/N$  and  $2(k+4)/N+o(1)$  respectively, with the circuit level reparametrization already made in Equations (440)–(441).

Other directional slips in that argument are read in the sense of the events defined there: after Equation (475) the inclusion is  $C_l \subset C_{l-4}$ ; at Equation (514) the ridge event is contained in the successful exploration event; and the good event at Equation (559) has *not too many* arms, as defined in Equation (562). The preceding  $\eta^{-2}$  bound can be retained in the combinatorial penalty of Equation (565), giving the required constant depending on  $(\rho, \varepsilon)$ . None of these readings changes the final order of limits or the multiplier.

Consequently  $\sigma^2 = 1/\arcsin(c/2)$ , under the full Gaussian convergence premise stated in [Theorem 6.14](#). The source GFF has logarithmic coefficient  $1/(2\pi)$ , so our coefficient is  $\kappa = \sigma^2/(2\pi) = 1/a^2$ . This explains both the slope convention and the conversion to  $H = K/2$ .

### A.3 The conformal growth estimate

The conformal mapping and coefficient-one Green identities used in [Theorems 9.2](#) and [9.4](#) are given in [1, Chapter 6, Sections 1.1 and 5.2]. Here is the short derivation of the growth inequality from published univalent-function inputs, so the numerical constant in [Equation \(9.4\)](#) does not depend on an unidentified draft edition. For a normalized univalent map  $f$  of the disk, the second coefficient obeys  $|a_2| \leq 2$  [11, Equation (2.68)]. Apply this to the normalized Koebe transform at  $z$  to obtain

$$\left| (1 - |z|^2) \frac{f''(z)}{f'(z)} - 2\bar{z} \right| \leq 4.$$

At  $z = re^{i\vartheta}$  this bounds the radial derivative of  $\log |f'(re^{i\vartheta})|$  by  $(4+2r)/(1-r^2)$ . Integration gives

$$|f'(re^{i\vartheta})| \leq \frac{1+r}{(1-r)^3}, \quad |f(re^{i\vartheta})| \leq \frac{r}{(1-r)^2}.$$

The last inequality follows by integrating along a radius. Together with the quarter theorem [11, Theorem 2.10.5], this gives precisely the conformal-radius bounds used in the flat-patch proof, for every proper simply connected domain, without boundary smoothness.

## References

- [1] Lars V. Ahlfors. *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable*. McGraw-Hill, New York, 3 edition, 1979.
- [2] Élie Aïdékon, Nathanaël Berestycki, Antoine Jégou, and Titus Lupu. Multiplicative chaos of the Brownian loop soup. *Proceedings of the London Mathematical Society*, 126(4):1254–1393, 2023. doi: 10.1112/plms.12511. URL <https://arxiv.org/abs/2107.13340v2>. Versioned source: arXiv:2107.13340v2, 20 October 2022.
- [3] Tomás Alcalde López and Avelio Sepúlveda. Excursion decomposition of the XOR-Ising model, 2026. URL <https://arxiv.org/abs/2602.06011v2>. arXiv:2602.06011v2, 25 March 2026.
- [4] Tomás Alcalde López, Lorca Heeney, and Marcin Lis. The Ising magnetisation field and the Gaussian free field, 2026. URL <https://arxiv.org/abs/2602.05886v1>. arXiv:2602.05886v1, 5 February 2026.
- [5] Juhan Aru and Avelio Sepúlveda. Two-valued local sets of the 2D continuum Gaussian free field: connectivity, labels, and induced metrics. *Electronic Journal of Probability*, 23:Paper No. 61, 35 pp., 2018. doi: 10.1214/18-EJP182. URL <https://arxiv.org/abs/1801.03828v2>. Versioned source: arXiv:1801.03828v2, 1 May 2018.

- [6] Juhan Aru, Avelio Sepúlveda, and Wendelin Werner. On bounded-type thin local sets of the two-dimensional Gaussian free field. *Journal of the Institute of Mathematics of Jussieu*, 18(3):591–618, 2019. doi: 10.1017/S1474748017000160. URL <https://arxiv.org/abs/1603.03362v3>. Versioned source: arXiv:1603.03362v3, 12 April 2017.
- [7] Juhan Aru, Titus Lupu, and Avelio Sepúlveda. First passage sets of the 2D continuum Gaussian free field. *Probability Theory and Related Fields*, 176(3–4):1303–1355, 2020. doi: 10.1007/s00440-019-00941-1. URL <https://arxiv.org/abs/1706.07737v3>. Versioned source: arXiv:1706.07737v3, 26 August 2019.
- [8] Juhan Aru, Titus Lupu, and Avelio Sepúlveda. The first passage sets of the 2D Gaussian free field: convergence and isomorphisms. *Communications in Mathematical Physics*, 375(3):1885–1929, 2020. doi: 10.1007/s00220-020-03718-z. URL <https://arxiv.org/abs/1805.09204v1>. Versioned source: arXiv:1805.09204v1, 23 May 2018.
- [9] Juhan Aru, Titus Lupu, and Avelio Sepúlveda. Excursion decomposition of the 2D continuum GFF. arXiv:2304.03150v2, 2023. URL <https://arxiv.org/abs/2304.03150v2>. Version of 3 October 2023.
- [10] J. Ashkin and E. Teller. Statistics of two-dimensional lattices with four components. *Physical Review*, 64(5–6):178–184, 1943. doi: 10.1103/PhysRev.64.178. URL <https://journals.aps.org/pr/abstract/10.1103/PhysRev.64.178>.
- [11] Kari Astala, Tadeusz Iwaniec, and Gaven Martin. *Elliptic Partial Differential Equations and Quasi-conformal Mappings in the Plane*, volume 48 of *Princeton Mathematical Series*. Princeton University Press, 2009.
- [12] Nathanaël Berestycki, Scott Sheffield, and Xin Sun. Equivalence of Liouville measure and Gaussian free field. arXiv:1410.5407v4, 2020. URL <https://arxiv.org/abs/1410.5407v4>. Version of 7 September 2020.
- [13] Hugo Duminil-Copin and Marcin Lis. On the double random current nesting field. *Probability Theory and Related Fields*, 175(3-4):937–955, 2019. doi: 10.1007/s00440-019-00899-0. URL <https://arxiv.org/abs/1712.02305v2>. Versioned source: arXiv:1712.02305v2.
- [14] Hugo Duminil-Copin, Karol Kajetan Kozłowski, Dmitry Krachun, Ioan Manolescu, and Tatiana Tikhonovskaia. On the Six-Vortex model’s free energy. *Communications in Mathematical Physics*, 395:1383–1430, 2022. doi: 10.1007/s00220-022-04459-x.
- [15] Hugo Duminil-Copin, Marcin Lis, and Wei Qian. Conformal invariance of double random currents I: Identification of the limit. *Proceedings of the London Mathematical Society*, 130(1):e70022, 2025. doi: 10.1112/plms.70022. URL <https://arxiv.org/abs/2107.12985v3>. Versioned source: arXiv:2107.12985v3, 6 January 2025.
- [16] Hugo Duminil-Copin, Karol Kajetan Kozłowski, Piet Lammers, and Ioan Manolescu. Gaussian free field convergence of the six-vertex model with  $-1 \leq \Delta \leq -\frac{1}{2}$ . arXiv:2603.06268v1, 2026. URL <https://arxiv.org/abs/2603.06268v1>. Version of 6 March 2026.
- [17] C. M. Fortuin, P. W. Kasteleyn, and J. Ginibre. Correlation inequalities on some partially ordered sets. *Communications in Mathematical Physics*, 22(2):89–103, 1971. doi: 10.1007/BF01651330.
- [18] Ronald C. Freiwald. *An Introduction to Set Theory and Topology*. Washington University in St. Louis, 2014. doi: 10.7936/K7D798QH. URL <https://www.math.wustl.edu/~freiwald/topology/Ch5.pdf>.
- [19] Sacha Friedli and Yvan Velenik. *Statistical Mechanics of Lattice Systems: A Concrete Mathematical Introduction*. Cambridge University Press, 2017. doi: 10.1017/9781316882603. URL [https://www.unige.ch/math/folks/velenik/smbook/Ising\\_Model.pdf](https://www.unige.ch/math/folks/velenik/smbook/Ising_Model.pdf). Proof locator in the author version dated 22 August 2017.

- [20] J. Fröhlich, R. Israel, E. H. Lieb, and B. Simon. Phase transitions and reflection positivity. I. general theory and long range lattice models. *Communications in Mathematical Physics*, 62(1):1–34, 1978. doi: 10.1007/BF01940327.
- [21] J. Fröhlich, R. Israel, E. H. Lieb, and B. Simon. Phase transitions and reflection positivity. II. lattice systems with short-range and Coulomb interactions. *Journal of Statistical Physics*, 22(3):297–347, 1980. doi: 10.1007/BF01014646.
- [22] John B. Garnett and Donald E. Marshall. *Harmonic Measure*, volume 2 of *New Mathematical Monographs*. Cambridge University Press, 2005. doi: 10.1017/CBO9780511546617.
- [23] Robert B. Griffiths. Correlations in Ising ferromagnets. I. *Journal of Mathematical Physics*, 8(3):478–483, 1967. doi: 10.1063/1.1705219.
- [24] Richard Holley. Remarks on the FKG inequalities. *Communications in Mathematical Physics*, 36(3):227–231, 1974. doi: 10.1007/BF01645980.
- [25] Antoine Jegou, Titus Lupu, and Wei Qian. Conformally invariant fields out of Brownian loop soups. arXiv:2307.10740v2, 2023. URL <https://arxiv.org/abs/2307.10740v2>. Version of 5 October 2023.
- [26] Marcin Lis. On boundary correlations in planar Ashkin–Teller models. *International Mathematics Research Notices*, 2022(13):9909–9940, 2022. doi: 10.1093/imrn/rnaa380. URL <https://arxiv.org/abs/2009.12131v2>. Proof locators refer to arXiv:2009.12131v2.
- [27] Marcin Lis. Spins, percolation and height functions. *Electronic Journal of Probability*, 27(35):1–21, 2022. doi: 10.1214/22-EJP761. URL <https://arxiv.org/abs/1909.07351v3>. Proof locators refer to arXiv:1909.07351v3.
- [28] Titus Lupu. From loop clusters and random interlacement to the free field. *The Annals of Probability*, 44(3):2117–2146, 2016. doi: 10.1214/15-AOP1019. URL <https://arxiv.org/abs/1402.0298v7>. Versioned source: arXiv:1402.0298v7, 9 July 2020.
- [29] OpenAI. The Gaussian free field limit of the balanced six-vertex model with variance multiplier  $1/\arcsin(c/2)$ . OpenAI Math Release preprint [OAI:The-Gaussian-free-field-limit-of-the-balanced-six-vertex-model-with-variance-multiplier-1-over-arcsin-c-over-2-September-23-2026](https://arxiv.org/abs/2609.11111), 2026.
- [30] OpenAI. The Gaussian free field limit of integer Lipschitz heights with two-arc boundary data. OpenAI Math Release preprint [OAI:The-Gaussian-free-field-limit-of-integer-Lipschitz-heights-with-two-arc-boundary-data-October-6-2026](https://arxiv.org/abs/2610.00000), 2026.
- [31] Franz Rys. Über ein zweidimensionales klassisches konfigurationsmodell. *Helvetica Physica Acta*, 36(5):537–559, 1963. doi: 10.5169/seals-113387.
- [32] Waclaw Sierpiński. Un théorème sur les continus. *Tôhoku Mathematical Journal, First Series*, 13:300–303, 1918. URL [https://www.jstage.jst.go.jp/article/tmj1911/13/0/13\\_0\\_300/\\_article/-char/en](https://www.jstage.jst.go.jp/article/tmj1911/13/0/13_0_300/_article/-char/en).
- [33] Kouta Usui. A note on reflection positivity and the Umezawa–Kamefuchi–Källén–Lehmann representation of two point correlation functions. arXiv:1201.3415v3, 2012. URL <https://arxiv.org/abs/1201.3415v3>. Version of 2 March 2012.
- [34] Dana P. Williams. Lecture notes on the spectral theorem, 2018. URL <https://www.math.dartmouth.edu/~dana/bookpapers/ln-spec-thm.pdf>. Lecture notes dated 10 August 2018.